QUT Centre for Robotics



ENN583 – Foundations of Robotic Vision

Sem 2, 2026

Prof Niko Suenderhauf

Welcome to Week 3!



CRICOS No.00213J

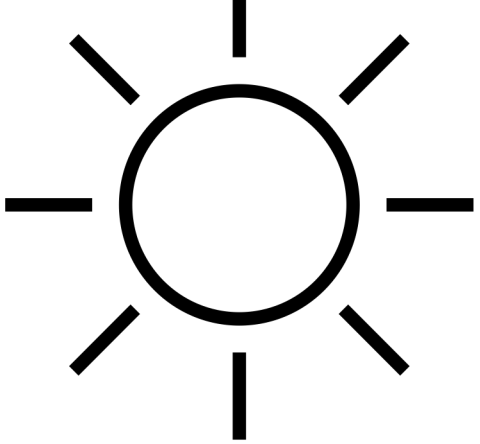
Cameras and how they see the World

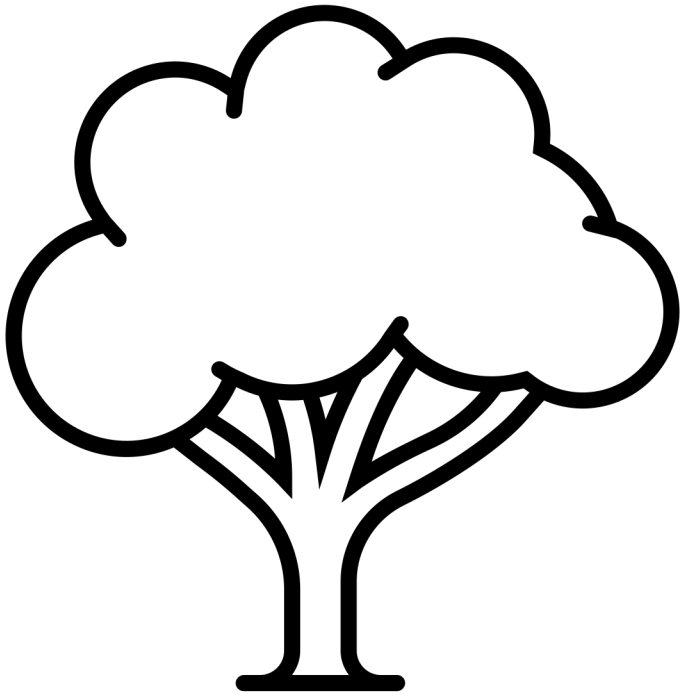
You carry multiple cameras with you! What do you already know about cameras?

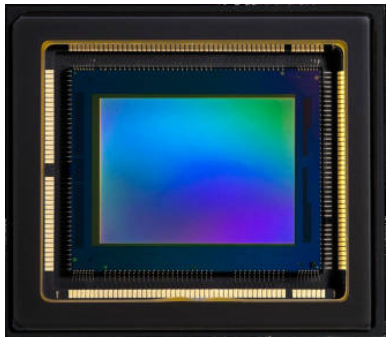
- What are the main **components** of a camera?
- How does a camera **capture an image**?
- Why are cameras **great sensors** for robots?



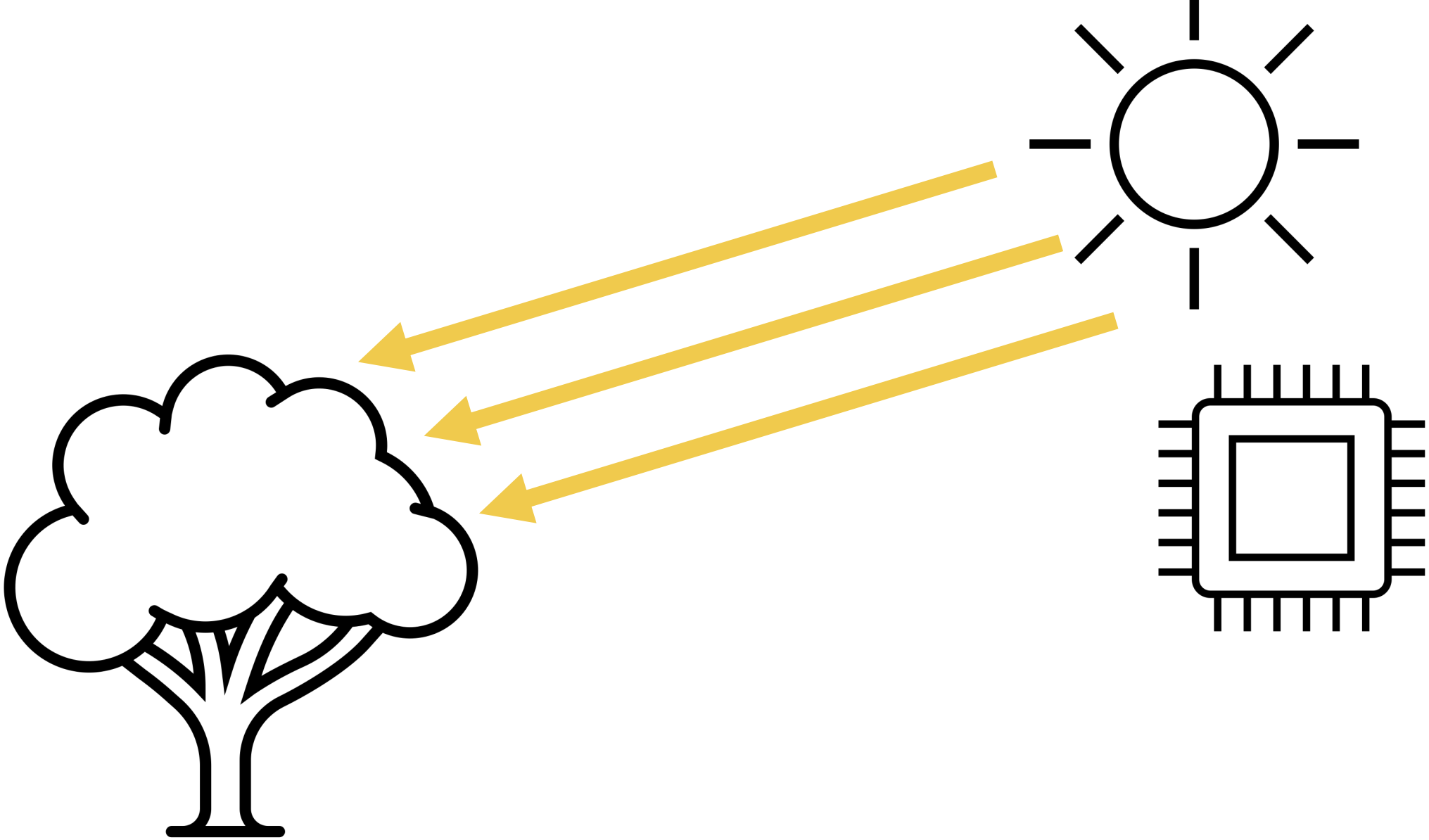
Let’s build the most simple camera!







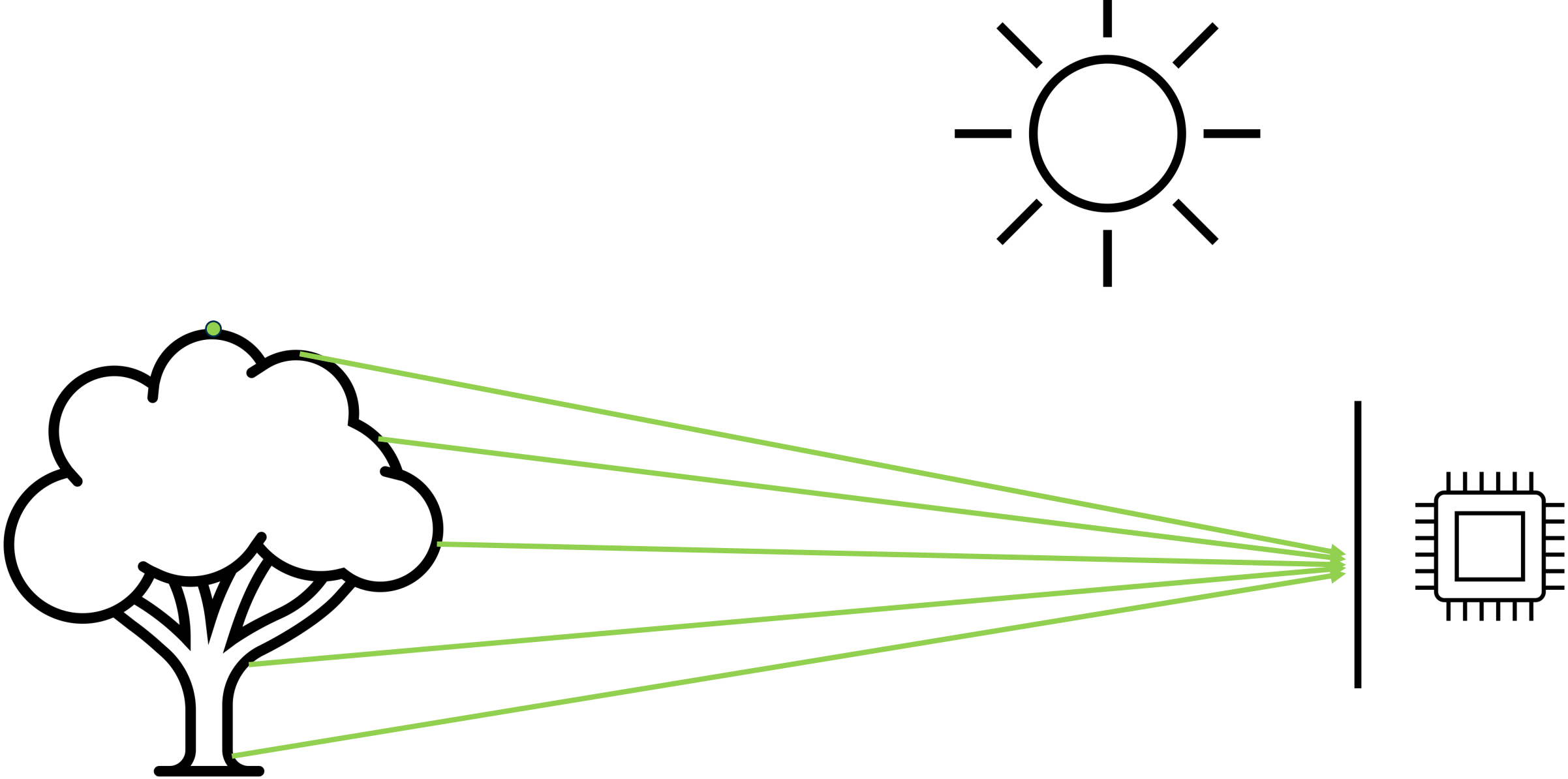
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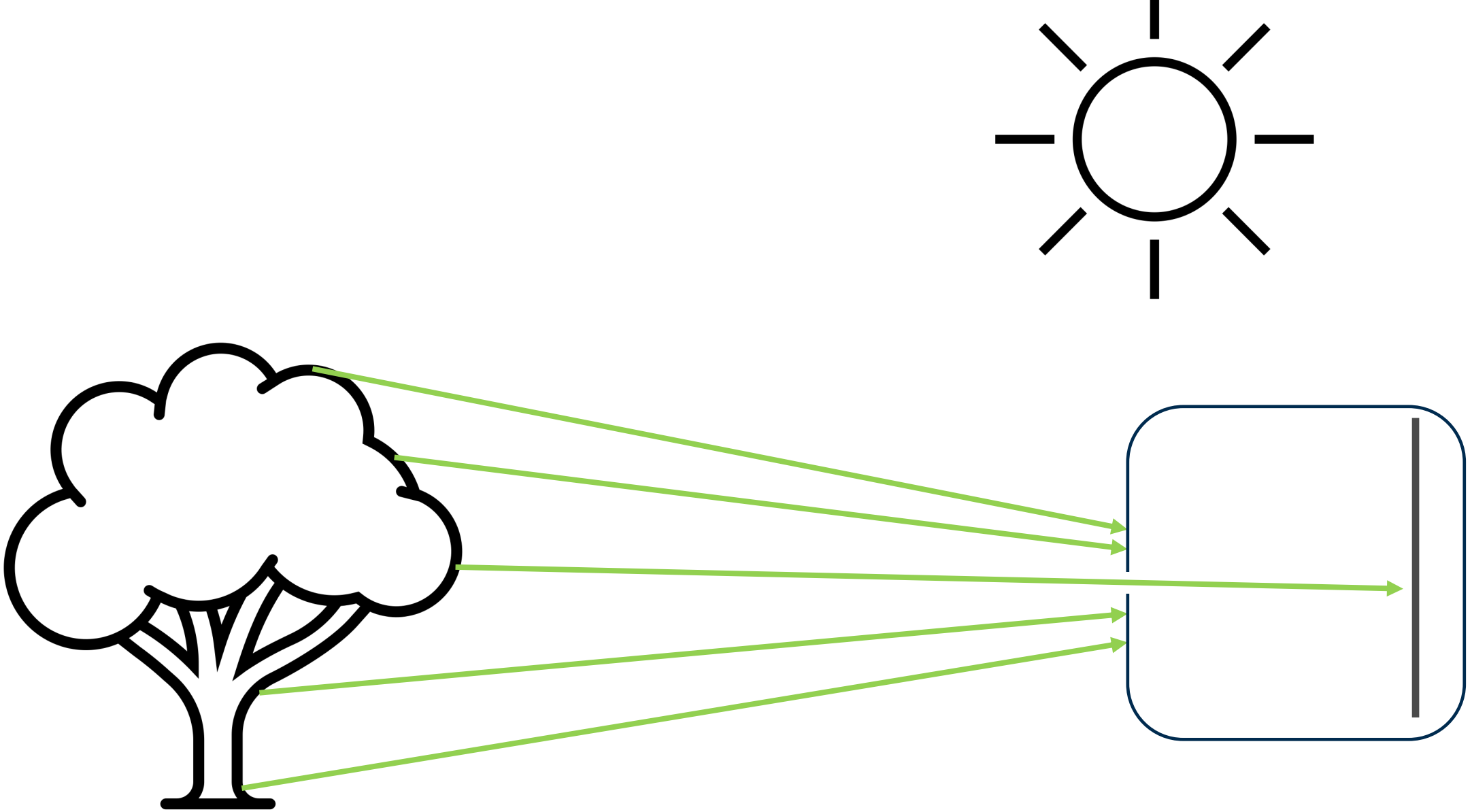
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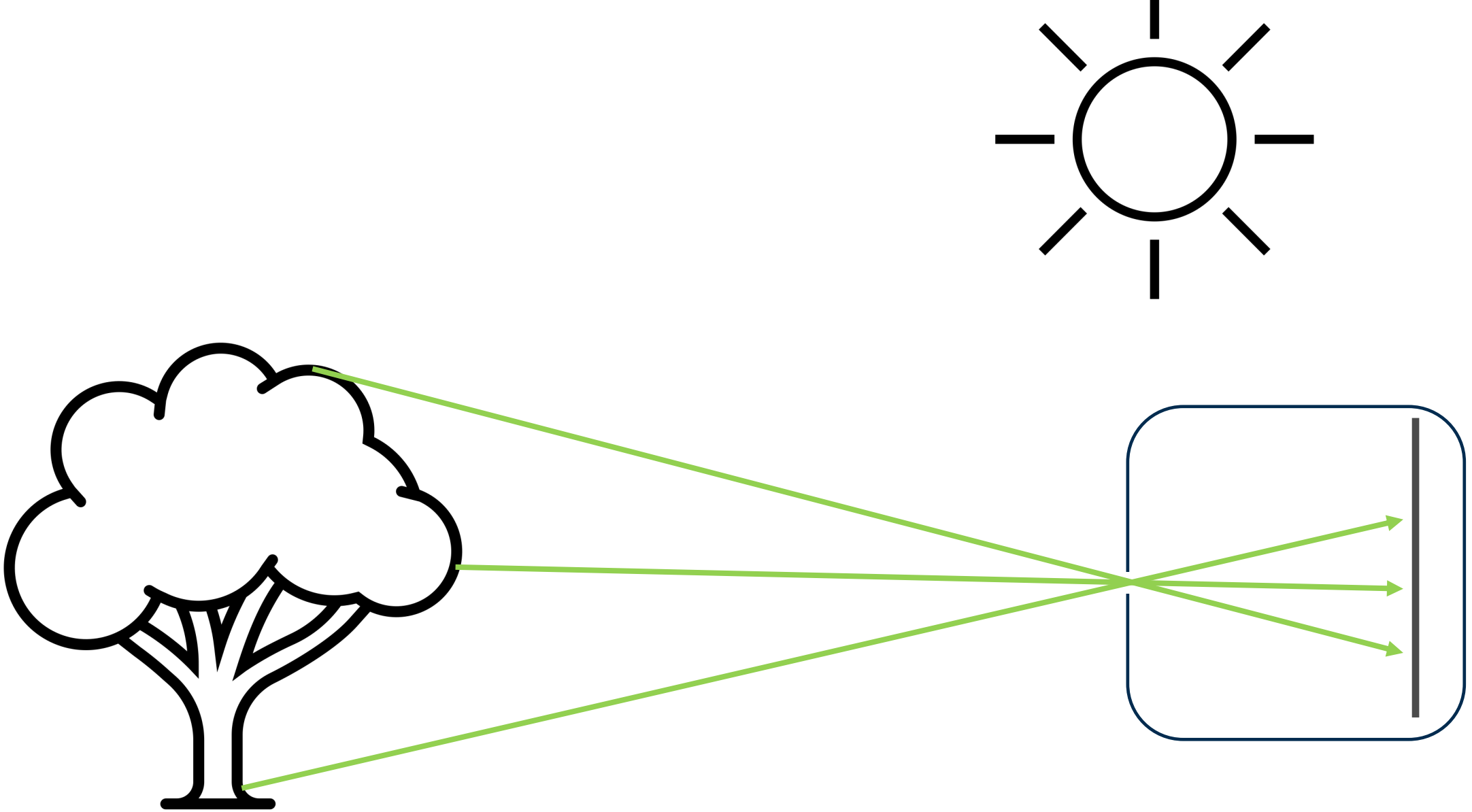
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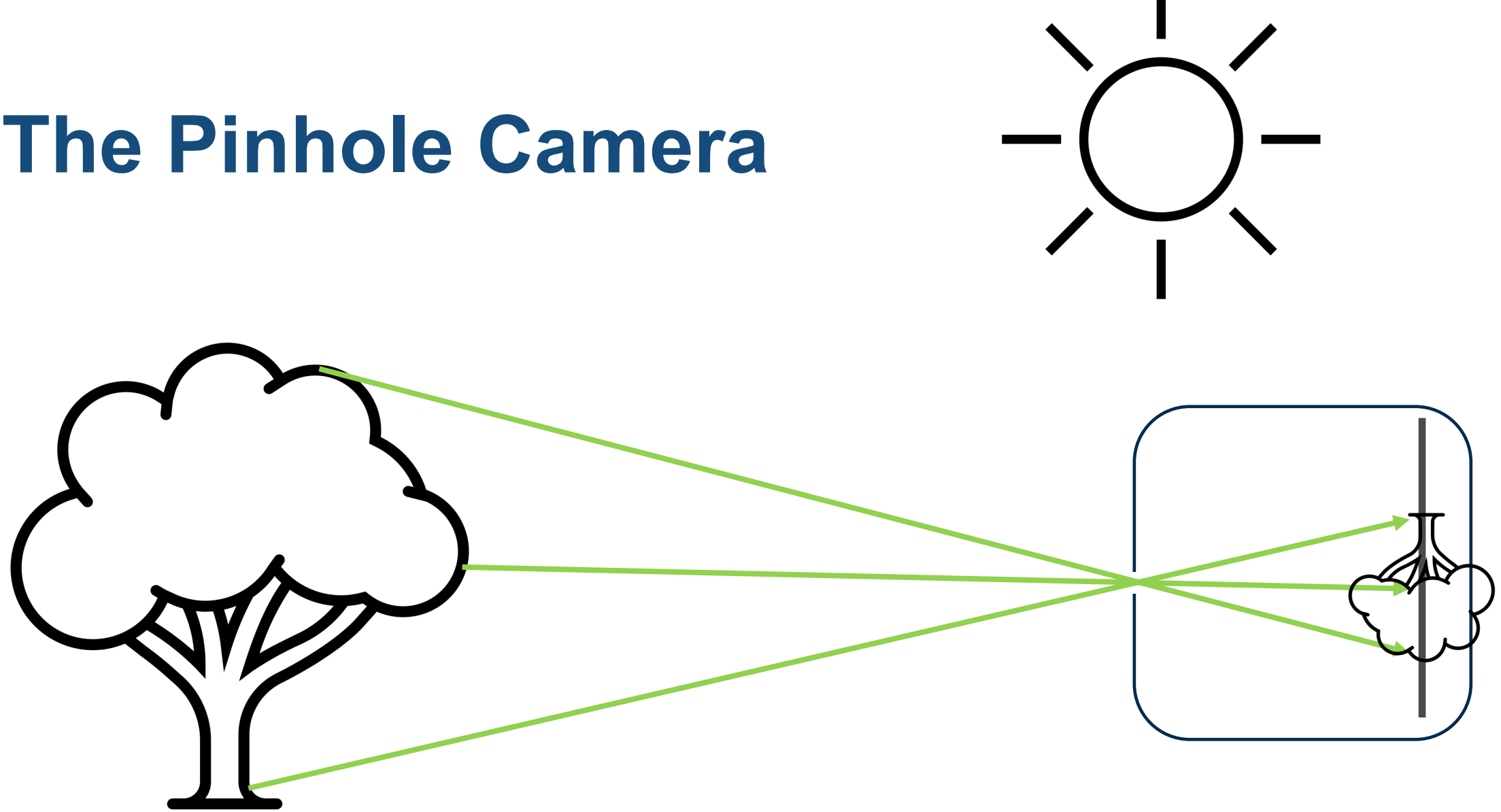


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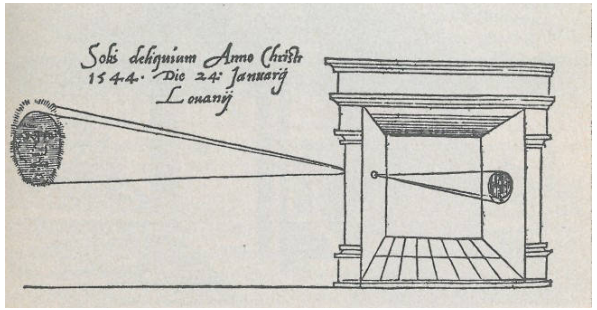
The Pinhole Camera



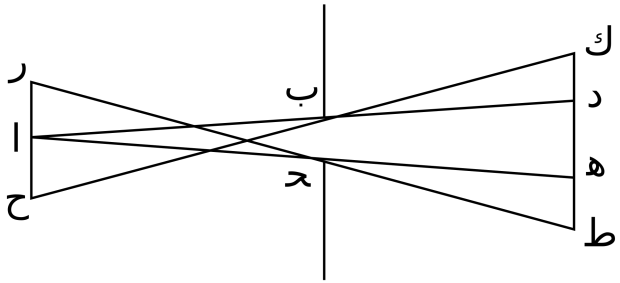
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Camera Obscura

- Earliest mention: **4th century BCE** (Mozi in China and Aristotle in Greece), potentially prehistoric use
- Rich history throughout renaissance (da Vinci and others) and 19th century
  - Check Wikipedia for more background
- Experiment: Making a Room-sized Camera Obscura (https://learn.zoner.com/experiment-making-a-room-sized-camera-obscura/)



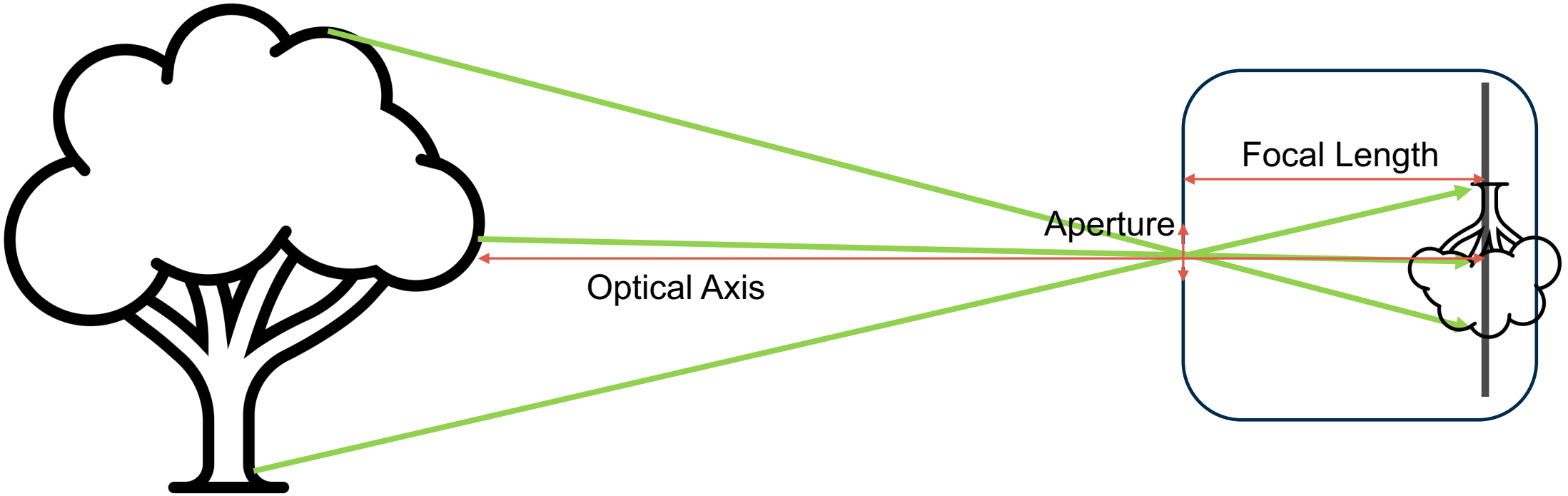
First published picture of camera obscura in Gemma Frisius' 1545 book De Radio Astronomica et Geometrica



11th century: Ibn al-Haytham's observations of light's behaviour through a pinhole

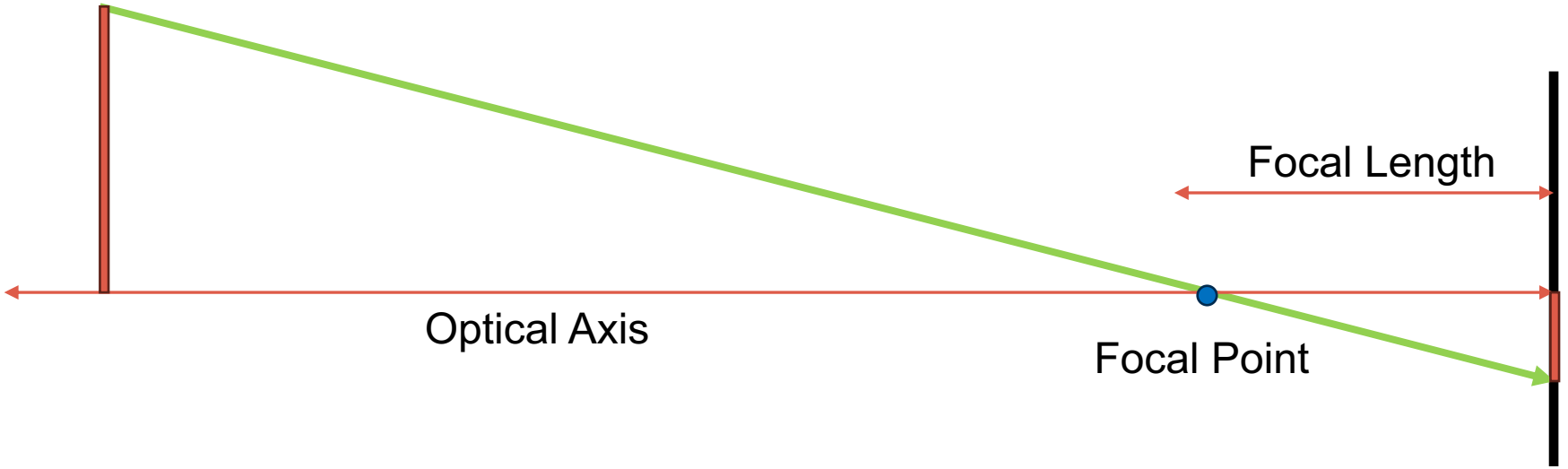
CRICOS No.00213J

The Pinhole Camera



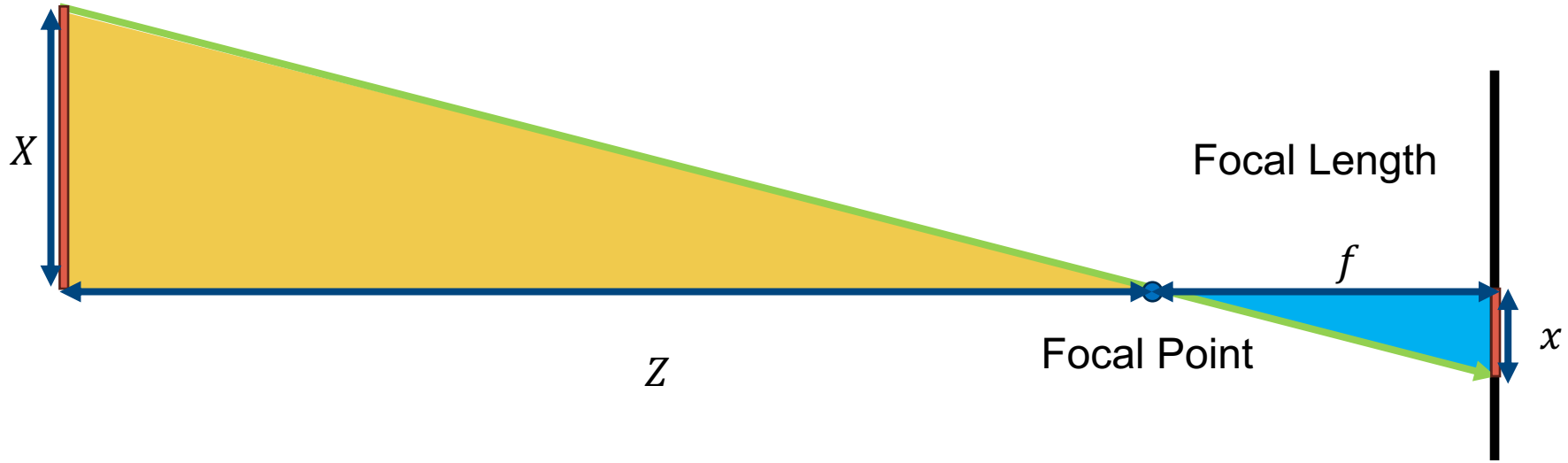
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The Pinhole Camera

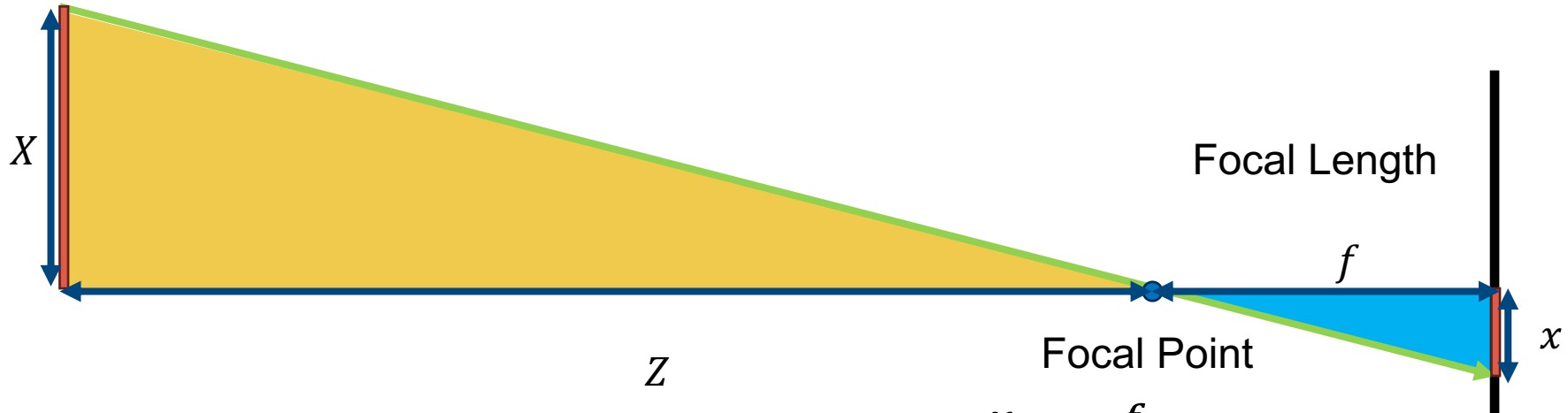


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The Pinhole Camera



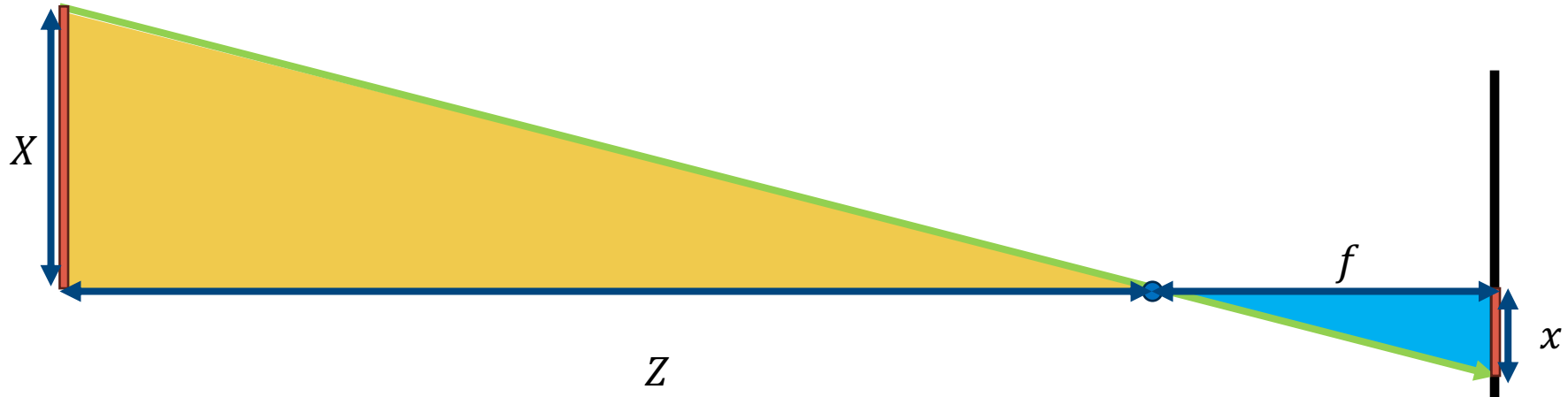
The Pinhole Camera



Relation from “similar triangles”: $\frac{x}{X} = \frac{f}{Z}$

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The Pinhole Camera

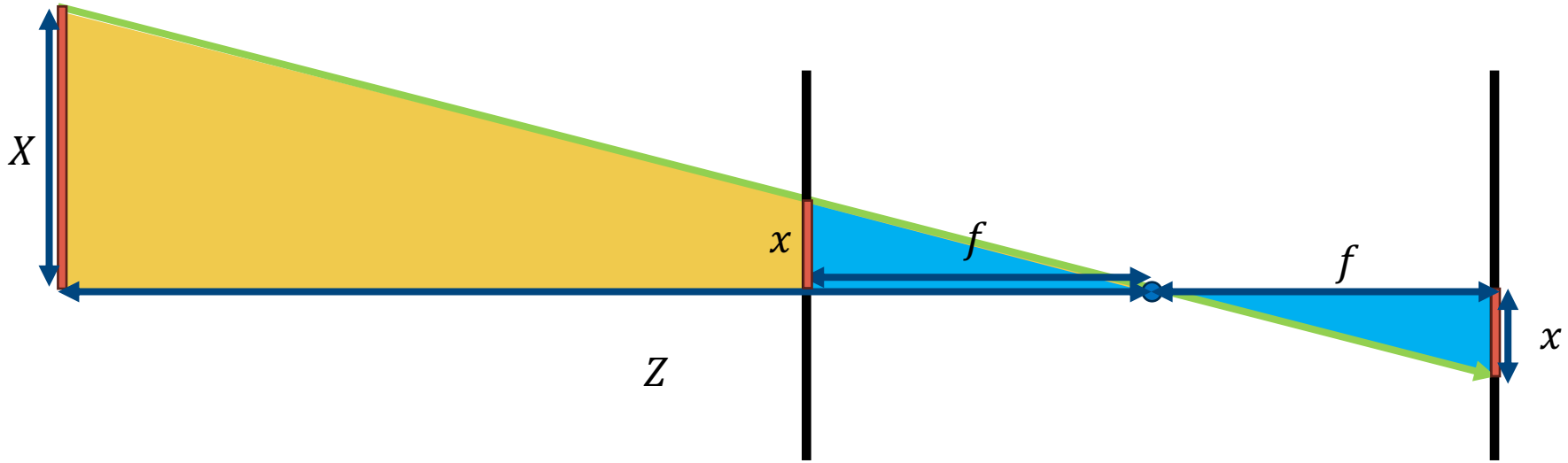


Relation from “similar triangles”: $\frac{x}{X} = \frac{f}{Z}$

$$x = X \cdot \frac{f}{Z} \quad \text{and} \quad y = Y \cdot \frac{f}{Z}$$

CRICOS No.00213J

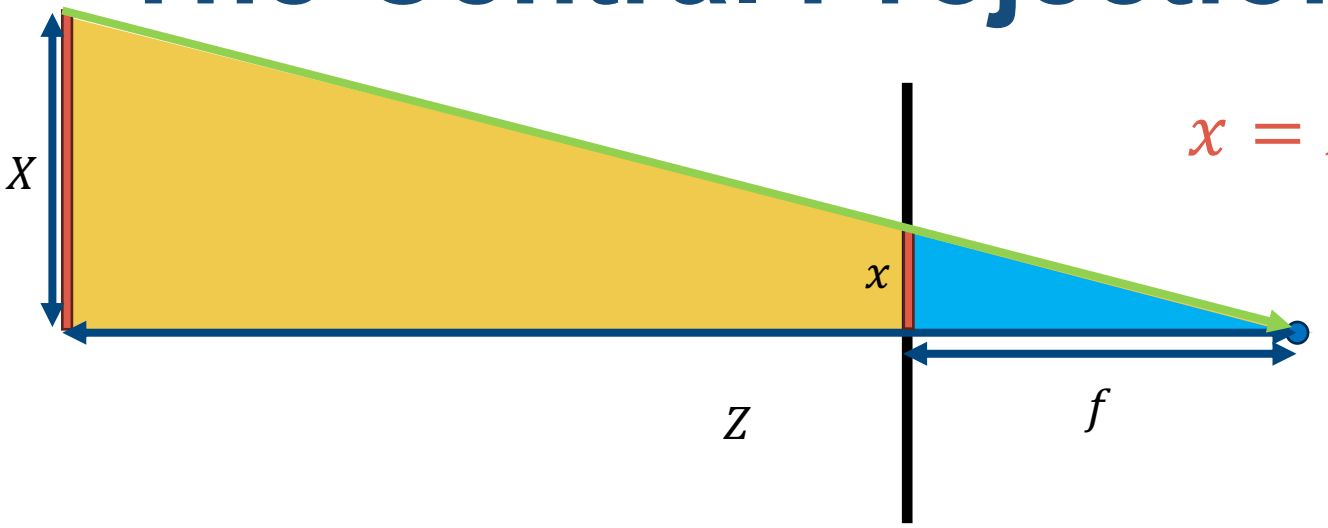
The Pinhole Camera



$$x = X \cdot \frac{f}{Z} \quad \text{and} \quad y = Y \cdot \frac{f}{Z}$$

CRICOS No.00213J

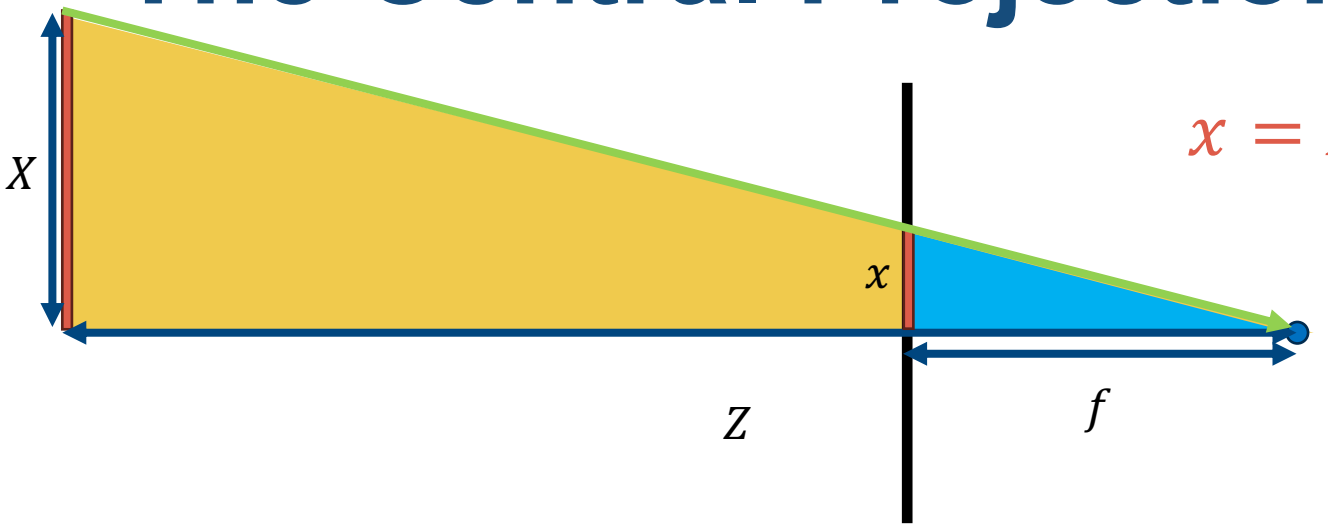
The Central Projection Model



$$x = X \cdot \frac{f}{Z} \quad \text{and} \quad y = Y \cdot \frac{f}{Z}$$

CRICOS No.00213J

The Central Projection Model

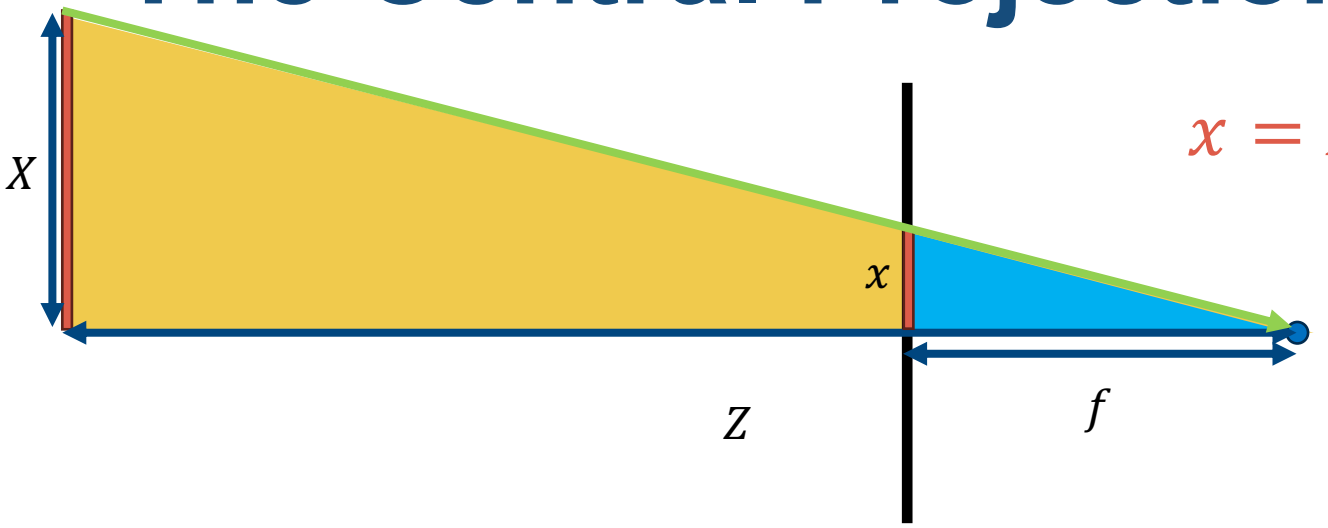


$$x = X \cdot \frac{f}{Z} \quad \text{and} \quad y = Y \cdot \frac{f}{Z}$$

$$\begin{pmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix} = \begin{pmatrix} f \cdot X \\ f \cdot Y \\ Z \end{pmatrix}$$

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The Central Projection Model



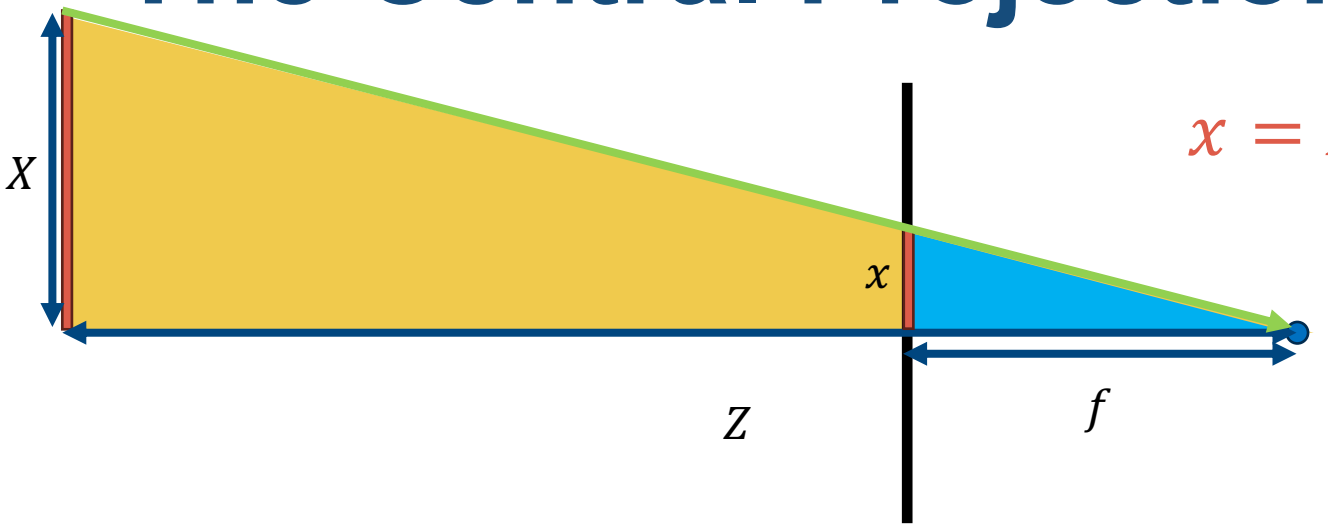
$$x = X \cdot \frac{f}{Z} \quad \text{and} \quad y = Y \cdot \frac{f}{Z}$$

These are 3D and 2D points in Homogeneous Coordinates!

$$\begin{pmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix} = \begin{pmatrix} f \cdot X \\ f \cdot Y \\ Z \end{pmatrix}$$

CRICOS No.00213J

The Central Projection Model



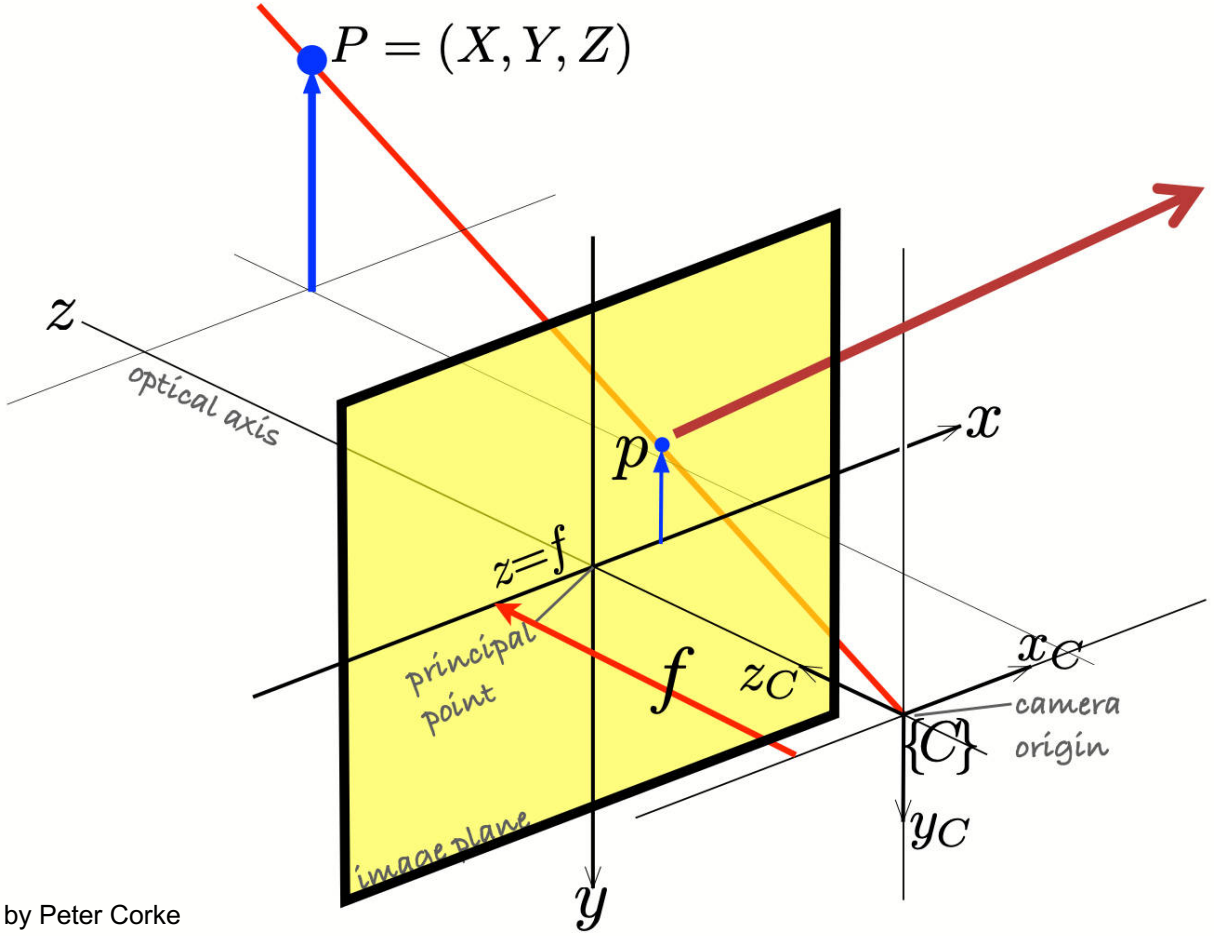
$$x = X \cdot \frac{f}{Z} \quad \text{and} \quad y = Y \cdot \frac{f}{Z}$$

To convert into a Carthesian 2D point: Divide by the 3rd element in the homogeneous vector.

$$\begin{pmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix} = \begin{pmatrix} f \cdot X \\ f \cdot Y \\ Z \end{pmatrix} \Longrightarrow \begin{pmatrix} \frac{f \cdot X}{Z} \\ \frac{f \cdot Y}{Z} \end{pmatrix}$$

CRICOS No.00213J

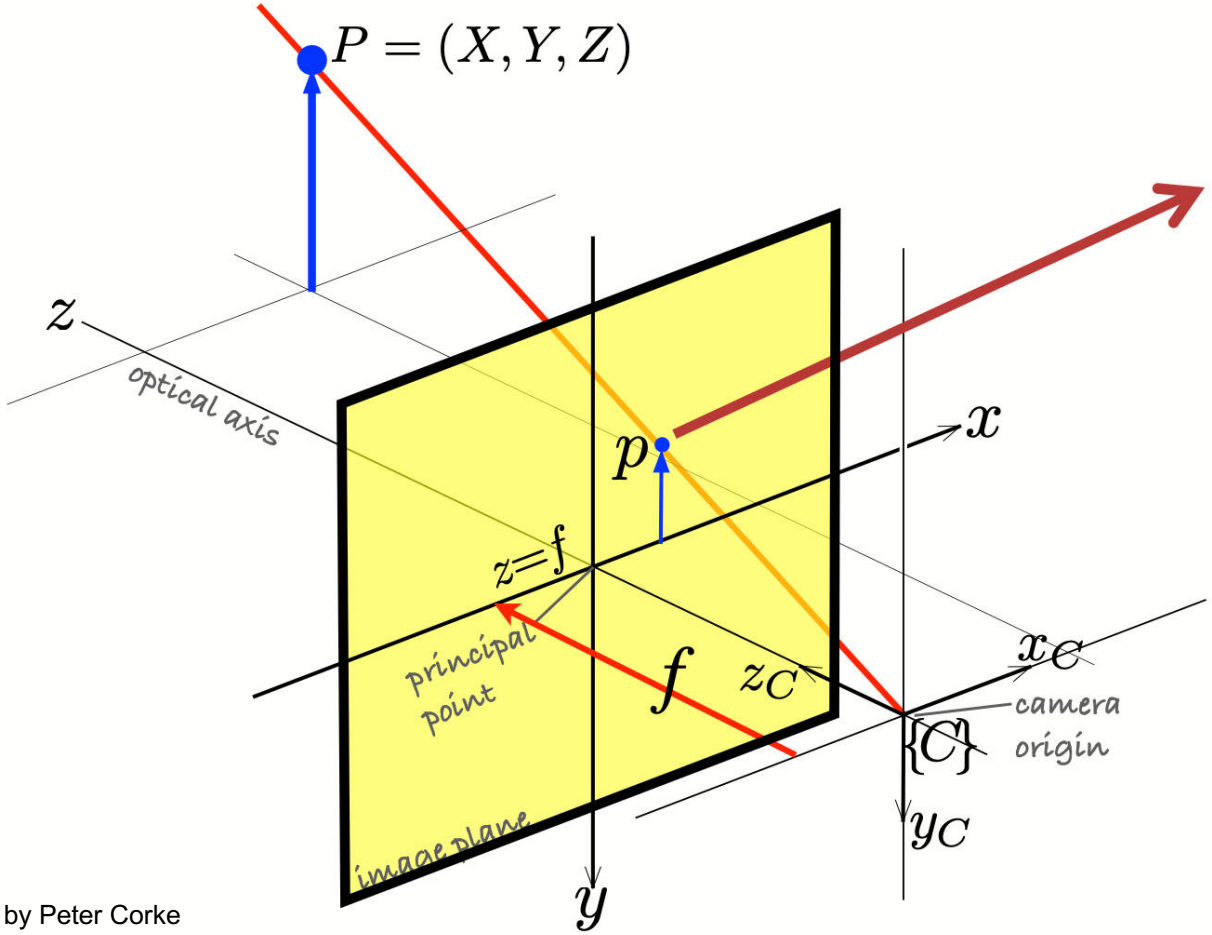
The Central Projection Model



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CRICOS No.0021

The Central Projection Model



$$\begin{pmatrix} \tilde{x} \\ \tilde{y} \\ \tilde{z} \end{pmatrix} = \begin{pmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

$$\tilde{x} = fX, \tilde{y} = fY, \tilde{z} = Z$$

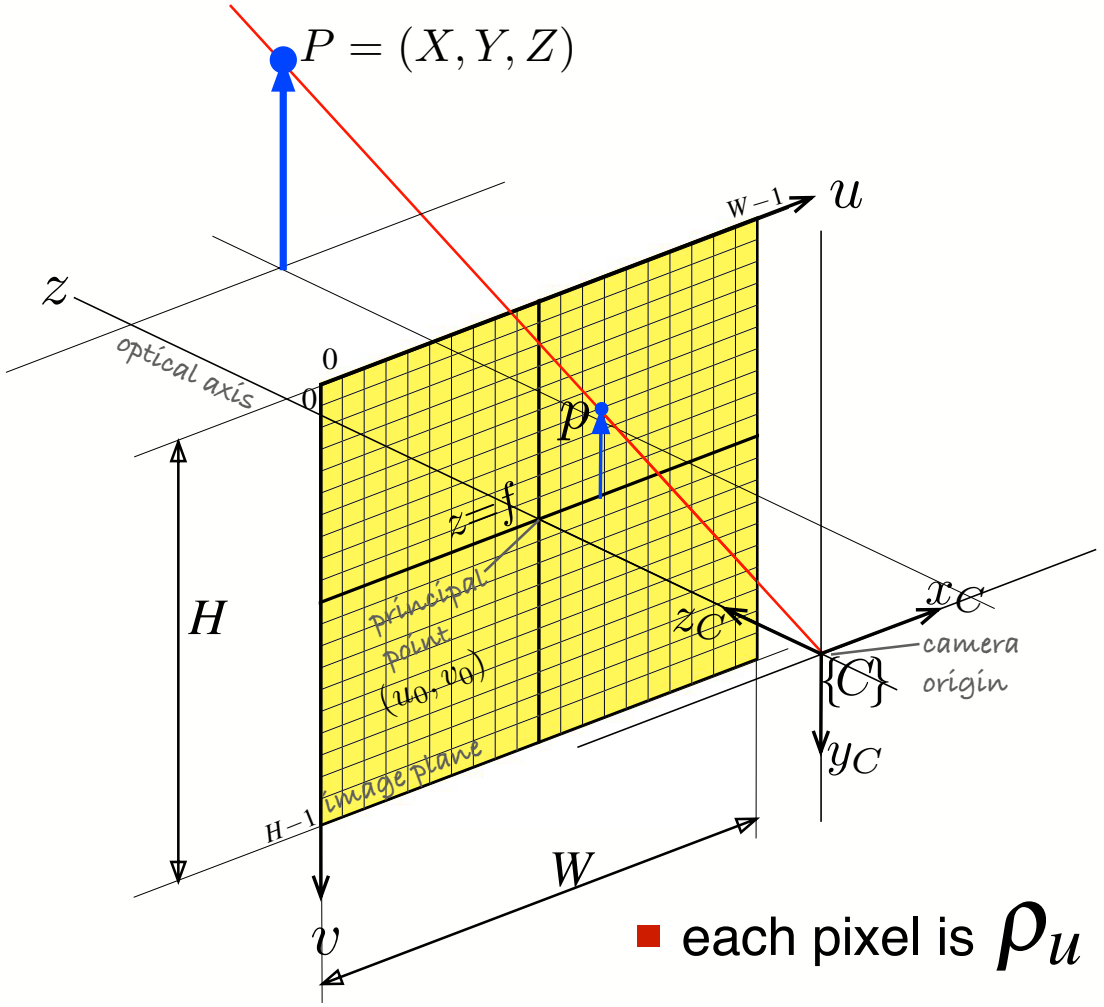
$$x = \frac{\tilde{x}}{\tilde{z}},\ y = \frac{\tilde{y}}{\tilde{z}}$$

$$\Rightarrow x = \frac{fX}{Z}, y = \frac{fY}{Z}$$

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CRICOS No.0021

Change of coordinates

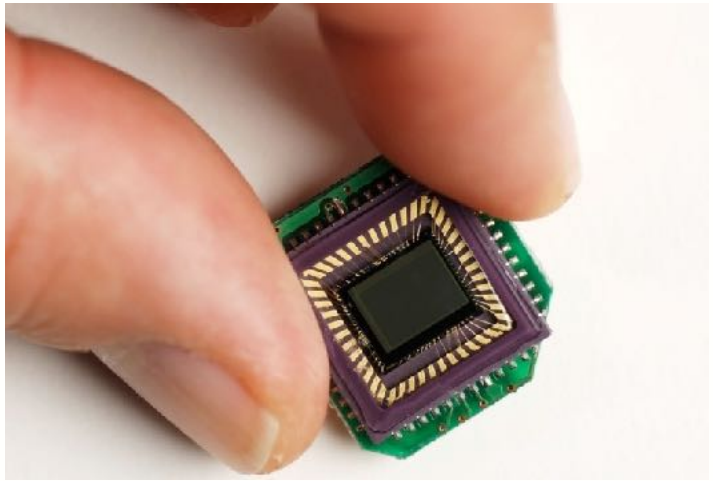


- each pixel is $\rho_u \times \rho_v$

- scale point from metres to pixels
- shift the origin to top left corner

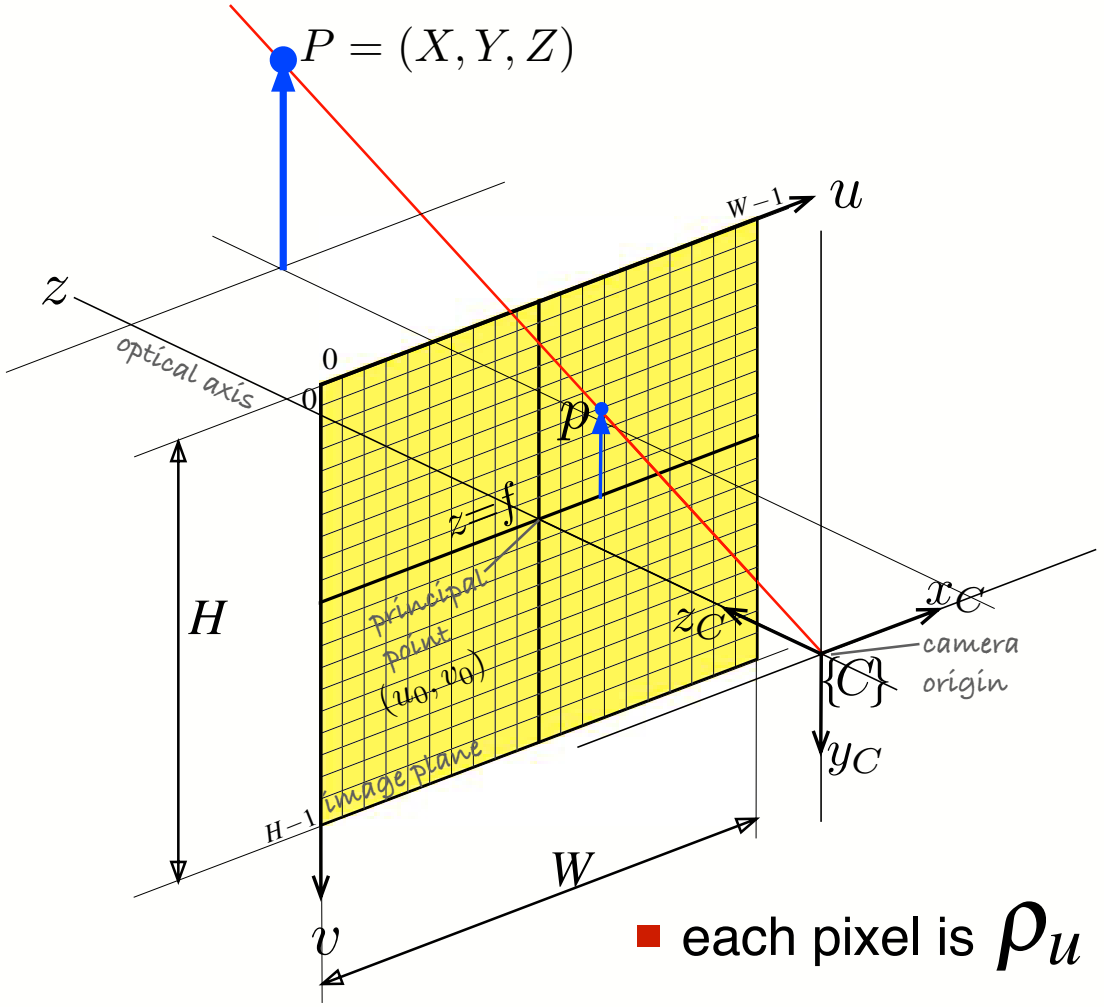
$$u = \frac{x}{\rho_u} + u_0$$

$$v = \frac{y}{\rho_v} + v_0$$



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Change of coordinates



- each pixel is $\rho_u \times \rho_v$

- scale point from metres to pixels
- shift the origin to top left corner

$$u = \frac{x}{\rho_u} + u_0$$

$$v = \frac{y}{\rho_v} + v_0$$

$$\begin{pmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{pmatrix} = \begin{pmatrix} \frac{1}{\rho_u} & 0 & u_0 \\ 0 & \frac{1}{\rho_v} & v_0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \tilde{x} \\ \tilde{y} \\ \tilde{z} \end{pmatrix}$$

$$p = \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} \tilde{u}/\tilde{w} \\ \tilde{v}/\tilde{w} \end{pmatrix}$$

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Complete camera model

$$\begin{pmatrix} \tilde{x} \\ \tilde{y} \\ \tilde{z} \end{pmatrix} = \begin{pmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{pmatrix} = \underbrace{\underbrace{\begin{pmatrix} \frac{1}{\rho_u} & 0 & u_0 \\ 0 & \frac{1}{\rho_v} & v_0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}}_{\mathbf{K}} \begin{pmatrix} \mathbf{R} & t \\ \mathbf{0}_{1\times 3} & 1 \end{pmatrix}^{-1}}_{\mathbf{C}} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

extrinsic parameters

intrinsic parameters

camera matrix

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Camera matrix

- Mapping points from **the world** to an **image (pixel) coordinate** is simply a **matrix multiplication** using **homogeneous coordinates**

$$\begin{pmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{pmatrix} = \begin{pmatrix} C_{11} & C_{12} & C_{13} & C_{14} \\ C_{21} & C_{22} & C_{23} & C_{24} \\ C_{31} & C_{32} & C_{33} & C_{34} \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

$$u = \frac{\tilde{u}}{\tilde{w}}, v = \frac{\tilde{v}}{\tilde{w}}$$

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Scale invariance

- Consider an arbitrary scalar scale factor

$$\begin{pmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{pmatrix} = \lambda \begin{pmatrix} C_{11} & C_{12} & C_{13} & C_{14} \\ C_{21} & C_{22} & C_{23} & C_{24} \\ C_{31} & C_{32} & C_{33} & C_{34} \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

- $\tilde{u}, \tilde{v}, \tilde{w}$ will all be scaled by λ
- but $u = \frac{\tilde{u}}{\tilde{w}}, v = \frac{\tilde{v}}{\tilde{w}}$
- so the result is unchanged

Normalized camera matrix

- Since scale factor is arbitrary we can fix the value of one element, typically C(3,4) to one.

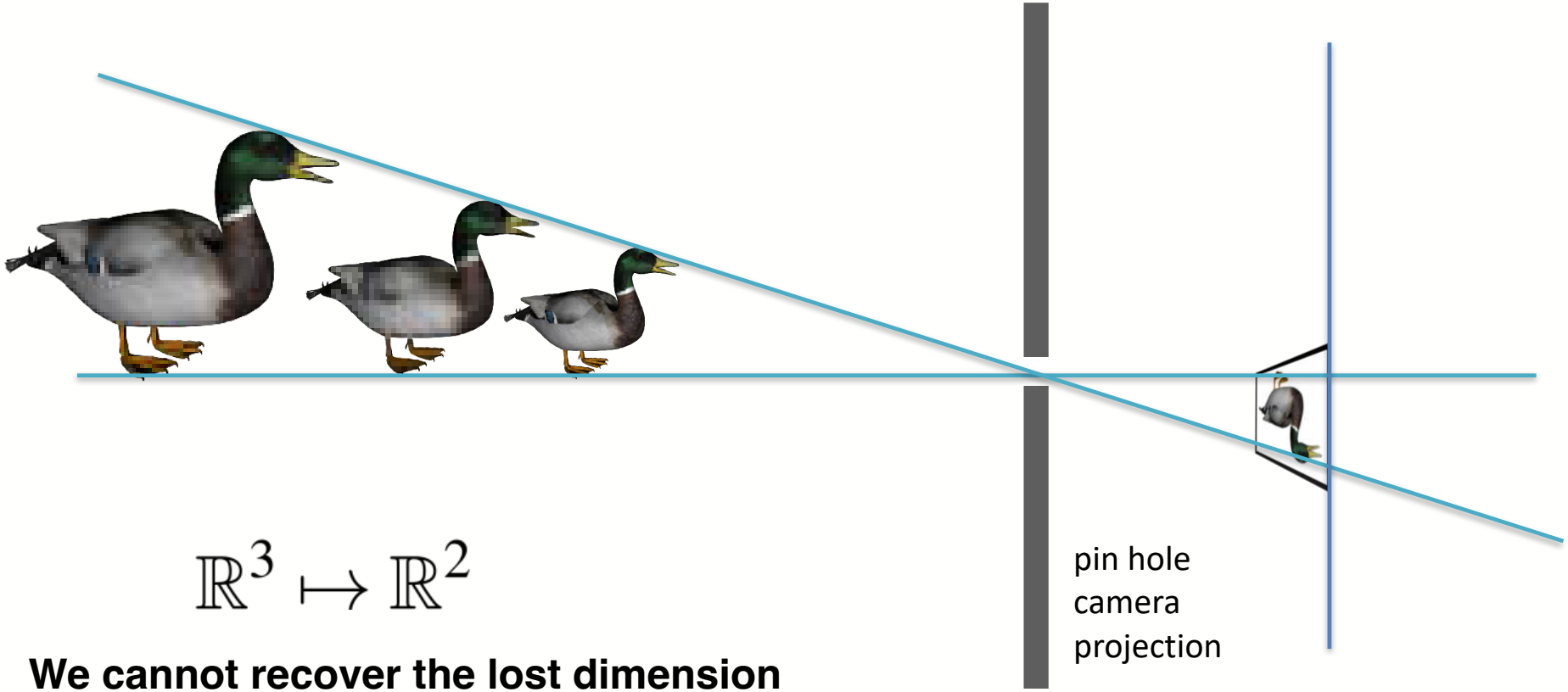
- focal length
- pixel size
- camera position
- & orientation

$$\begin{pmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{pmatrix} = \begin{pmatrix} C_{11} & C_{12} & C_{13} & C_{14} \\ C_{21} & C_{22} & C_{23} & C_{24} \\ C_{31} & C_{32} & C_{33} & 1 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

$$u = \frac{\tilde{u}}{\tilde{w}}, v = \frac{\tilde{v}}{\tilde{w}}$$

Effects of Pinhole Cameras on the Scene

No unique inverse



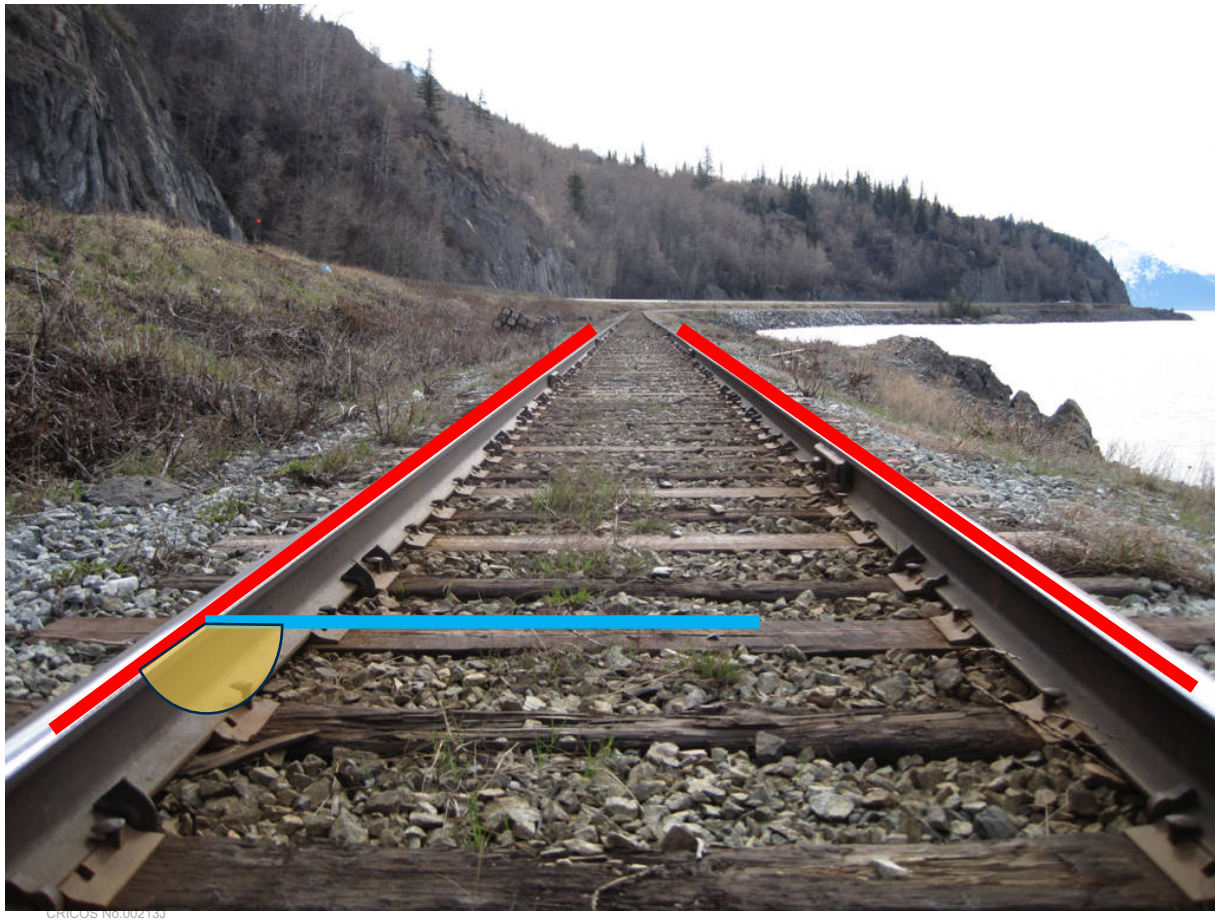
$$\mathbb{R}^3 \mapsto \mathbb{R}^2$$

We cannot recover the lost dimension

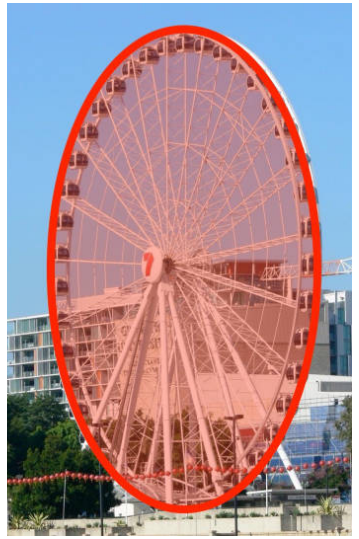
- Any 2D image could be generated by one of an infinite number of possible 3D worlds

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Perspective Projection



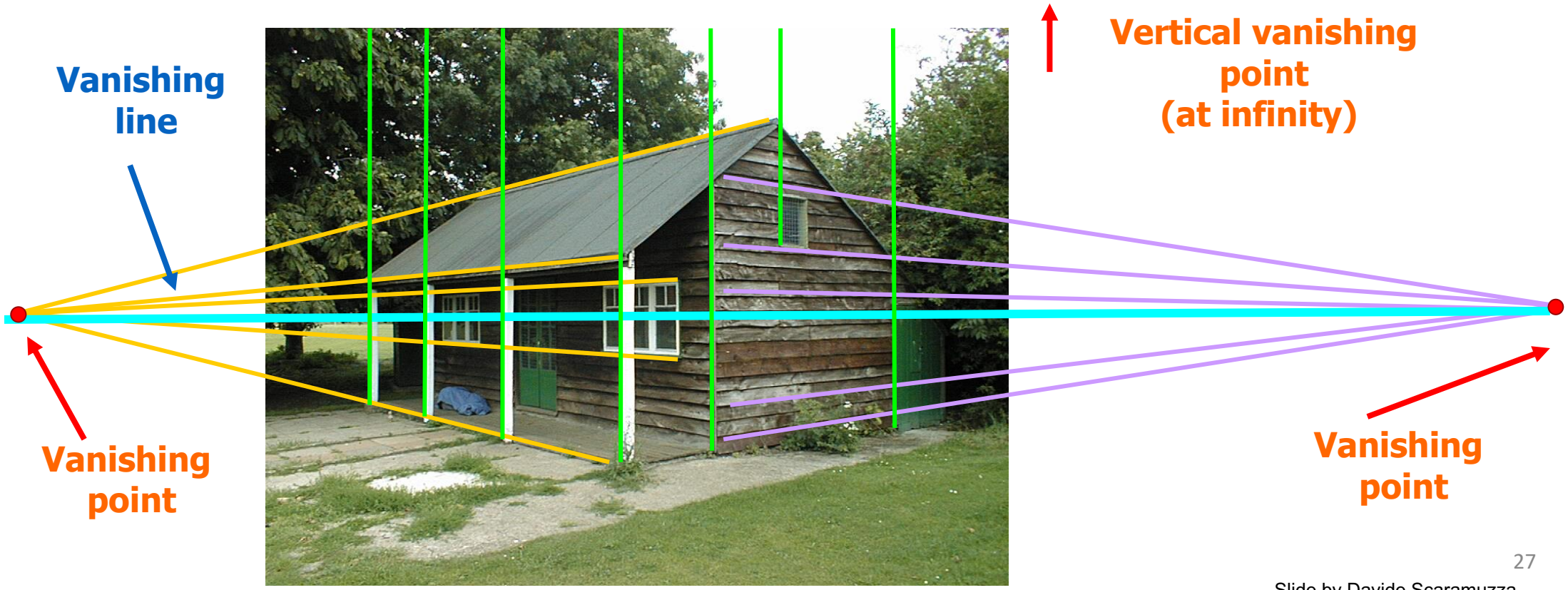
- Lengths and angles are not preserved
- Parallelism is lost
- Perpendicularity is lost
- Straight lines are still straight



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Vanishing points and lines

- **Parallel lines** intersect at a "**vanishing point**" in the image
- **Parallel planes** intersect at a "**vanishing line**" in the image
- Notice that vanishing points can fall both inside or outside the image



Slide by Davide Scaramuzza

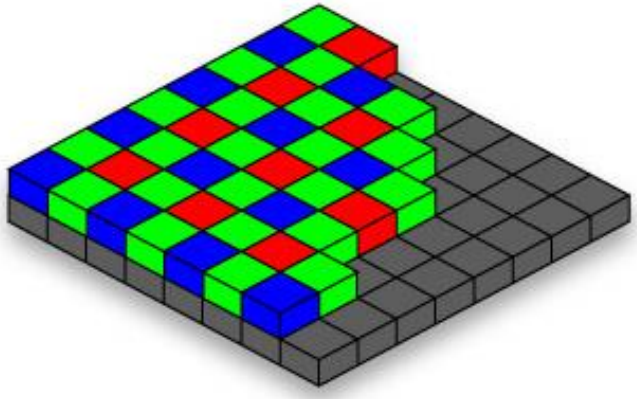
Color sensing in digital cameras

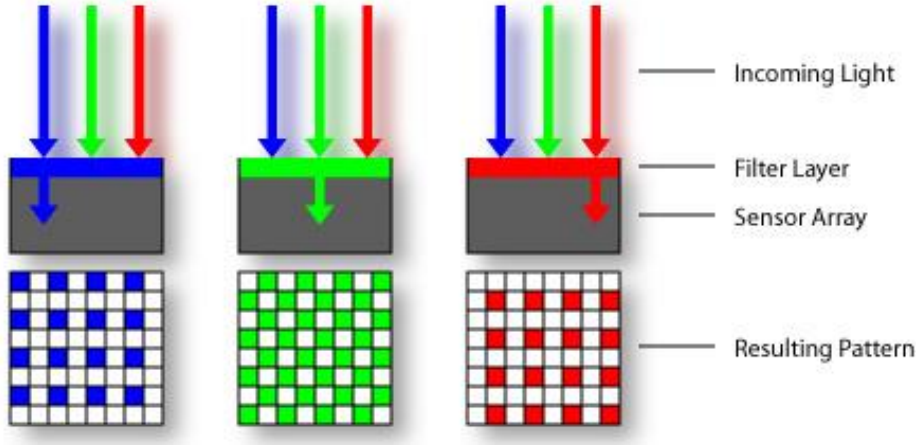
- The **Bayer pattern** (invented by Bayer in 1976, who worked at Kodak) places interleaved RGB filters over the pixel array
- The reason why the number of green filters is twice as that of red and blue ones is because the **luminance** signal is mostly **determined by green** values and the **human visual system** is much more **sensitive** to **spatial differences in luminance** than in chrominance.



For each pixel, the missing color components can be estimated from neighboring values by interpolation (**demosaicing**)

Bayer grid





Slide by Davide Scaramuzza

Rolling Shutter vs Global Shutter Camera

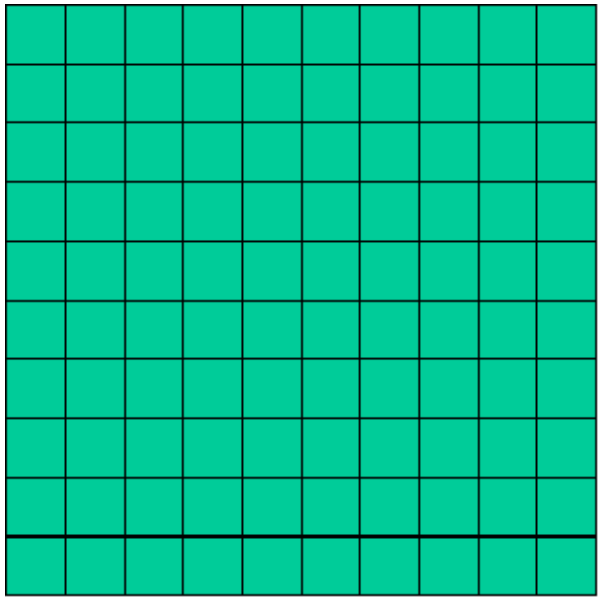
Rolling Shutter

- **Rows** of pixels are **exposed and read at different times,** one after the other
- May **distort** (skew) **moving objects**
- **Cheap**



Global Shutter

- **All pixels** are **exposed simultaneously**
- **No distortion** of moving objects
- More **costly**



Slide by Davide Scaramuzza

Rolling Shutter vs Global Shutter Camera

- Rolling Shutter cameras may distort (skew) moving objects
- Global Shutter cameras do not have this problem



Rolling shutter

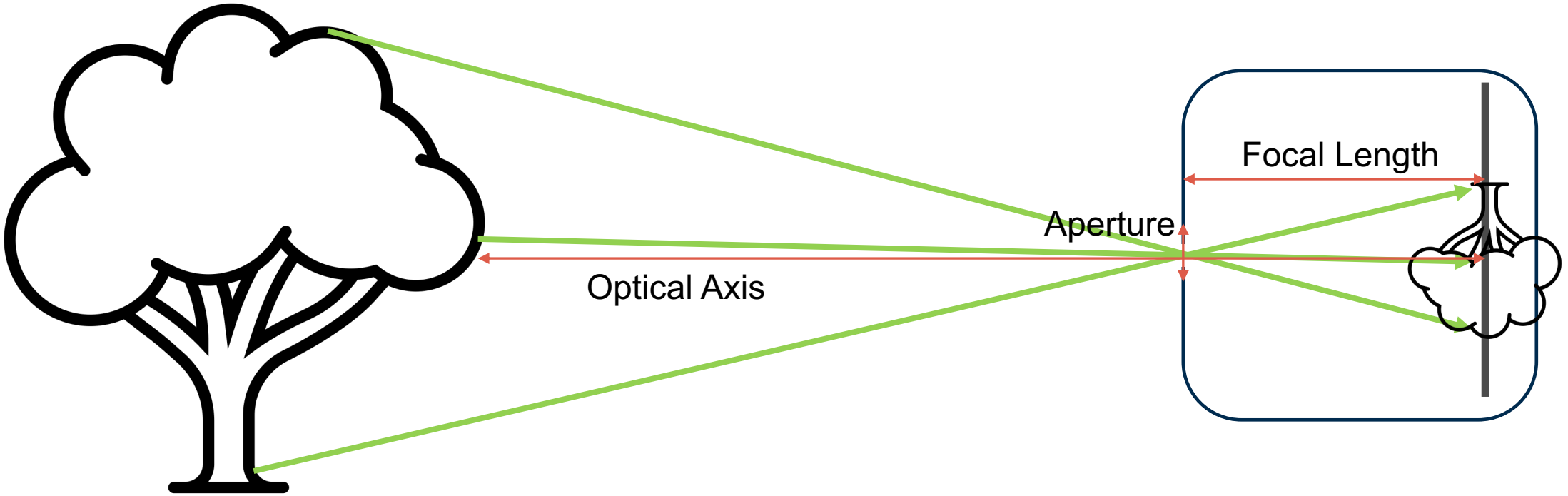


Global shutter

Slide by Davide Scaramuzza

Pinhole vs Lenses

The Pinhole Camera



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Effect of Aperture Size

- Large aperture → blurry image!
  - A **cone** of light is let through from every world point
- Ideal aperture: a pinhole
  - Lets through only one **ray** of light from each world point
  - But disadvantages:
    less light → increase exposure time → motion blur
  - Too small aperture → diffraction → burry image

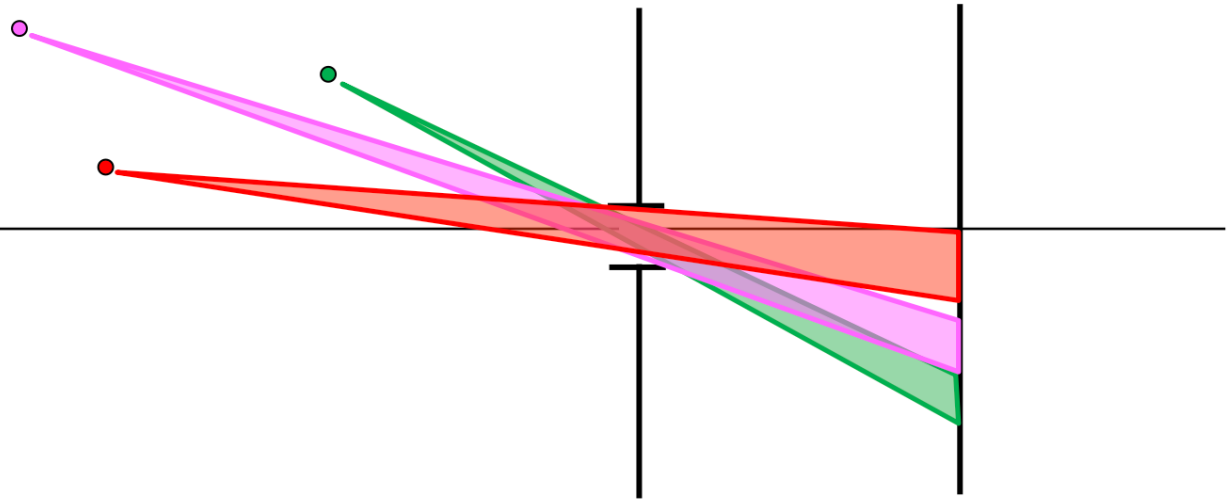


Illustration by Davide Scaramuzza

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Use Lenses!

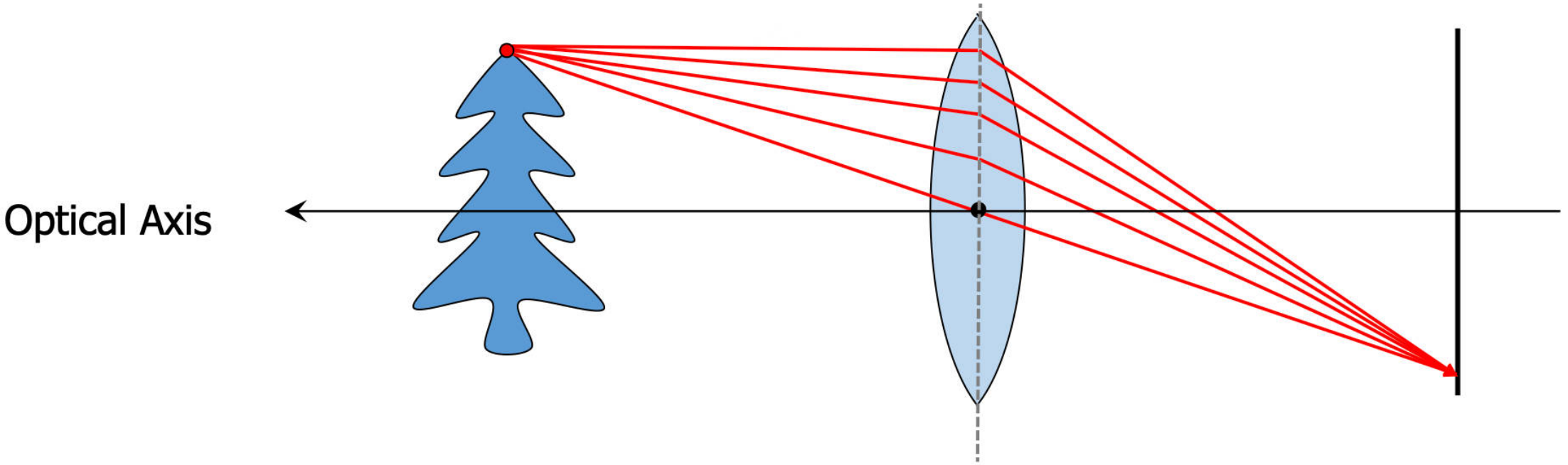
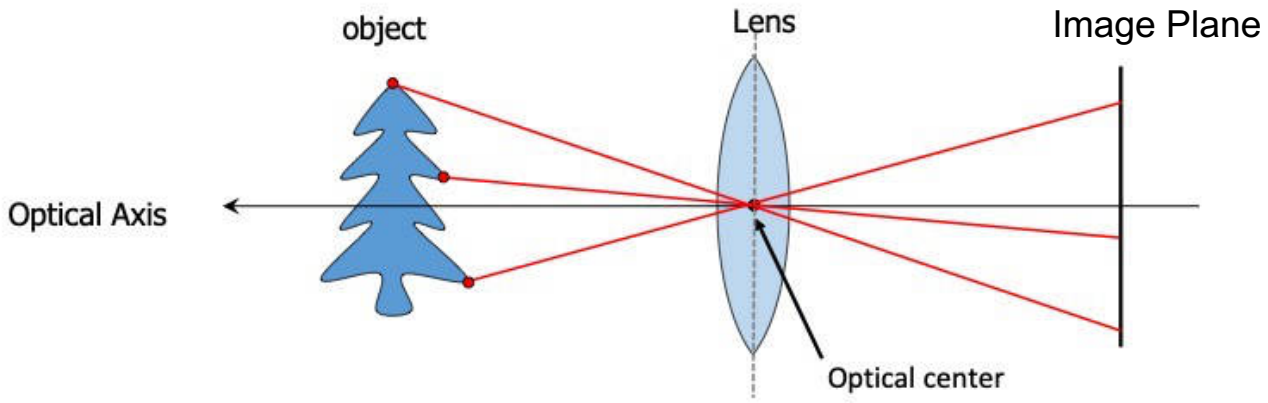


Illustration by Davide Scaramuzza

Thin Converging Lens Model

Rays through the optical center are not deviated, they continue in a straight line
Rays parallel to the optical axis converge at the focal point



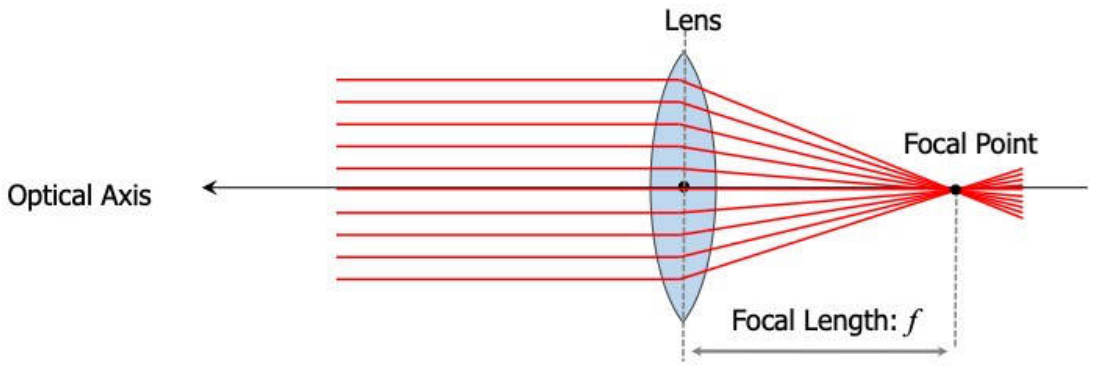


Illustration by Davide Scaramuzza

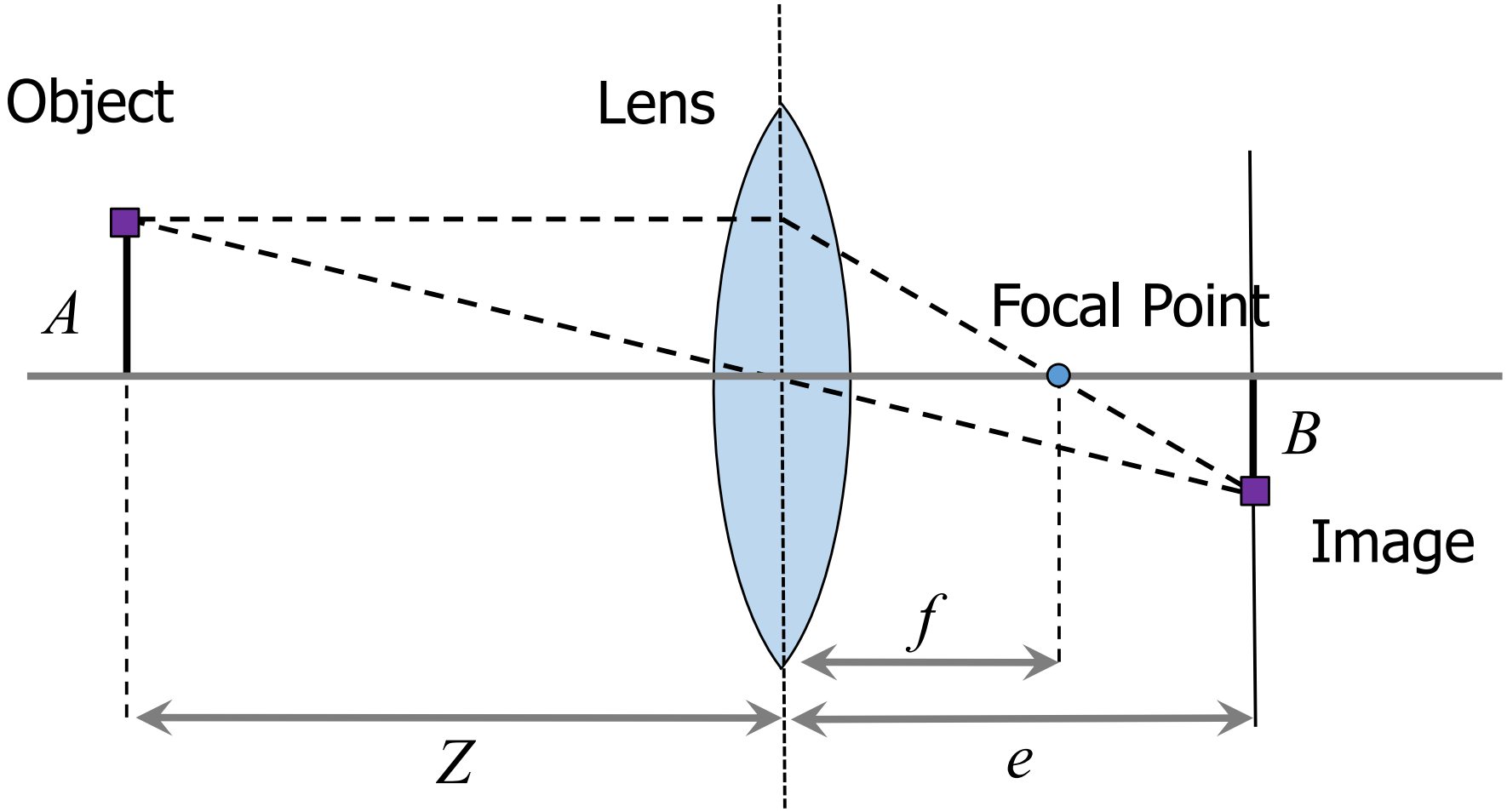
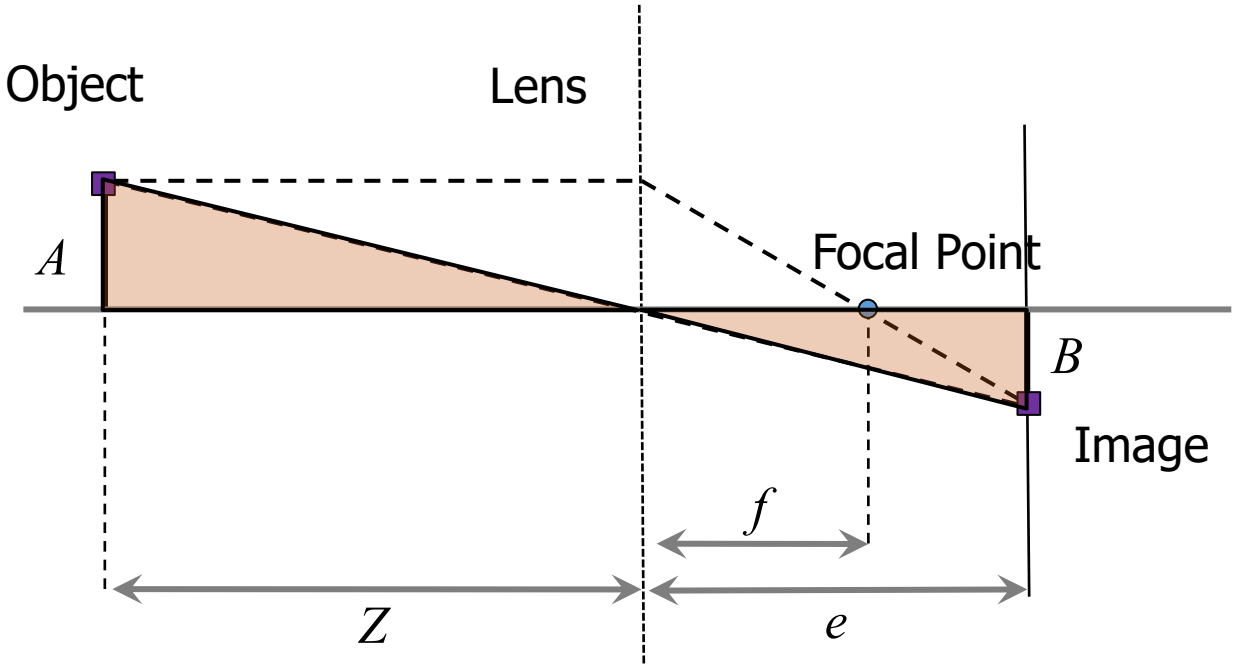


Illustration by Davide Scaramuzza

CRICOS No.00213J

Thin lens equation

- Find a relationship between f, Z, and e



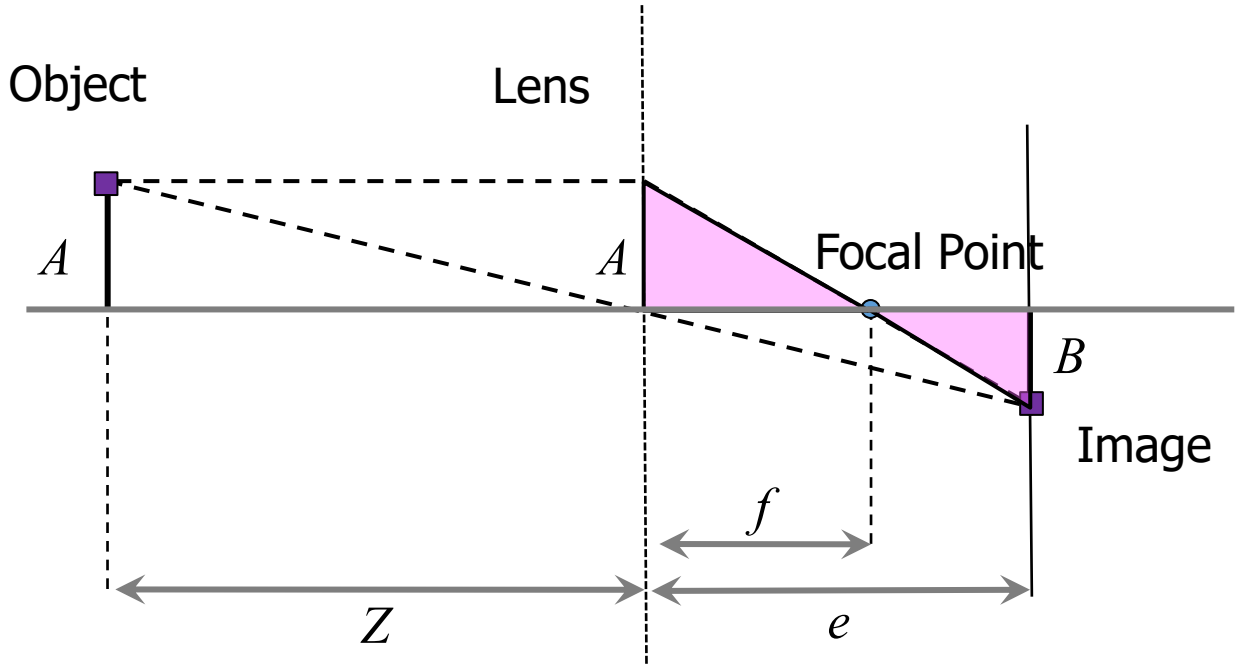
Similar Triangles:

$$\frac{B}{A} = \frac{e}{Z}$$

Slide by Davide Scaramuzza

Thin lens equation

- Find a relationship between f, Z, and e



Similar Triangles:

$$\frac{B}{A} = \frac{e}{Z}$$

$$\frac{B}{A} = \frac{e-f}{f} = \frac{e}{f} - 1$$

$$\left.\begin{array}{l} \frac{B}{A} = \frac{e}{Z} \\ \frac{B}{A} = \frac{e-f}{f} = \frac{e}{f} - 1 \end{array}\right\} \frac{e}{f} - 1 = \frac{e}{Z} \Rightarrow \boxed{\frac{1}{f} = \frac{1}{Z} + \frac{1}{e}}$$

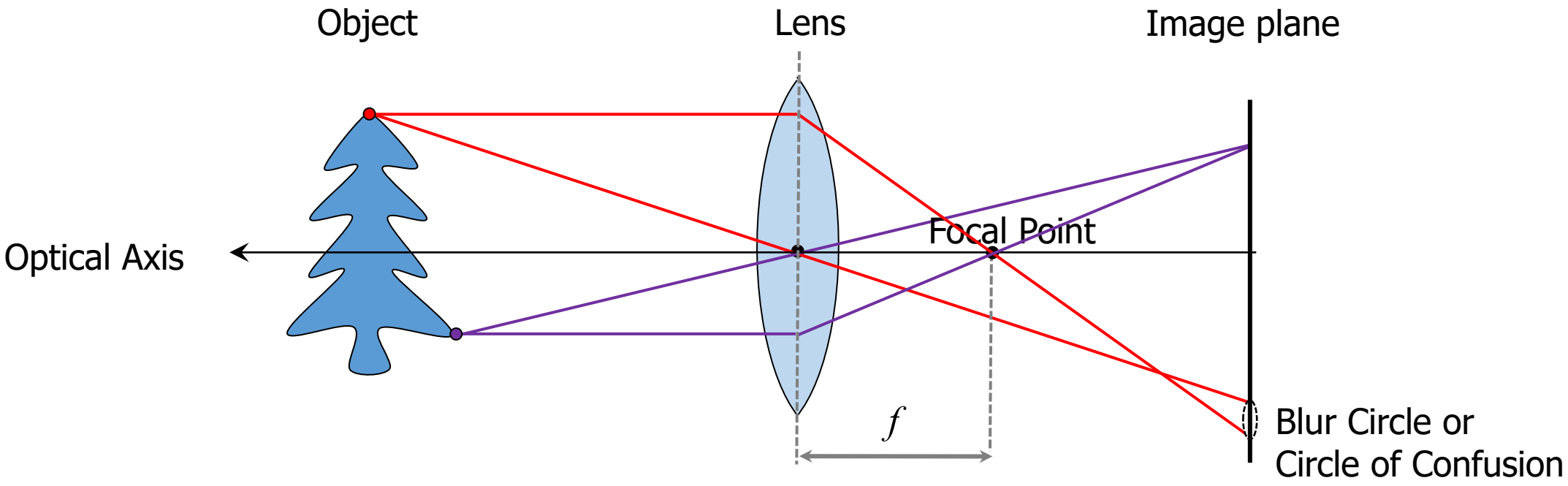
"Thin lens equation"

- Any object point satisfying this equation is in focus
- Can I use this to measure distances?

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“In focus”

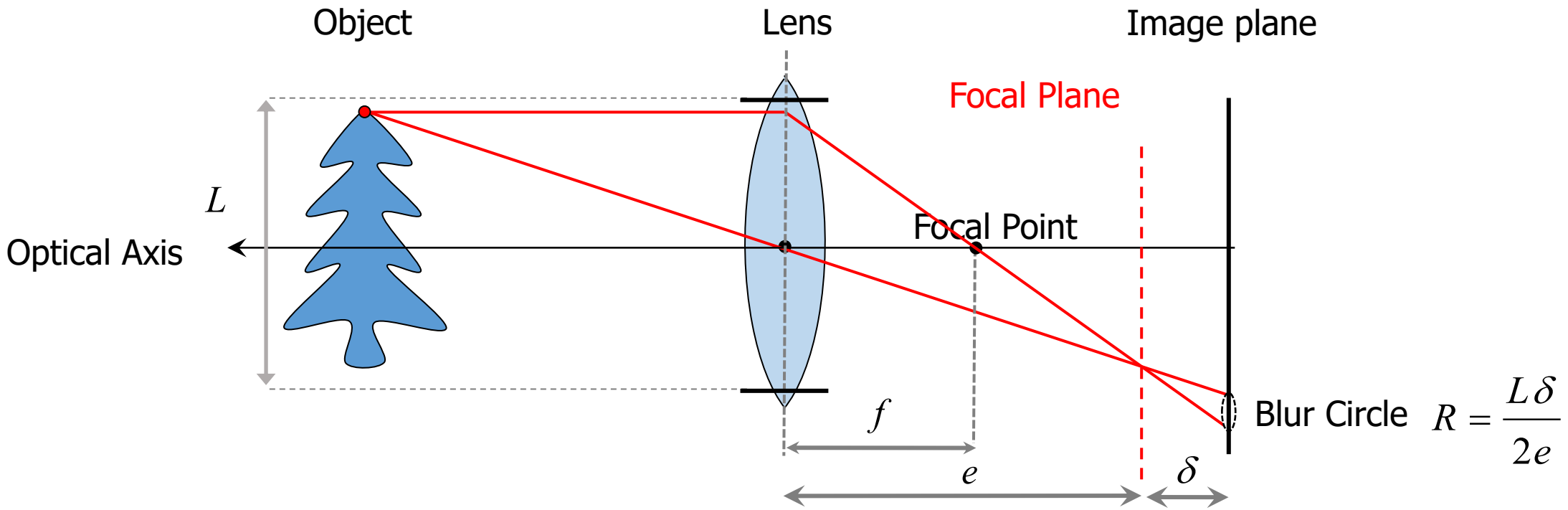
- For a given point on the object, there is a specific distance between the lens and the film, at which the object appears **in focus** in the image
- Other points project to a **blur circle** in the image



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Blur Circle

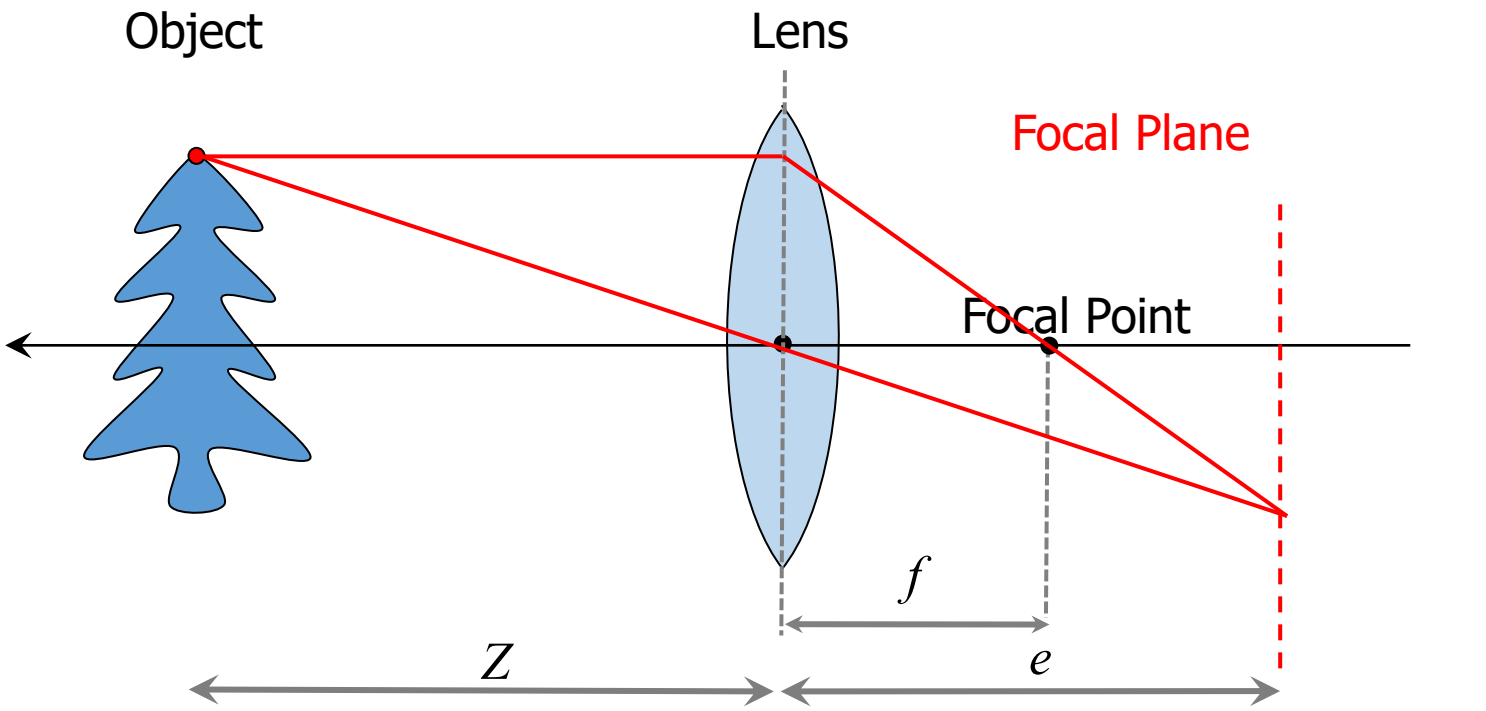
- The blur circle has radius: $\boldsymbol{R = L\delta/(2e)}$
  - Observe how both **small aperture (L) and δ** yield a **small blur circle (R)**
  - To capture a sharp image, we must **adjust the camera settings** such that R remains **smaller** than the **1 pixel**



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The Pin-hole approximation

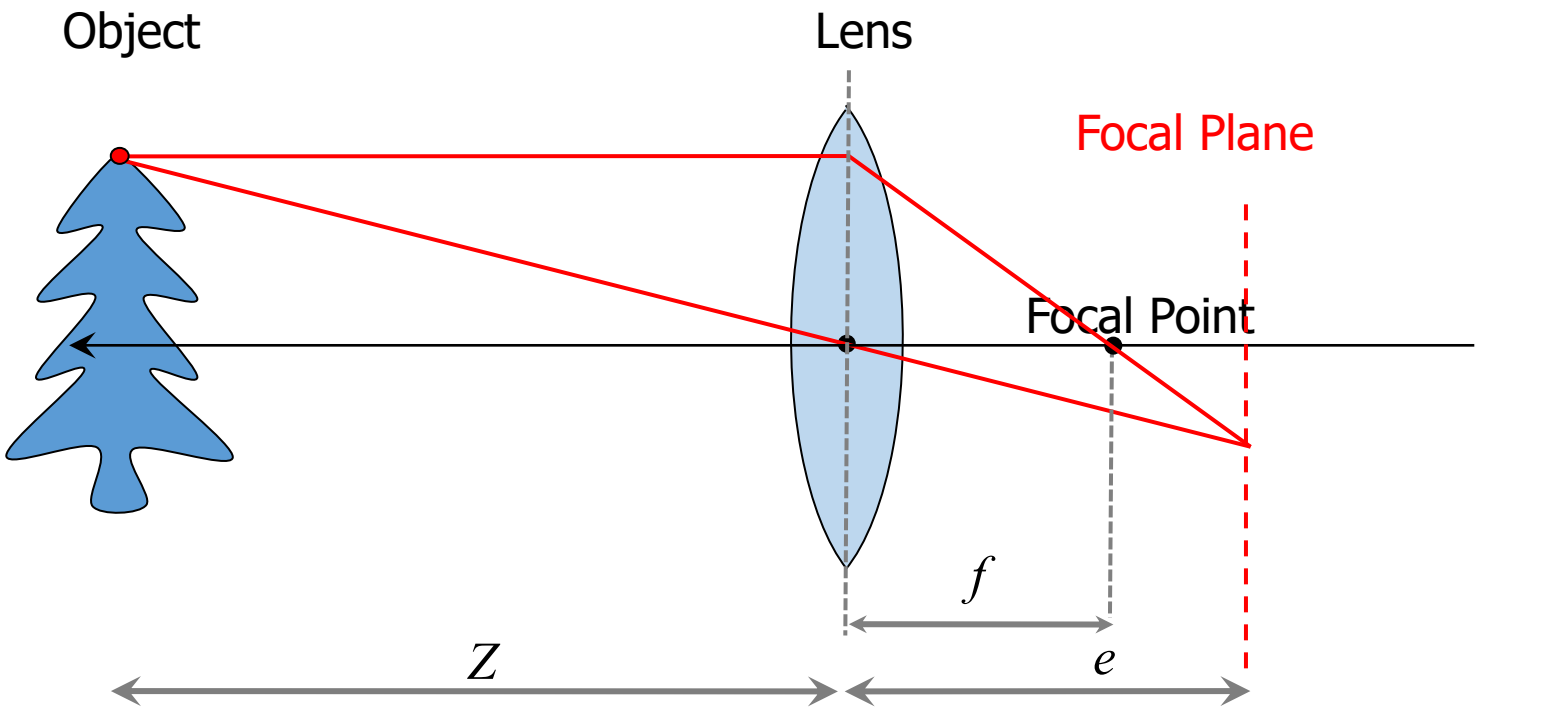
- What happens if $z \gg f$ and $z \gg L$?



Slide by Davide Scaramuzza

The Pin-hole approximation

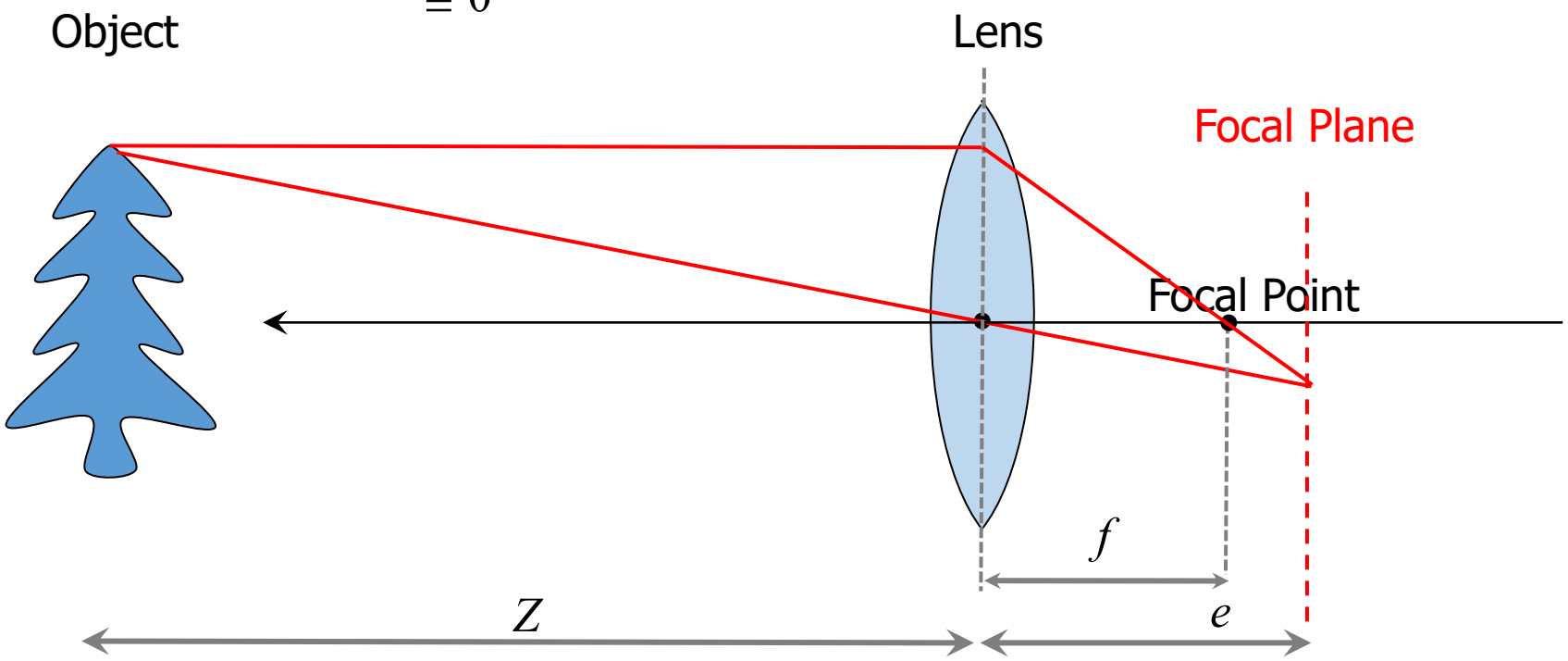
- What happens if $z \gg f$ and $z \gg L$?



Slide by Davide Scaramuzza

The Pin-hole approximation

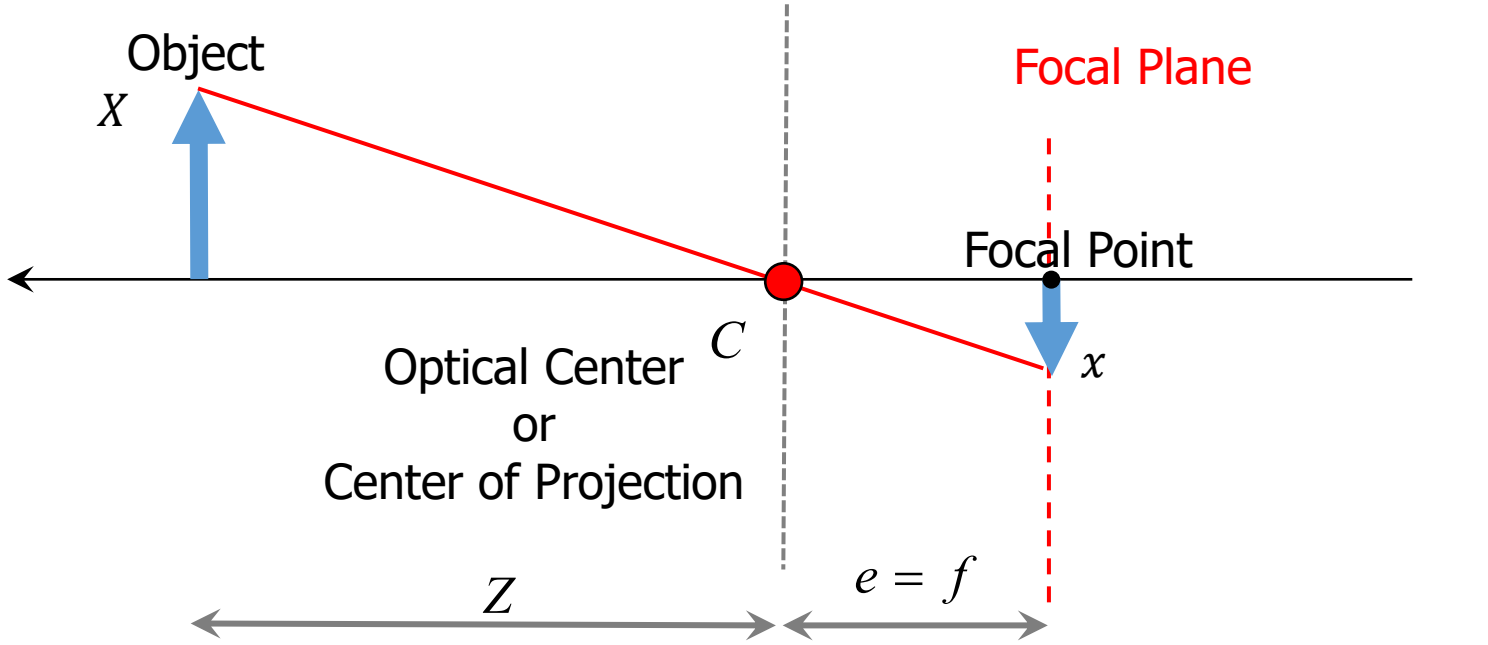
- What happens if $z \gg f$ and $z \gg L$?
- We observe that $\frac{1}{f} = \underbrace{\frac{1}{z}}_{\cong 0} + \frac{1}{e} \Rightarrow \frac{1}{f} \approx \frac{1}{e} \Rightarrow f \approx e \Rightarrow$ focal plane approaches the focal point



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The Pin-hole approximation

- This is known as **Pinhole Approximation**
- The relation between the image and object becomes: $-\frac{x}{X} = \frac{f}{Z} \Rightarrow \boxed{x = -f\frac{X}{Z}}$
- This is called **Perspective Projection**



Slide by Davide Scaramuzza



Stereo Cameras: Triangulation and Rectification

Stereo Vision



Slide / Illustrations by Peter Corke

Stereo Vision









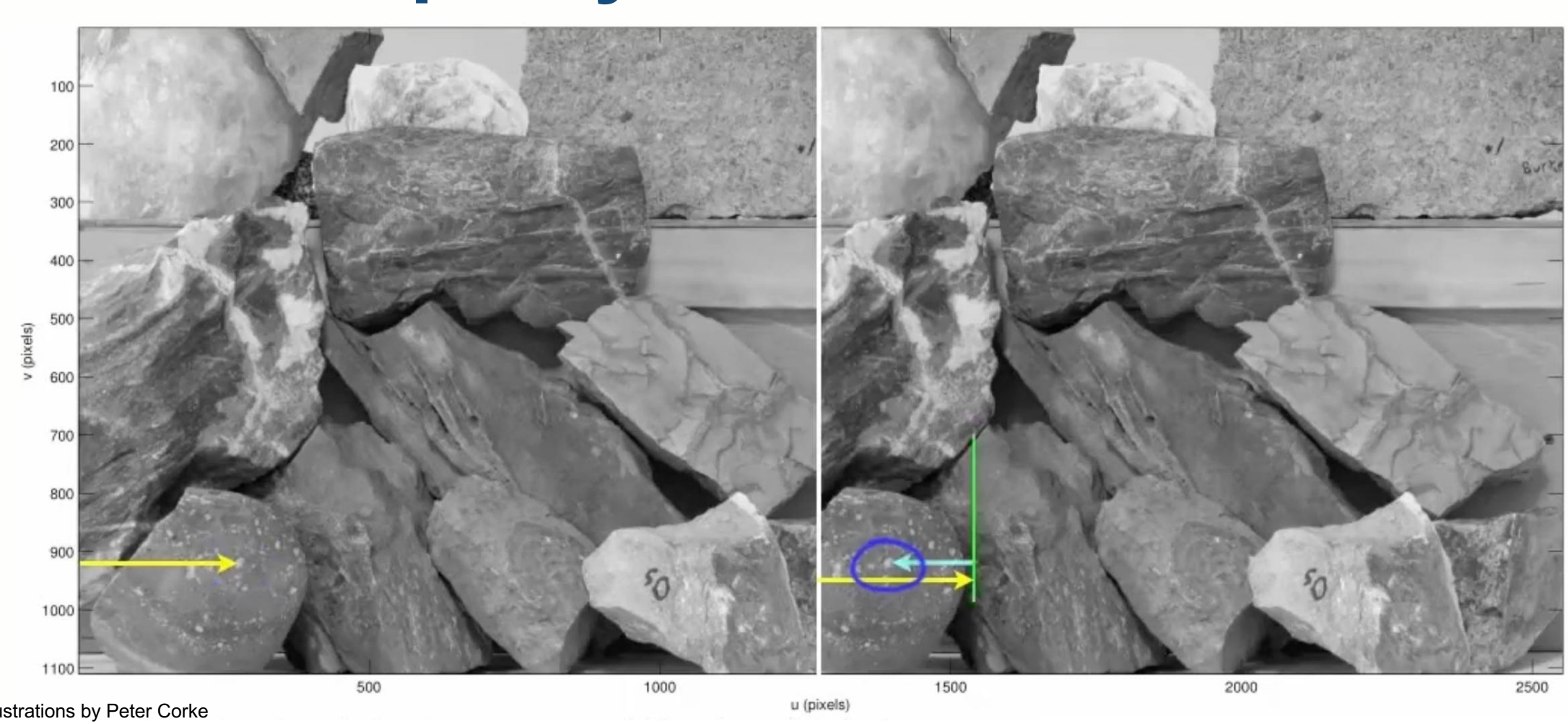
CRICOS No.00213J

Stereo Disparity



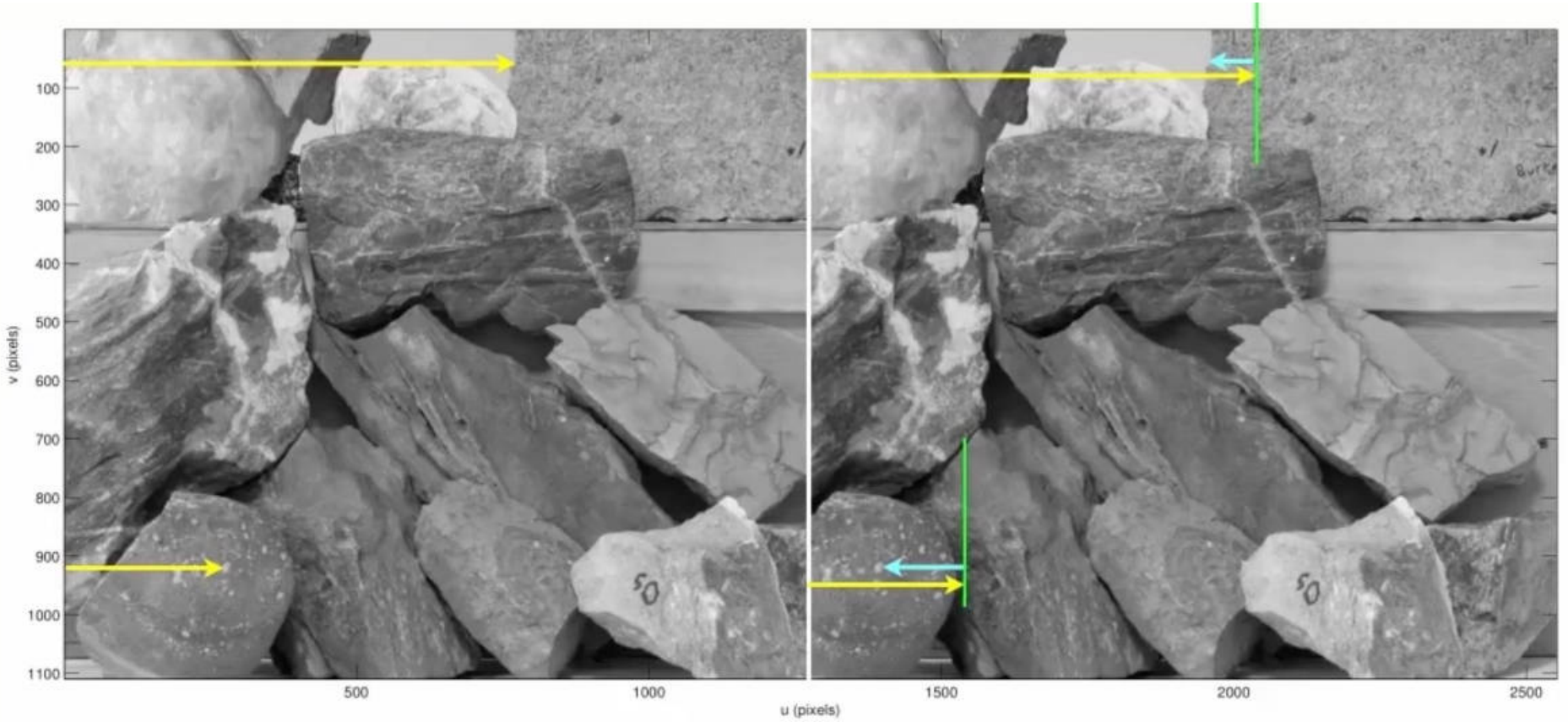
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Stereo Disparity



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Stereo Disparity



- Points in the right image appear shifted to the left
- The shift is less for more distant points

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From Disparity to 3D

Disparity

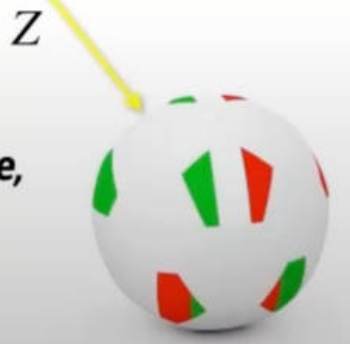
- The horizontal displacement in an image point due to horizontal translation of the camera

$$d \propto \frac{fb}{Z}$$



Kodak Stereo Camera 2008
John Kratz from Bulington NJ, USA | CC BY-SA 2.0 via Wikimedia Commons

- If we know f, b and d then we can estimate Z



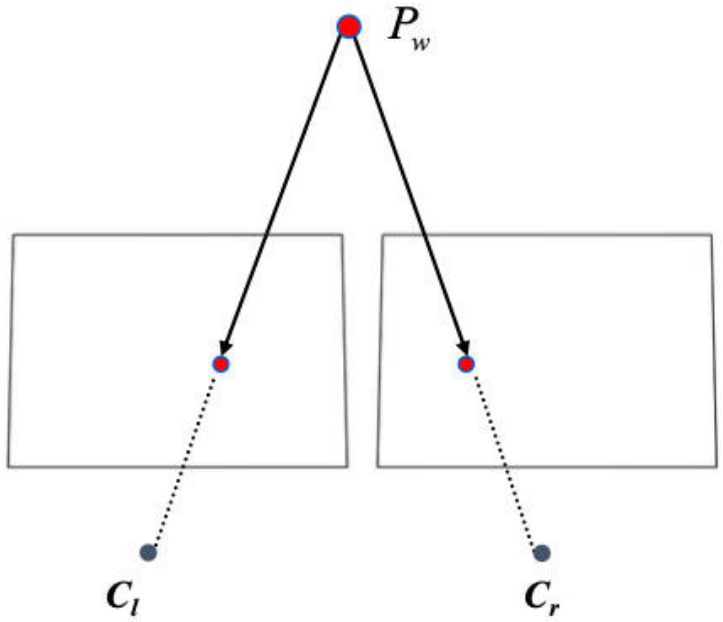
f is focal length, b is baseline, Z is depth.

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Finding the Disparity

Simplified case

(identical cameras and aligned)



General case

(non identical cameras and not aligned)

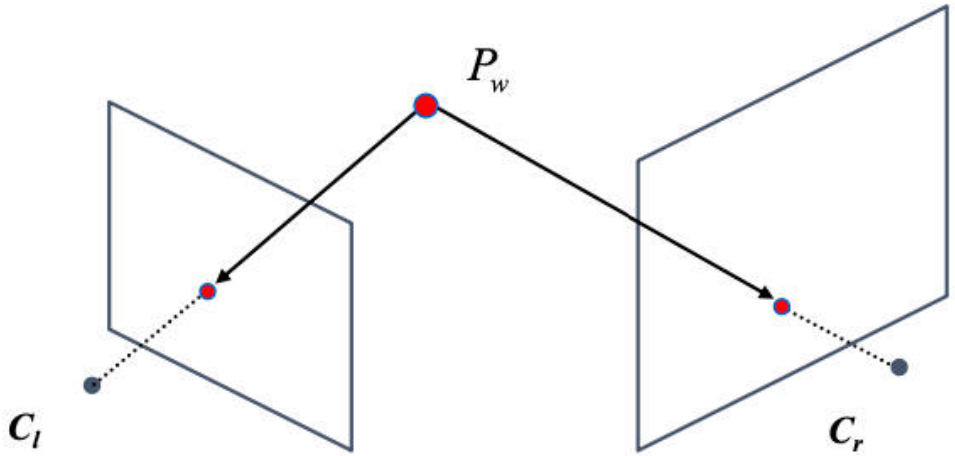
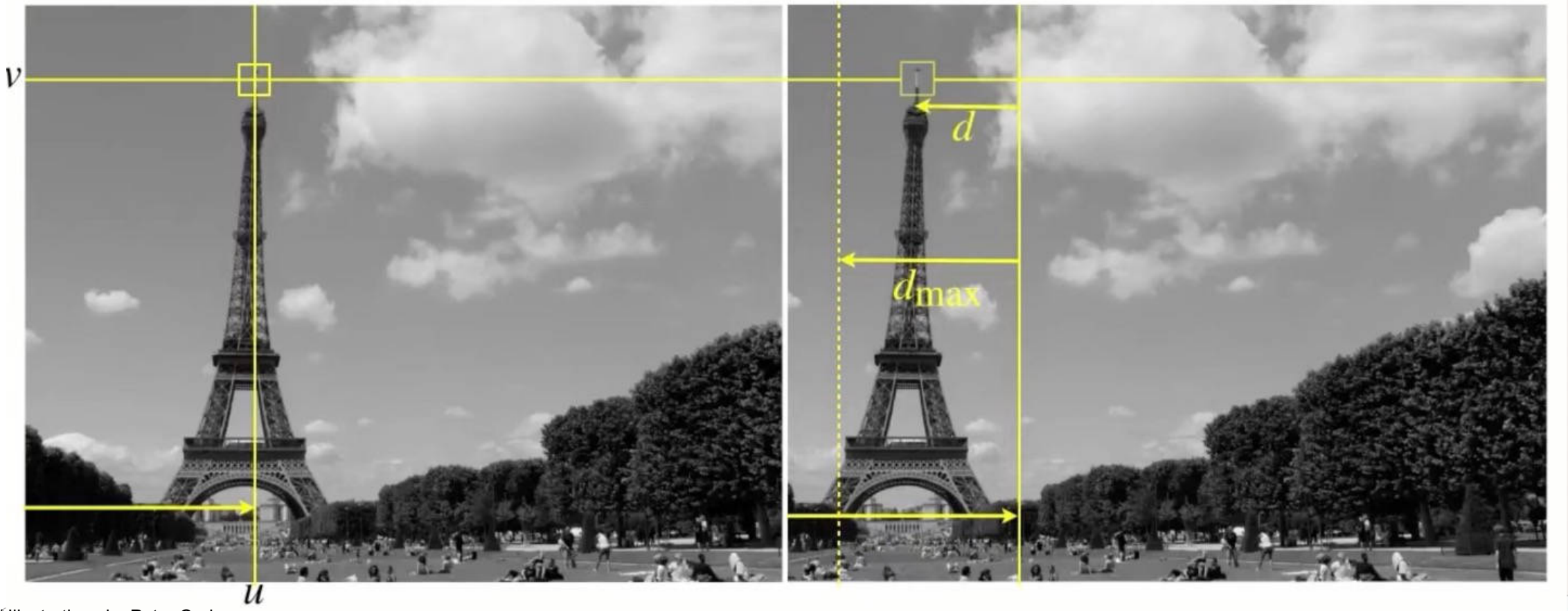


Illustration courtesy of Davide Scaramuzza

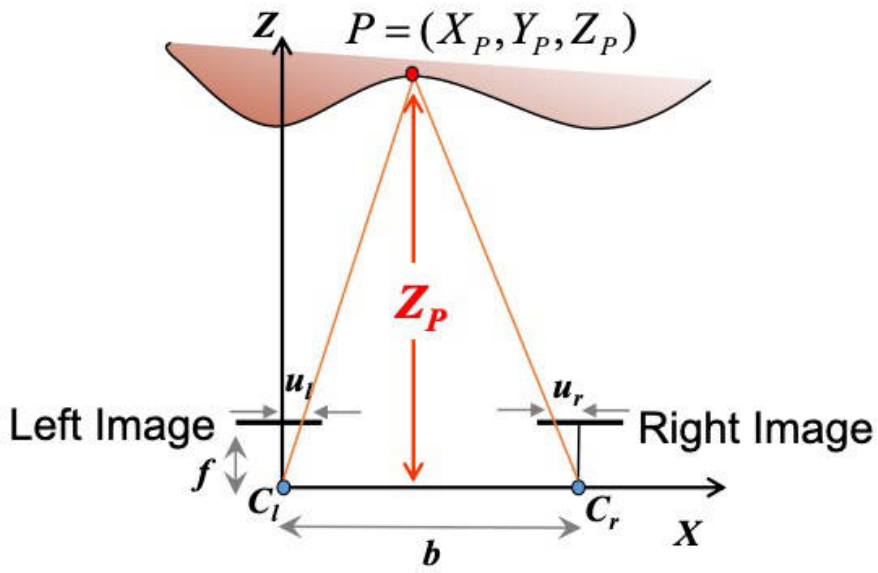
Finding the Disparity: Block Matching



Slide / Illustrations by Peter Corke

Triangulating 3D Points

Both cameras are **identical** (i.e., same intrinsics) and are **aligned** to the x-axis



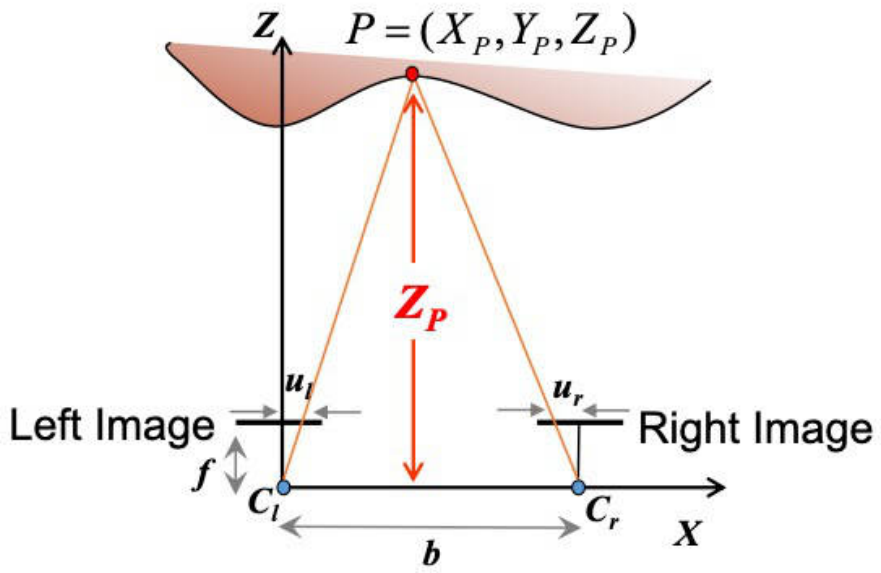
Baseline = distance between the optical centers of the two cameras

CRICOS No.00213J

Illustration courtesy of Davide Scaramuzza

Triangulating 3D Points

Both cameras are **identical** (i.e., same intrinsics) and are **aligned** to the x-axis



Baseline = distance between the optical centers of the two cameras

From Similar Triangles:

$$\frac{f}{Z_P}=\frac{u_l}{X_P}$$

$$\frac{f}{Z_P}=\frac{-u_r}{b-X_P}$$

$$Z_P=\frac{bf}{u_l-u_r}$$

Disparity

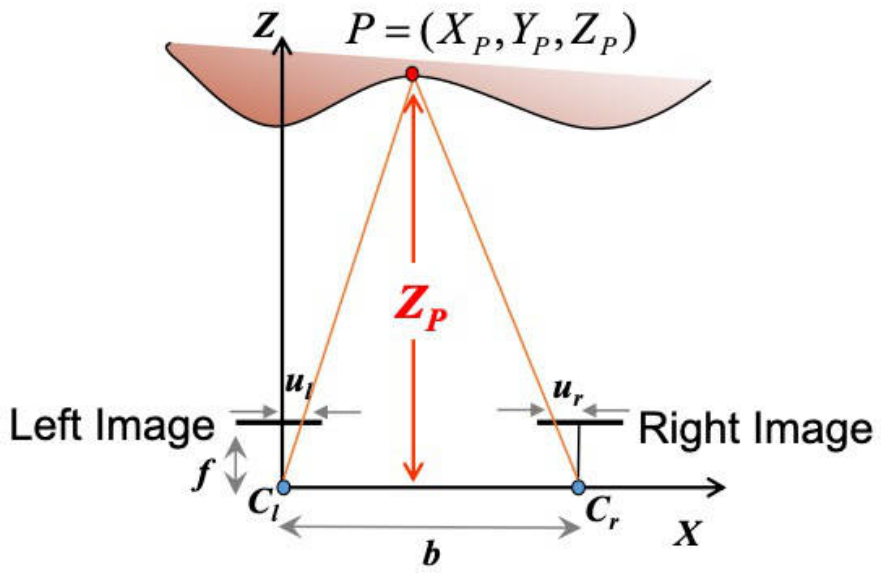
difference in image location of the projection of a 3D point on two image planes

CRICOS No.00213J

Illustration courtesy of Davide Scaramuzza

Triangulating 3D Points

Both cameras are **identical** (i.e., same intrinsics) and are **aligned** to the x-axis



Baseline = distance between the optical centers of the two cameras

From Similar Triangles:

$$\frac{f}{Z_P}=\frac{u_l}{X_P}$$

$$\frac{f}{Z_P}=\frac{-u_r}{b-X_P}$$

$$Z_P=\frac{bf}{u_l-u_r}$$

$$X_P=\frac{Z_P\cdot u_l}{f}$$

$$Y_P=\frac{Z_P\cdot v_l}{f}$$

Disparity

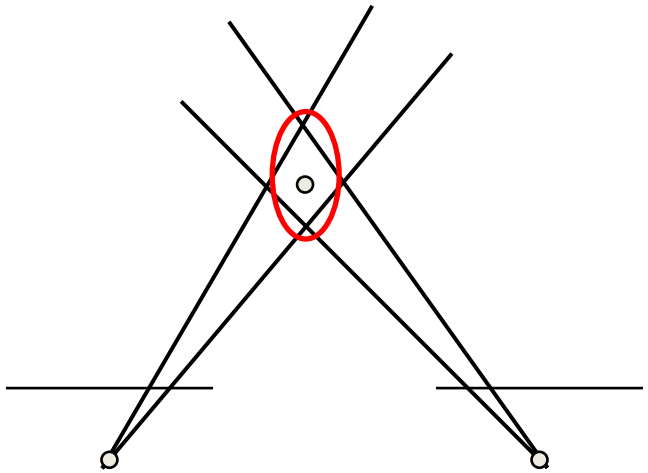
difference in image location of the projection of a 3D point on two image planes

Illustration courtesy of Davide Scaramuzza

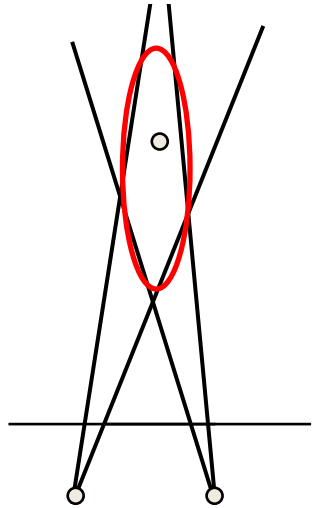
What is the best Baseline?

What's the optimal baseline?

- **Large baseline:**
  - **Small depth error but**…
  - **Minimum** measurable **depth increases**
  - **Difficult search** problem for **close objects**
- **Small baseline:**
  - **Large depth error** but…
  - Minimum measureable depth decreases
  - Easier search problem for close objects



Large Baseline



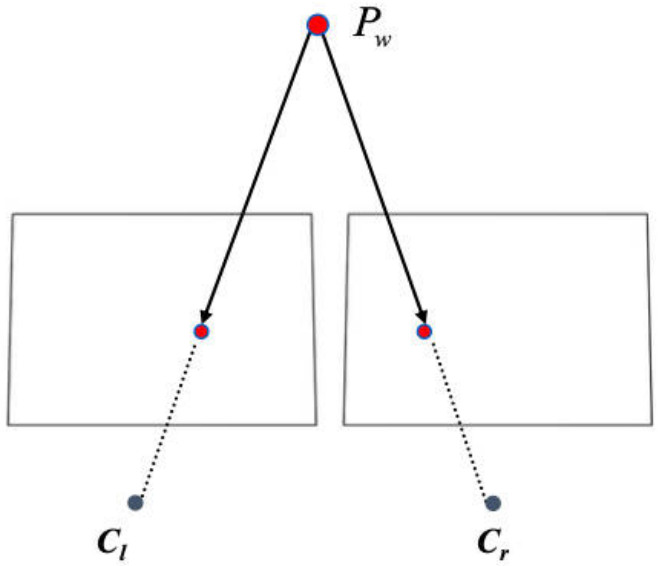
Small Baseline

Illustration courtesy of Davide Scaramuzza

Stereo Rectification

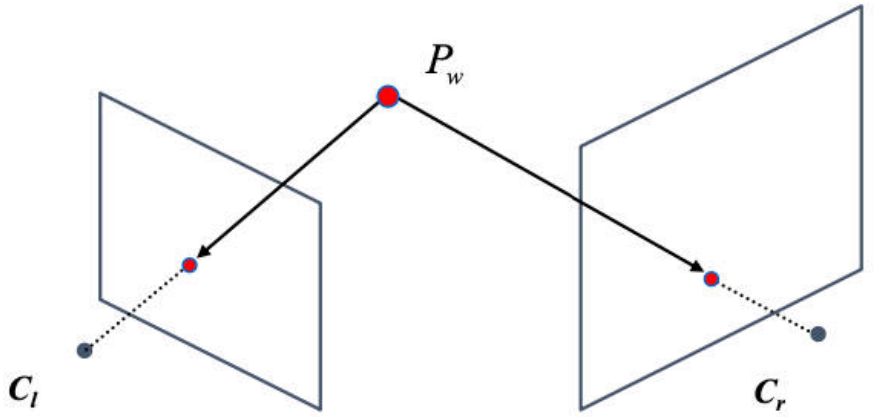
Simplified case

(identical cameras and aligned)



General case

(non identical cameras and not aligned)



CRICOS No.00213J

Illustration courtesy of Davide Scaramuzza

Stereo Rectification

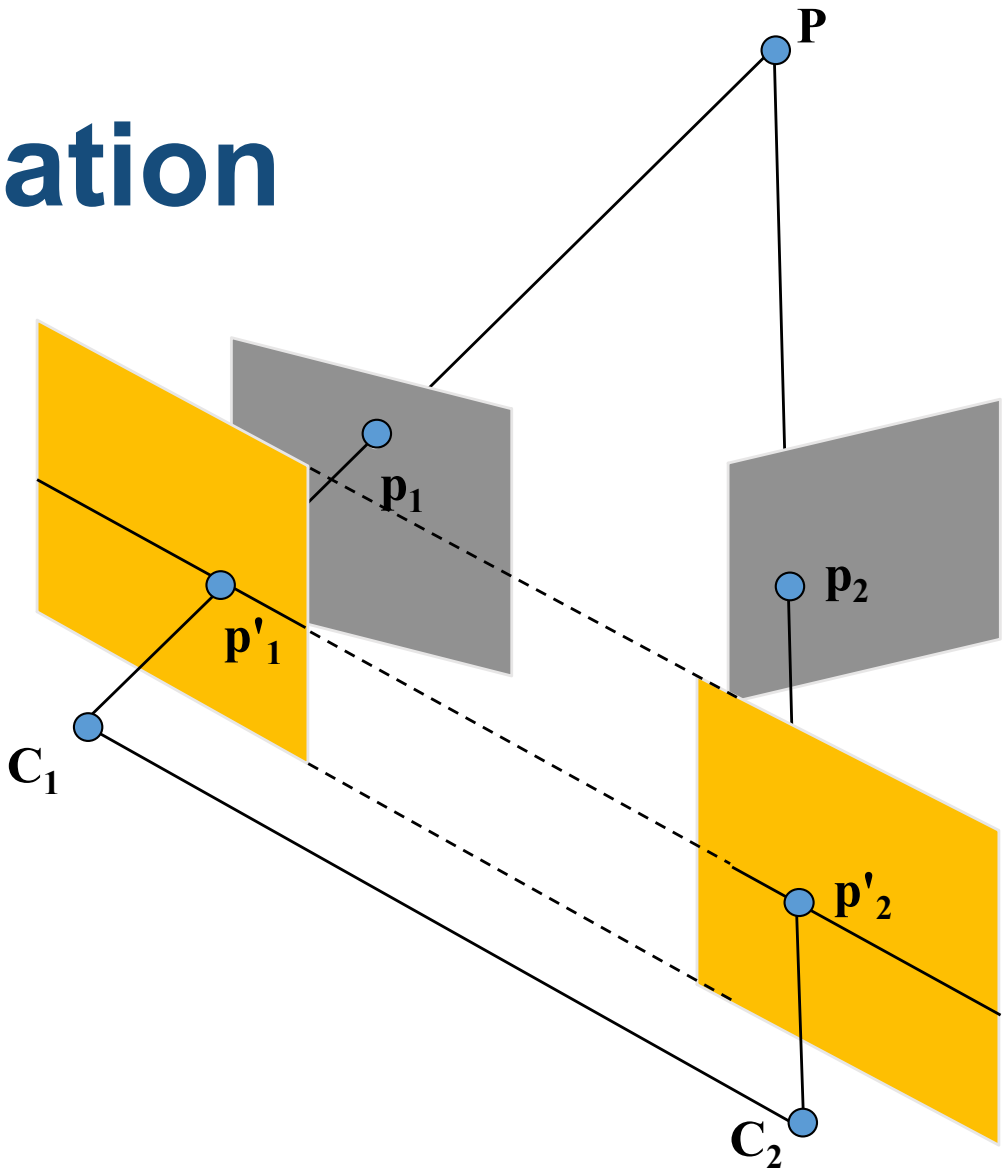
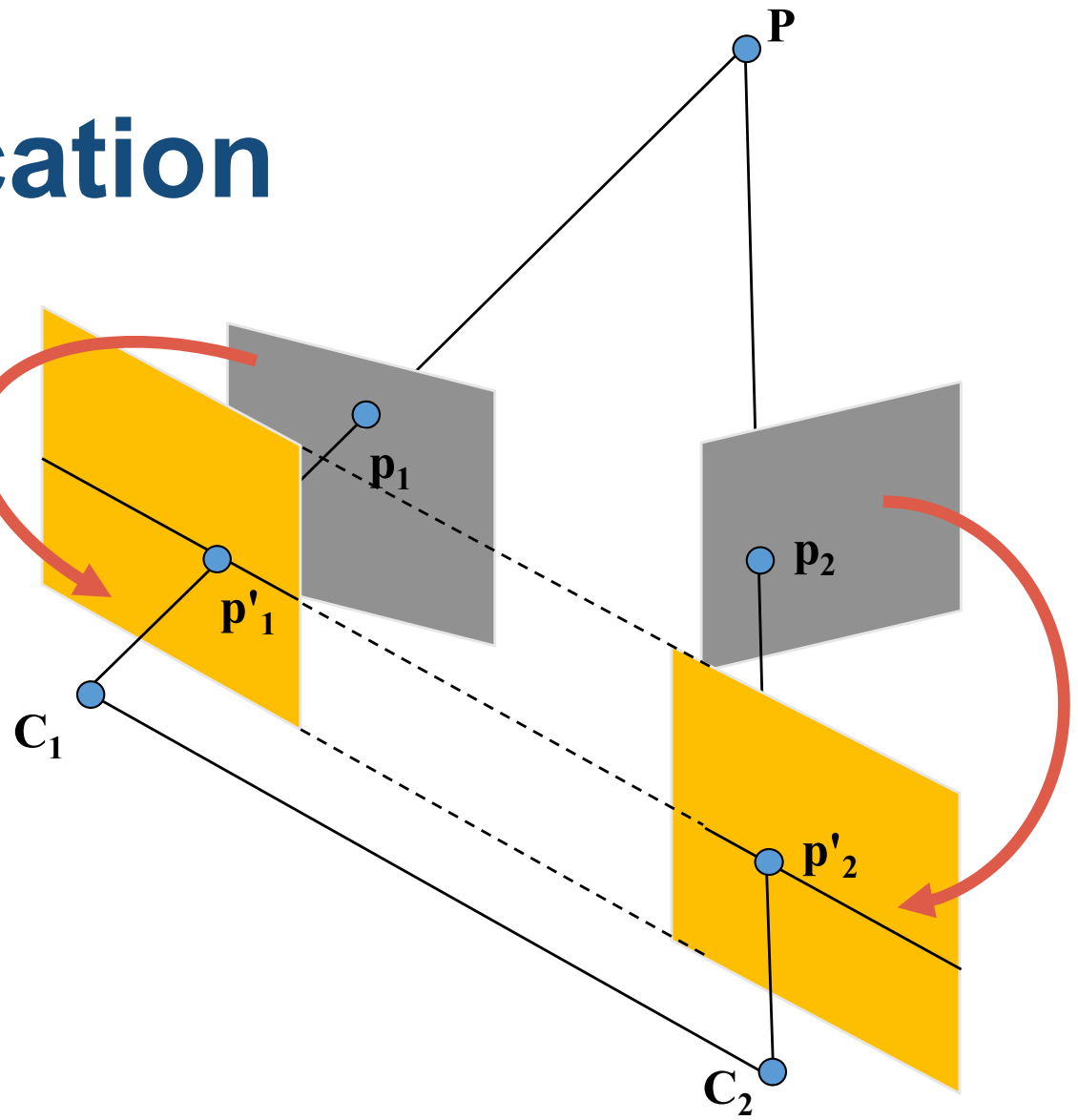


Illustration courtesy of Davide Scaramuzza

Stereo Rectification

$$p'_1 = H_1 p_1$$



$$p'_2 = H_2 p_2$$

Illustration adopted from Davide Scaramuzza



Camera Calibration

How can we determine the intrinsic parameters?

$$\begin{pmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{pmatrix} = \underbrace{\begin{pmatrix} \frac{1}{\rho_u} & 0 & u_0 \\ 0 & \frac{1}{\rho_v} & v_0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}}_{\mathbf{K}} \underbrace{\begin{pmatrix} \mathbf{R} & t \\ \mathbf{0}_{1\times 3} & 1 \end{pmatrix}^{-1}}_{\mathbf{C}} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

intrinsic parameters

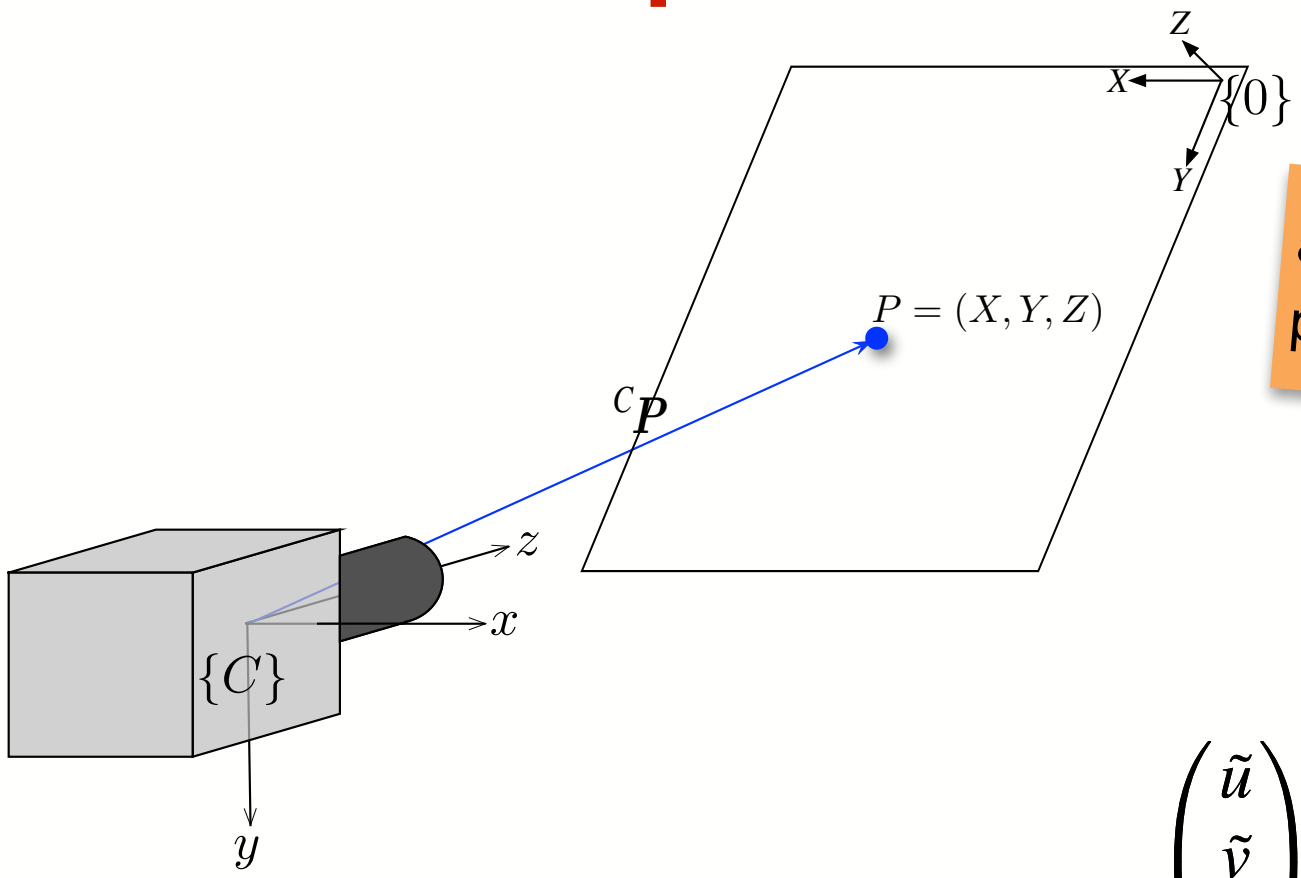
camera matrix

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A Special Case: Points on a Plane

Points on a plane



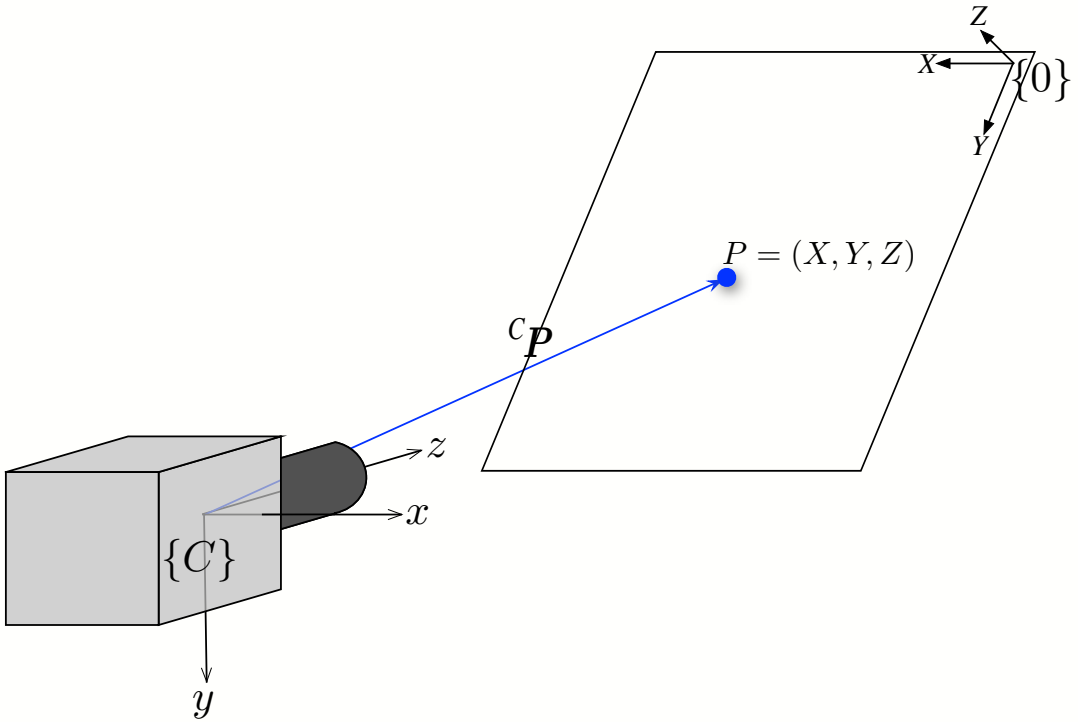
all points on the plane have z=0

3 x 3 matrix

$$\begin{pmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{pmatrix} = \begin{pmatrix} C_{11} & C_{12} & C_{13} & C_{14} \\ C_{21} & C_{22} & C_{23} & C_{24} \\ C_{31} & C_{32} & C_{33} & 1 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

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Planar homography



homography matrix

$$\begin{pmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{pmatrix} = \begin{pmatrix} H_{11} & H_{12} & H_{13} \\ H_{21} & H_{22} & H_{23} \\ H_{31} & H_{32} & 1 \end{pmatrix} \begin{pmatrix} X \\ Y \\ 1 \end{pmatrix}$$

- Once again the scale factor is arbitrary
- 8 unique numbers in the homography matrix
- Can be estimated from 4 world points and their corresponding image points

$$\mathbf{H} = \mathbf{R} + \frac{\boldsymbol{t}}{d}\boldsymbol{n}^T$$

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How can we determine the intrinsic parameters?

$$\begin{pmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{pmatrix} = \underbrace{\begin{pmatrix} \frac{1}{\rho_u} & 0 & u_0 \\ 0 & \frac{1}{\rho_v} & v_0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}}_{\mathbf{K}} \underbrace{\begin{pmatrix} \mathbf{R} & t \\ \mathbf{0}_{1\times 3} & 1 \end{pmatrix}^{-1}}_{\mathbf{C}} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

intrinsic parameters

camera matrix

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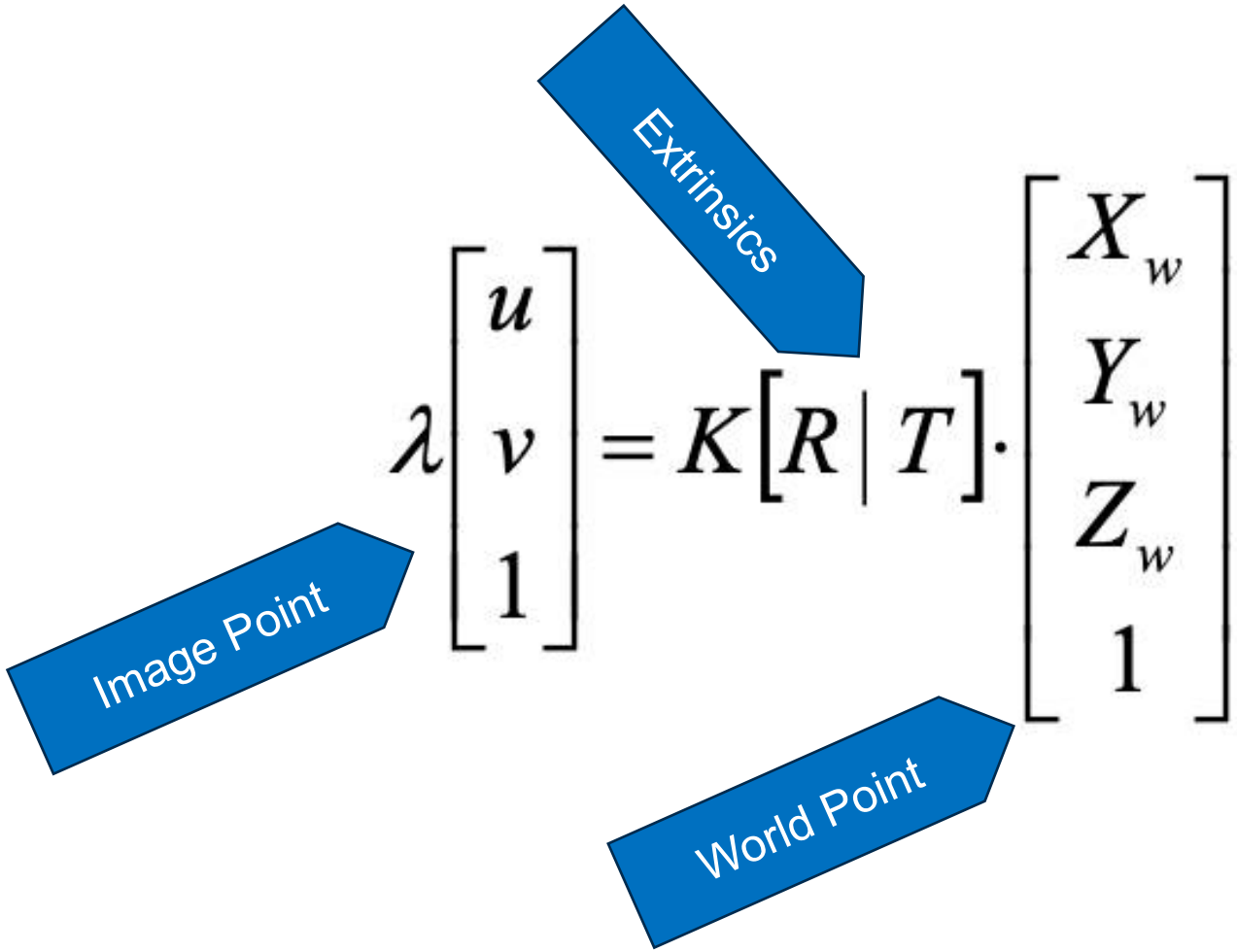
CRICOS No.00213J

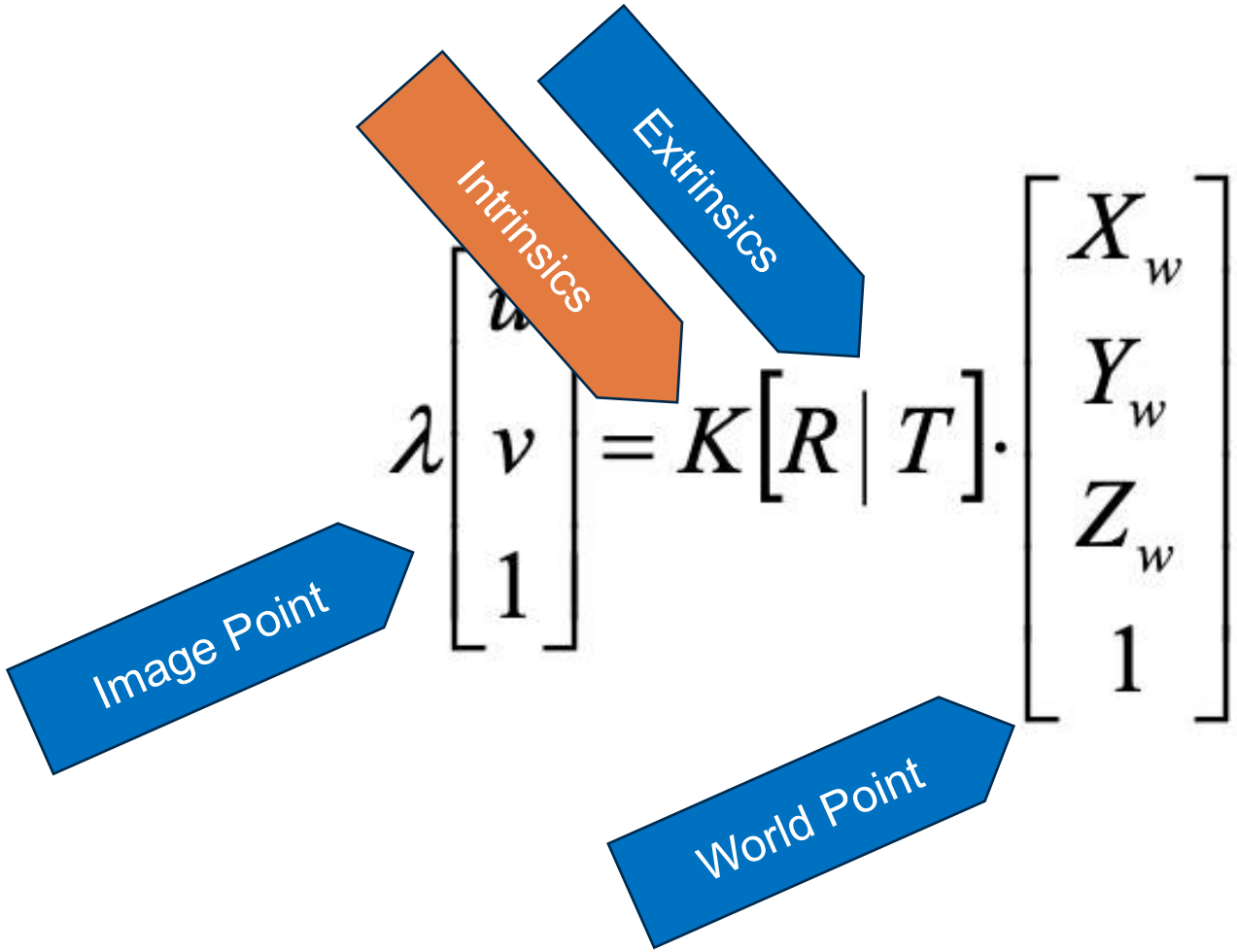
$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K \left[R \mid T \right] \cdot \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix}$$

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K[R \mid T] \cdot \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix}$$

Image Point

World Point


Extrinsics
$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K \left[R \mid T \right] \cdot \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix}$
Image Point
World Point



CRICOS No.00213J

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K \left[R \mid T \right] \cdot \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix}$$

Intrinsic Parameters

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha_u & 0 & u_0 \\ 0 & \alpha_v & v_0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t_3 \end{bmatrix} \cdot \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix}$$

Image Point

Intrinsics

Extrinsics

World Point

CRICOS No.00213J

How to Determine Intrinsic Parameters?

Focal Length Principal Point

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha_u & 0 & u_0 \\ 0 & \alpha_v & v_0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t_3 \end{bmatrix} \cdot \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix}$$

How to Determine Intrinsic Parameters?

Unknowns

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha_u & 0 & u_0 \\ 0 & \alpha_v & v_0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t_3 \end{bmatrix} \cdot \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix}$$

How to Determine Intrinsic Parameters?

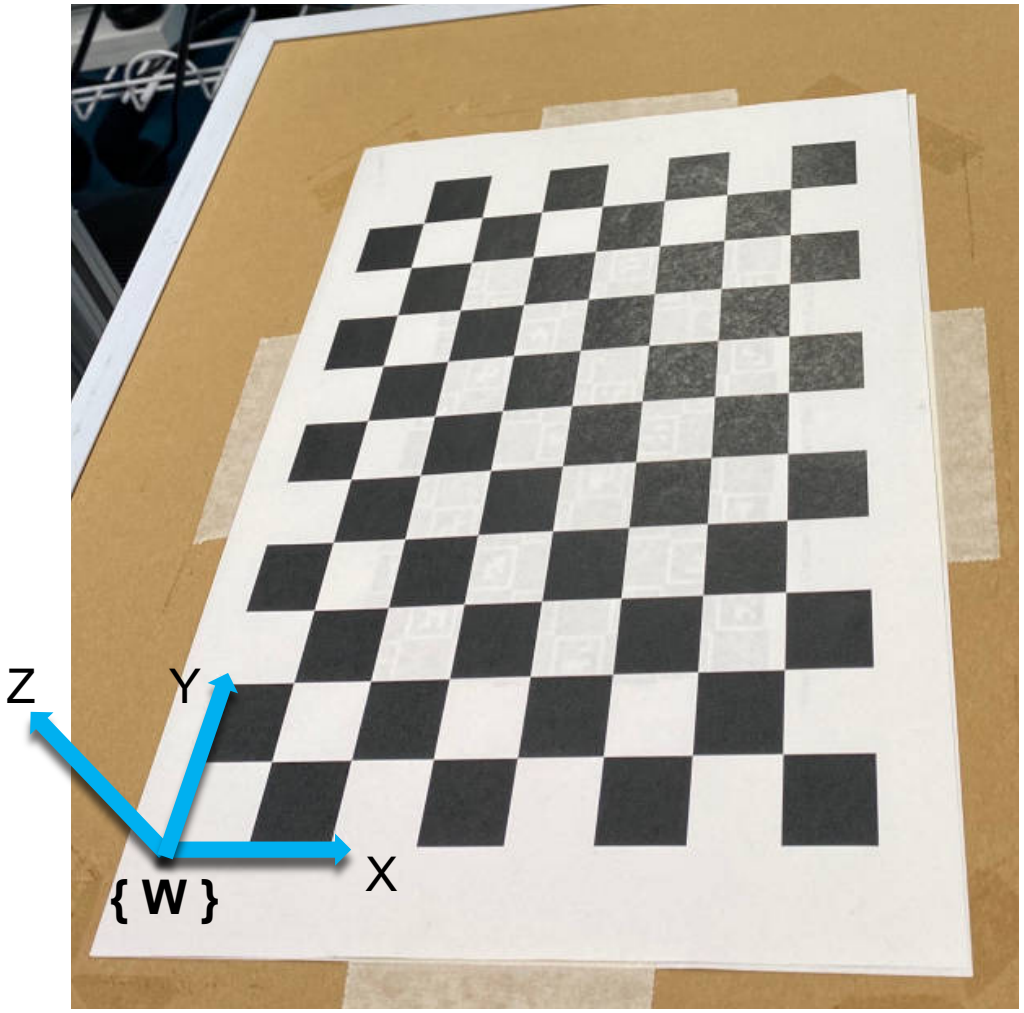
Known Unknowns Known

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha_u & 0 & u_0 \\ 0 & \alpha_v & v_0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t_3 \end{bmatrix} \cdot \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix}$$

Calibration Patterns

Known Unknowns Known

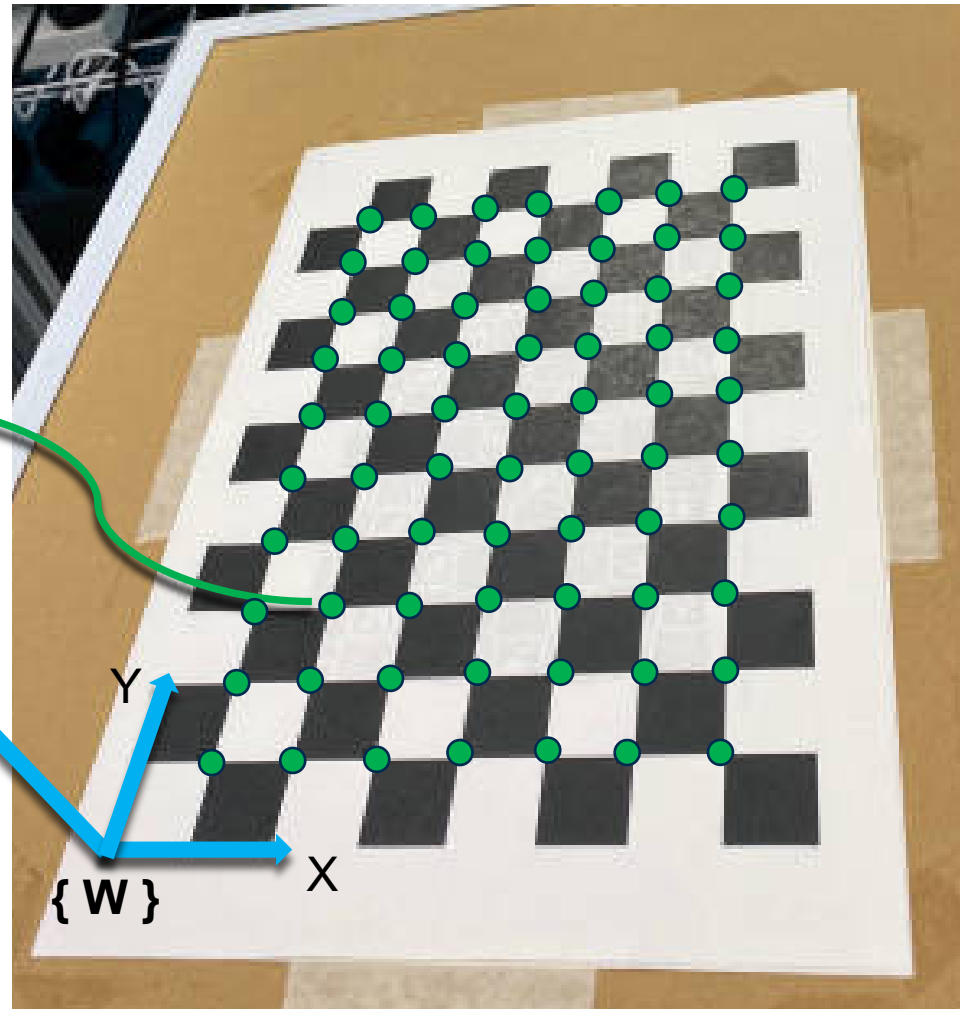
$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha_u & 0 & u_0 \\ 0 & \alpha_v & v_0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t_3 \end{bmatrix} \cdot \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix}$$



Calibration Patterns

Known Unknowns Known

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha_u & 0 & u_0 \\ 0 & \alpha_v & v_0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t_3 \end{bmatrix} \cdot \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix}$$



Calibration Patterns

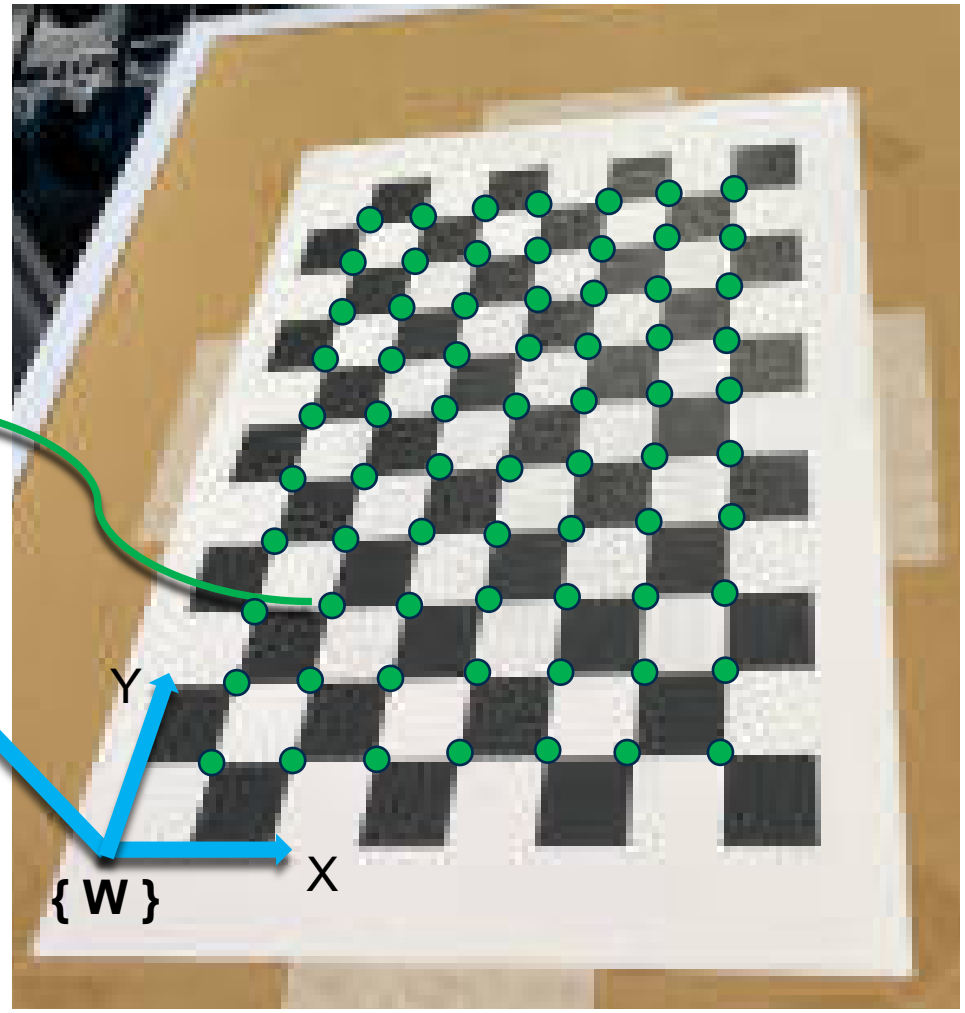
Known

Unknowns

Known

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha_u & 0 & u_0 \\ 0 & \alpha_v & v_0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t \end{bmatrix} \cdot \begin{bmatrix} X_w \\ Y_w \\ 0 \\ 1 \end{bmatrix}$$

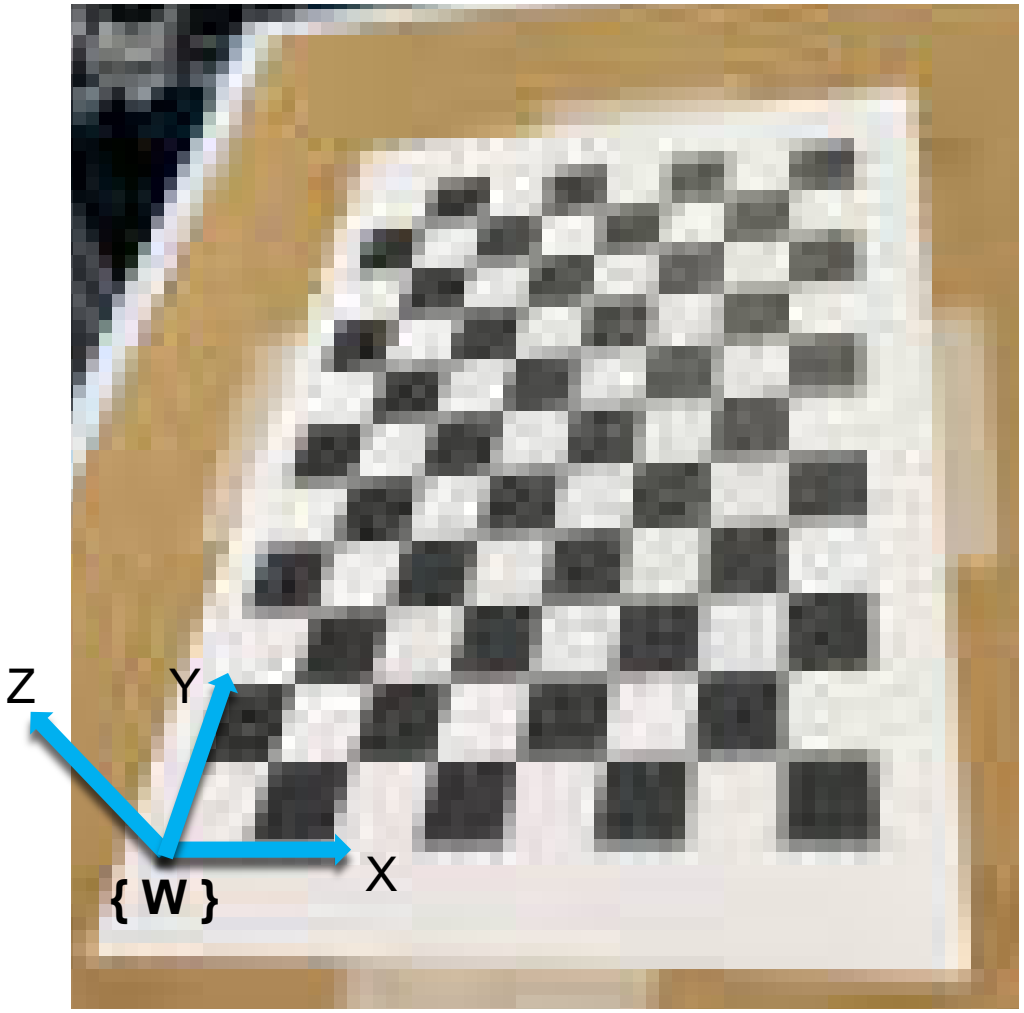
This is 0!



Calibration Patterns

Known Unknowns Known

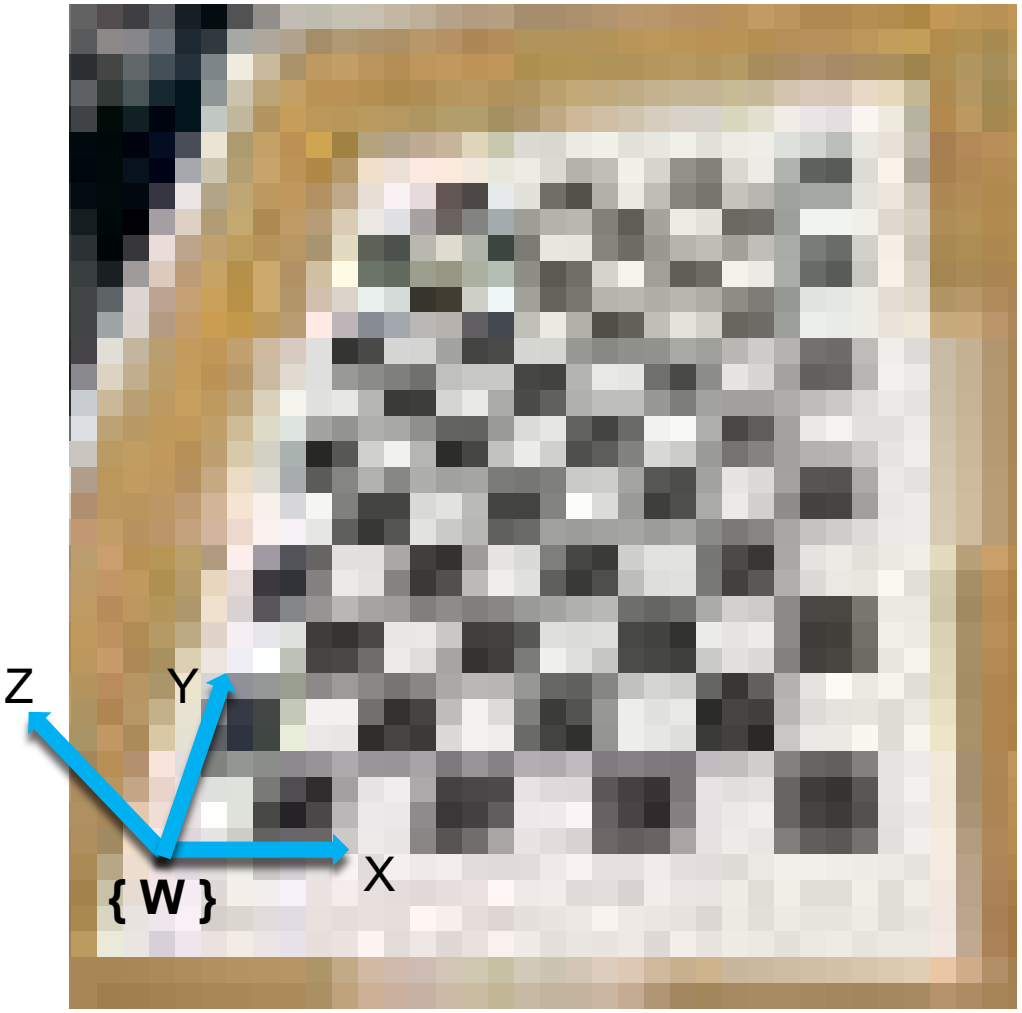
$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha_u & 0 & u_0 \\ 0 & \alpha_v & v_0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t_3 \end{bmatrix} \cdot \begin{bmatrix} X_w \\ Y_w \\ 0 \\ 1 \end{bmatrix}$$



Calibration Patterns

Known Unknowns Known

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \cdot \begin{bmatrix} X_w \\ Y_w \\ 1 \end{bmatrix}$$

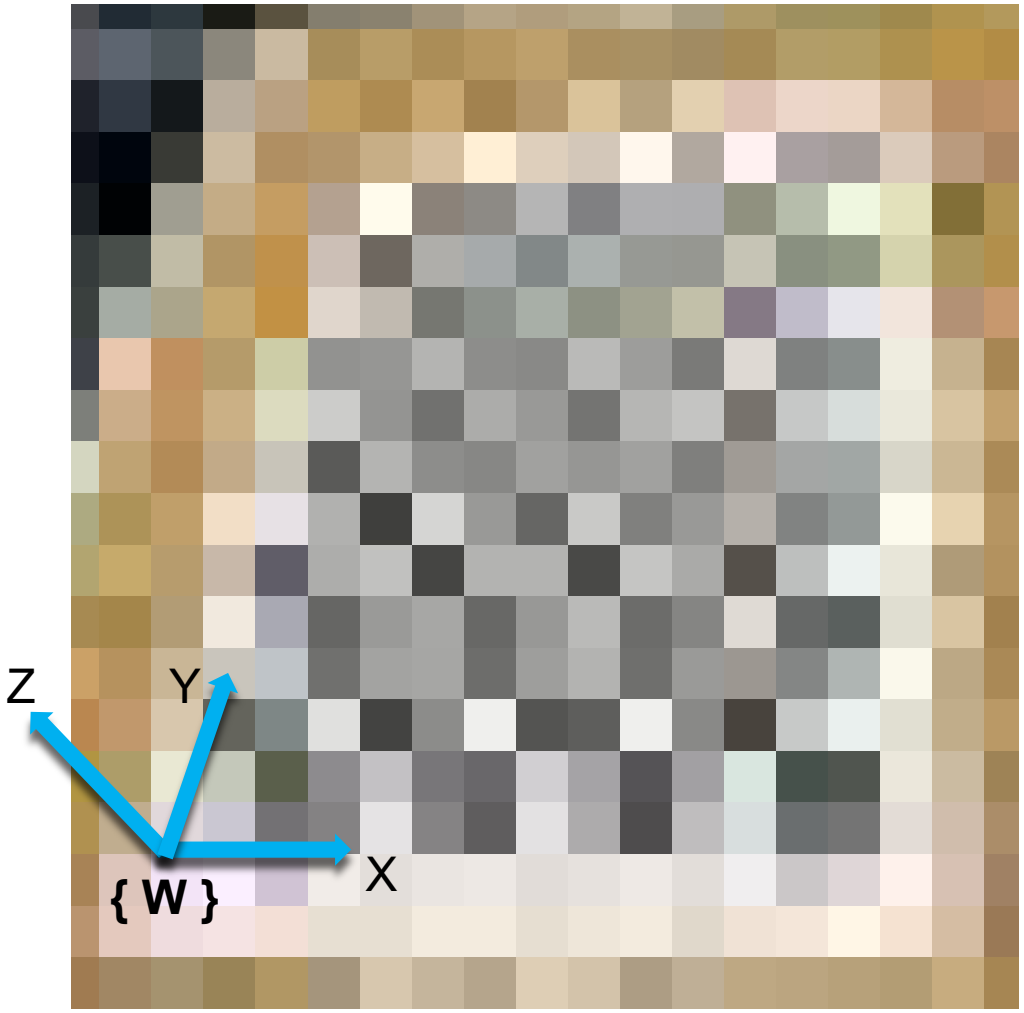


Calibration Patterns

Known Unknowns Known

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \cdot \begin{bmatrix} X_w \\ Y_w \\ 1 \end{bmatrix}$$

Homography



Applying the Direct Linear Transform (DLT) algorithm

- By re-arranging the terms, we obtain:

$$\begin{aligned}(h_1^{\mathrm{T}} - u_i h_3^{\mathrm{T}}) \cdot P_i = 0 \\ (h_2^{\mathrm{T}} - v_i h_3^{\mathrm{T}}) \cdot P_i = 0\end{aligned} \Rightarrow \begin{aligned}P_i^{\mathrm{T}} \cdot h_1 + 0 \cdot h_2^{\mathrm{T}} - u_i P_i^{\mathrm{T}} \cdot h_3^{\mathrm{T}} = 0 \\ 0 \cdot h_1^{\mathrm{T}} + P_i^{\mathrm{T}} \cdot h_2^{\mathrm{T}} - v_i P_i^{\mathrm{T}} \cdot h_3^{\mathrm{T}} = 0\end{aligned} \Rightarrow \begin{pmatrix} P_i^{\mathrm{T}} & 0^{\mathrm{T}} & -u_1 P_i^{\mathrm{T}} \\ 0^{\mathrm{T}} & P_i^{\mathrm{T}} & -v_1 P_i^{\mathrm{T}} \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \\ h_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

- For n points (from a **single view**), we can stack all these equations into a big matrix:

$$\begin{pmatrix} P_1^{\mathrm{T}} & 0^{\mathrm{T}} & -u_1 P_1^{\mathrm{T}} \\ 0^{\mathrm{T}} & P_1^{\mathrm{T}} & -v_1 P_1^{\mathrm{T}} \\ \cdots & \cdots & \cdots \\ P_n^{\mathrm{T}} & 0^{\mathrm{T}} & -u_n P_n^{\mathrm{T}} \\ 0^{\mathrm{T}} & P_n^{\mathrm{T}} & -v_n P_n^{\mathrm{T}} \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \\ h_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 0 \end{pmatrix} \Rightarrow \mathrm{Q} \cdot \mathrm{H} = 0$$

Q (this matrix is **known**) H (this matrix is **unknown**)

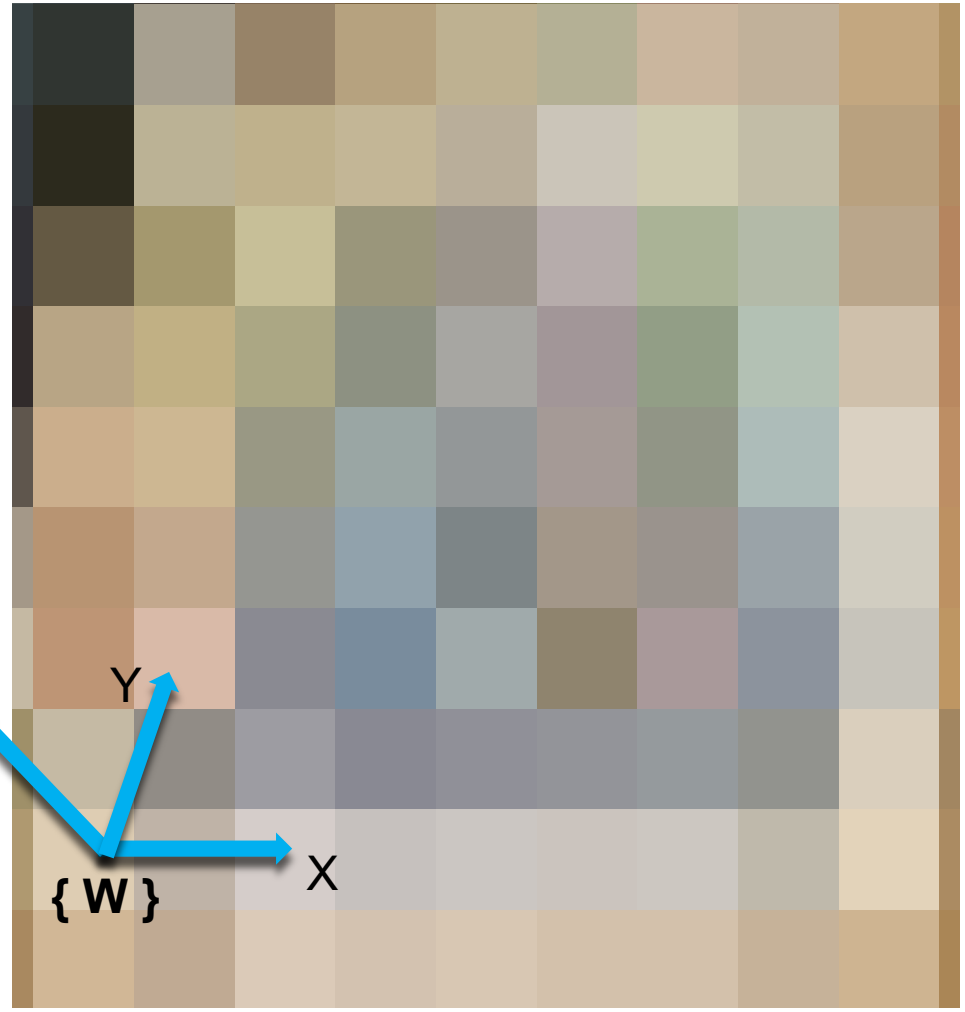
Slide courtesy of Davide Scaramuzza

Calibration Patterns

Known Unknowns Known

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \cdot \begin{bmatrix} X_w \\ Y_w \\ 1 \end{bmatrix}$$

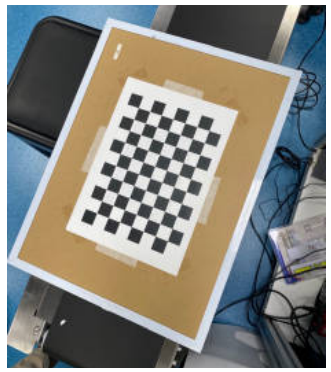
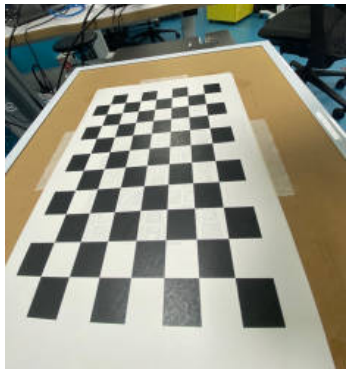
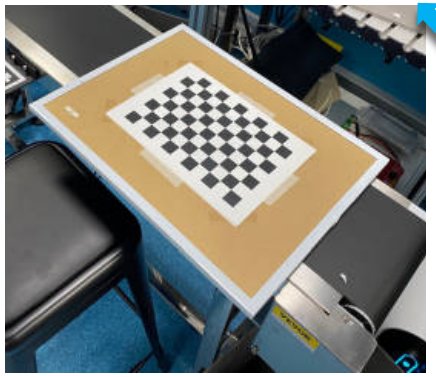
$$\begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} = \begin{bmatrix} \alpha_u & 0 & u_0 \\ 0 & \alpha_v & v_0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} r_{11} & r_{12} & t_1 \\ r_{21} & r_{22} & t_2 \\ r_{31} & r_{32} & t_3 \end{bmatrix}$$

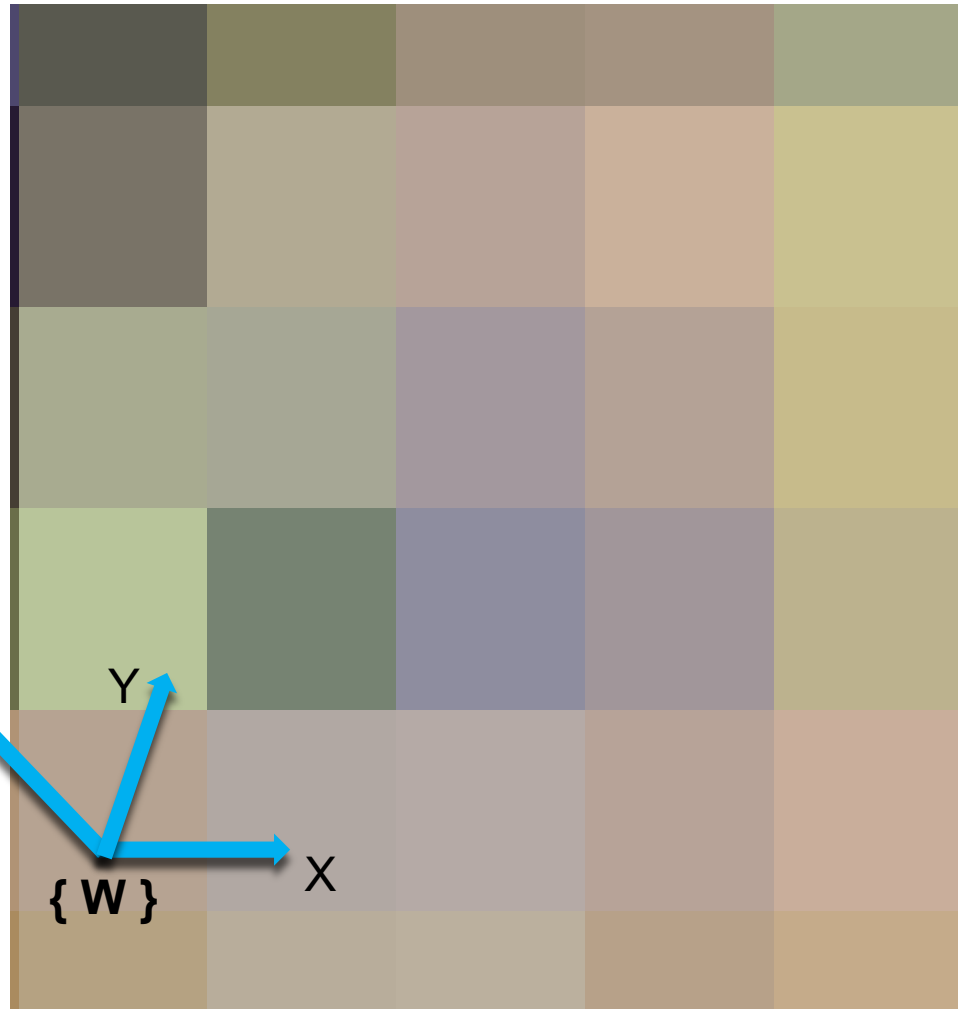


Calibration Patterns

Same Different

$$\begin{bmatrix} h_{11}^j & h_{12}^j & h_{13}^j \\ h_{21}^j & h_{22}^j & h_{23}^j \\ h_{31}^j & h_{33}^j & h_{33}^j \end{bmatrix} = \begin{bmatrix} \alpha_u & 0 & u_0 \\ 0 & \alpha_v & v_0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} r_{11}^j & r_{12}^j & t_1^j \\ r_{21}^j & r_{22}^j & t_2^j \\ r_{31}^j & r_{32}^j & t_3^j \end{bmatrix}$$



How to recover K, R, T from H and from multiple views?

1. Estimate the homography H_i for each i-th view using the DLT algorithm.

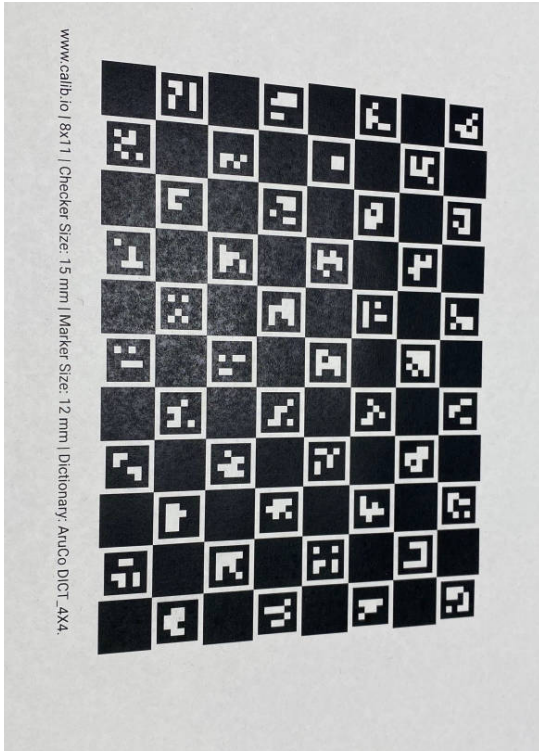
2. Determine the intrinsics K of the camera from a set of homographies:
   1. Each homography $H_i \sim K(\boldsymbol{r}_1, \boldsymbol{r}_2, \boldsymbol{t})$ provides two *linear* equations in the 6 entries of the matrix $\mathrm{B} \coloneqq K^{-\mathsf{T}}K^{-1}$. Letting $\boldsymbol{w}_1 \coloneqq K\boldsymbol{r}_1$, $\boldsymbol{w}_2 \coloneqq K\boldsymbol{r}_2$, the rotation constraints $\boldsymbol{r}_1^{\mathsf{T}}\boldsymbol{r}_1 = \boldsymbol{r}_2^{\mathsf{T}}\boldsymbol{r}_2 = 1$ and $\boldsymbol{r}_1^{\mathsf{T}}\boldsymbol{r}_2 = 0$ become $\boldsymbol{w}_1^{\mathsf{T}}B\boldsymbol{w}_1 - \boldsymbol{w}_2^{\mathsf{T}}B\boldsymbol{w}_2 = 0$ and $\boldsymbol{w}_1^{\mathsf{T}}B\boldsymbol{w}_2 = 0$.
   2. Stack 2*N* equations from *N* views, to yield a linear system $A\boldsymbol{b} = \boldsymbol{0}$. Solve for $\boldsymbol{b}$ (i.e., *B*) using the Singular Value Decomposition (SVD).
   3. Use Cholesky decomposition to obtain *K* from *B*.

3. The extrinsic parameters for each view can be computed using *K*:
   $\boldsymbol{r}_1 \sim \lambda K^{-1}H_i(:,1)$, $\boldsymbol{r}_2 \sim \lambda K^{-1}H_i(:,2)$, $\boldsymbol{r}_3 = \boldsymbol{r}_1 \times \boldsymbol{r}_2$ and $T_i = \lambda K^{-1}H_i(:,3)$, with $\lambda = 1/K^{-1}H_i(:,1)$.
   Finally, build $R_i = (\boldsymbol{r}_1, \boldsymbol{r}_2, \boldsymbol{r}_3)$ and enforce rotation matrix constraints.

Slide courtesy of Davide Scaramuzza

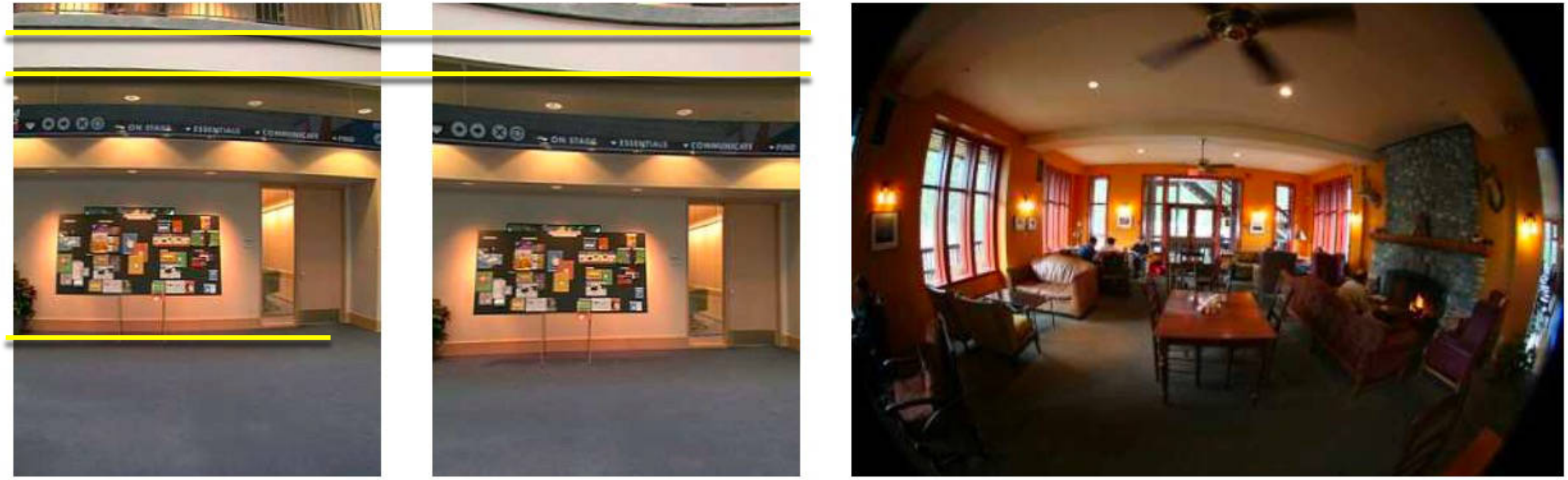
Calibration Patterns







Lens Distortion



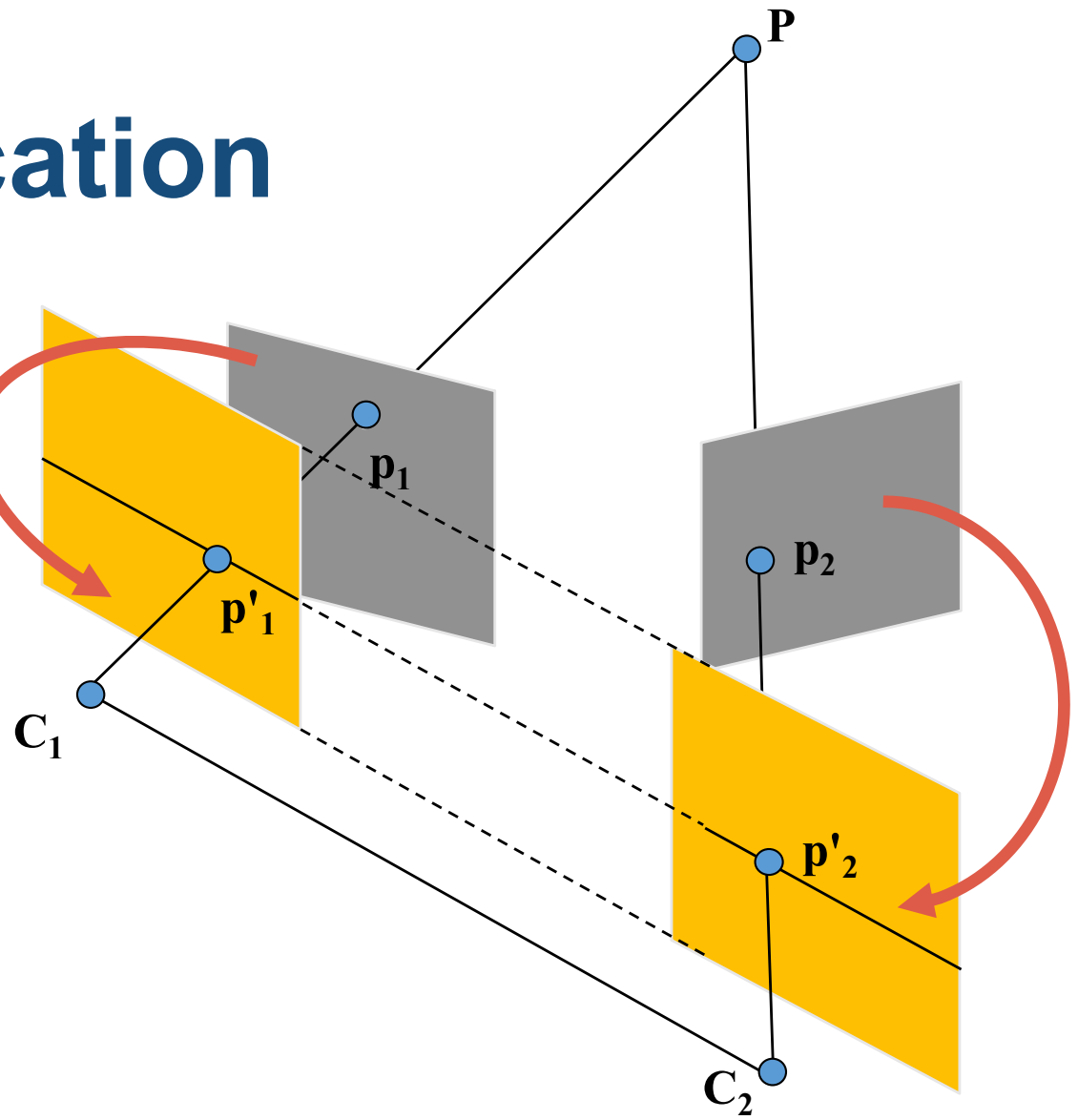
Radial lens distortions: (left) barrel, (middle) pincushion, and (right) fisheye.

Szeliski, Computer Vision: Algorithms and Applications (2nd Edition), Springer Verlag

Stereo Rectification

Stereo Rectification

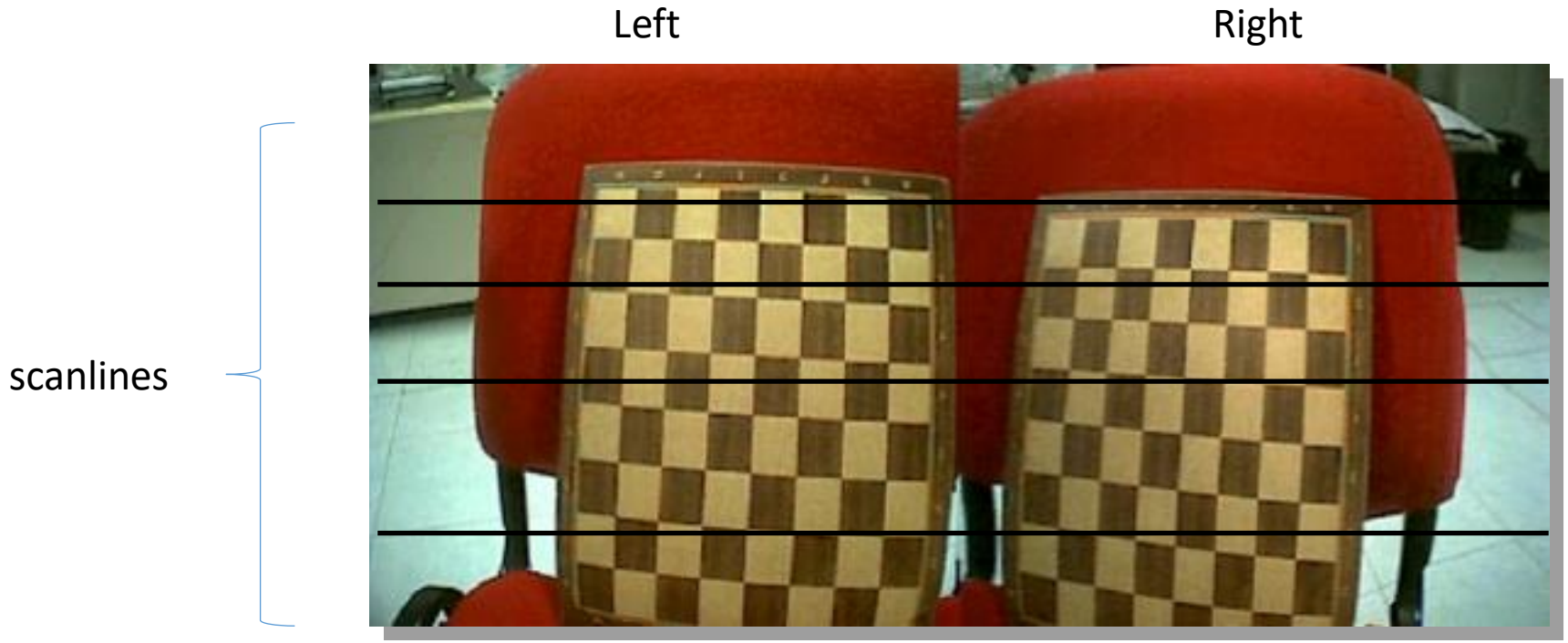
$$p'_1 = H_1 p_1$$



$$p'_2 = H_2 p_2$$

Illustration adopted from Davide Scaramuzza

Stereo Rectification: Example

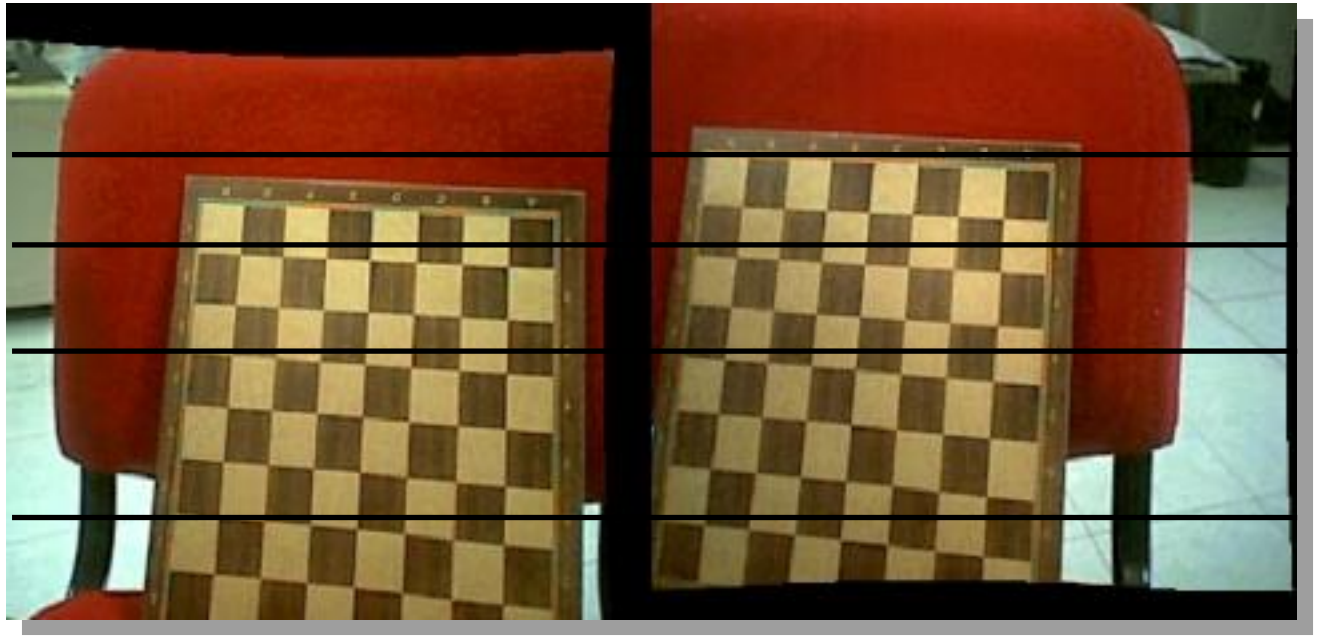


Slide courtesy of Davide Scaramuzza

Stereo Rectification: Example

- First, **compute lens distortion**

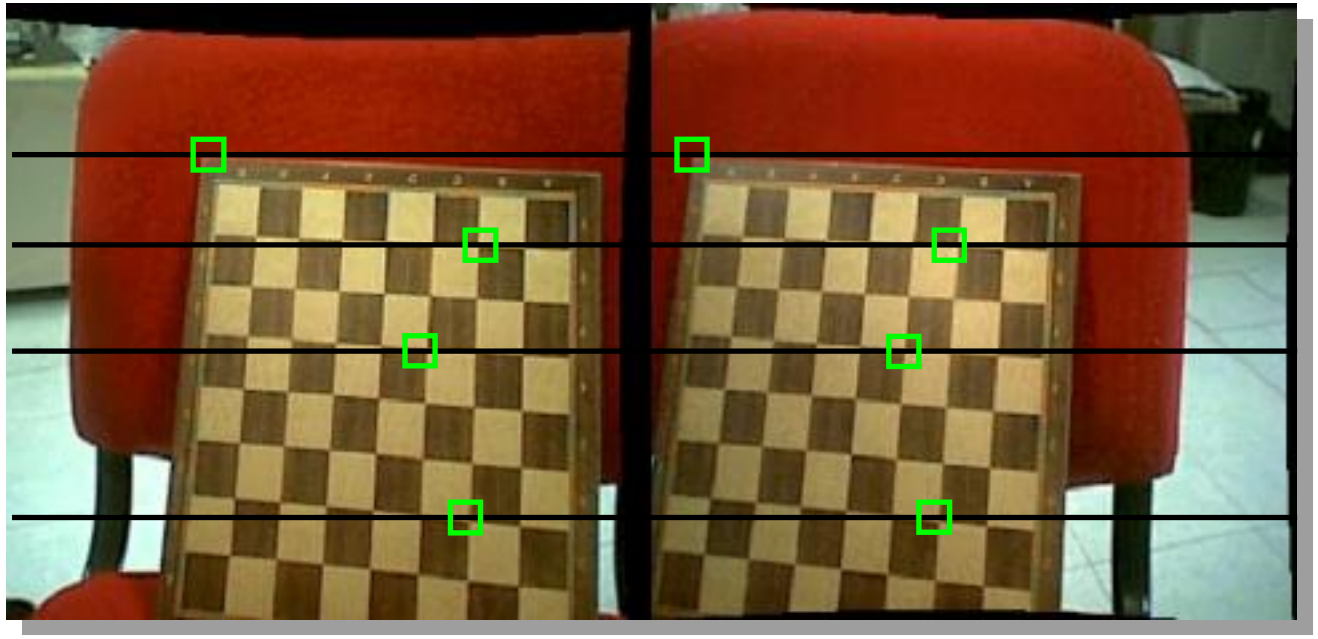
Left Right



Slide courtesy of Davide Scaramuzza

Stereo Rectification: Example

- First, **compute lens distortion**
- Then, **compute homographies and rectify**



Slide courtesy of Davide Scaramuzza

Stereo Rectification (2/5)

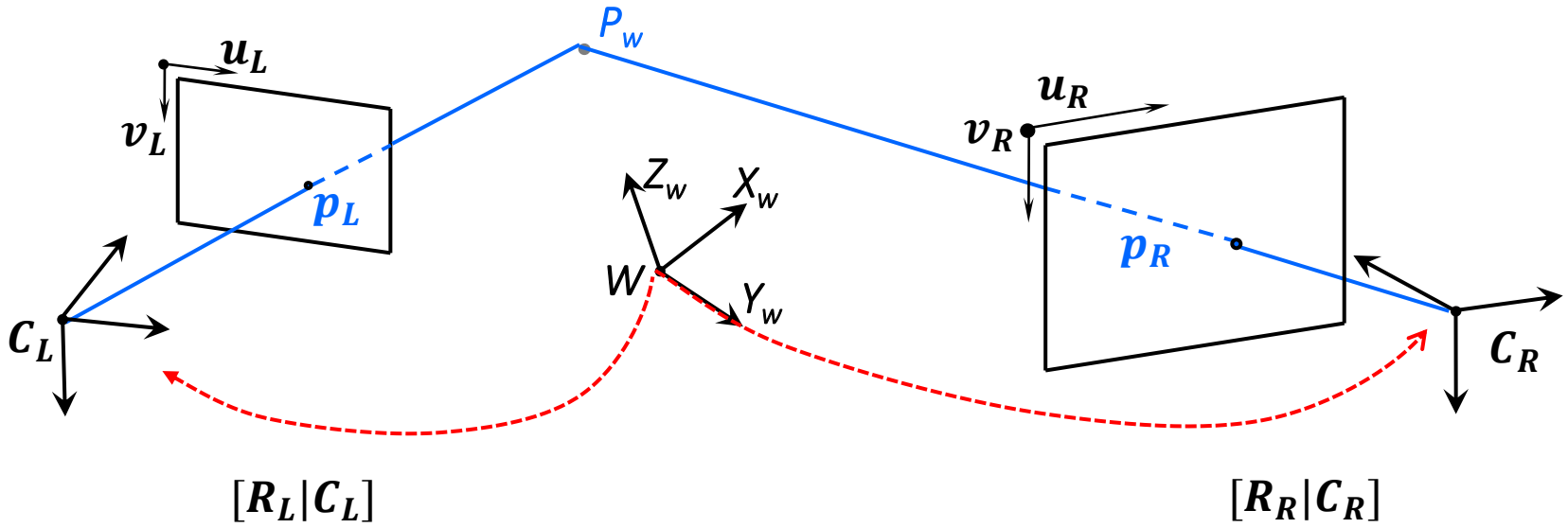
We can now write the Perspective Equation for the Left and Right cameras. For generality, we assume that Left and Right cameras have different intrinsic parameter matrices, K_L, K_R:

Left camera

$$\lambda_L \begin{bmatrix} u_L \\ v_L \\ 1 \end{bmatrix} = K_L {R_L}^{-1} \left(\begin{bmatrix} X_w \\ Y_w \\ Z_w \end{bmatrix} - C_L \right)$$

Right camera

$$\lambda_R \begin{bmatrix} u_R \\ v_R \\ 1 \end{bmatrix} = K_R {R_R}^{-1} \left(\begin{bmatrix} X_w \\ Y_w \\ Z_w \end{bmatrix} - C_R \right)$$



Slide courtesy of Davide Scaramuzza

Stereo Rectification (3/5)

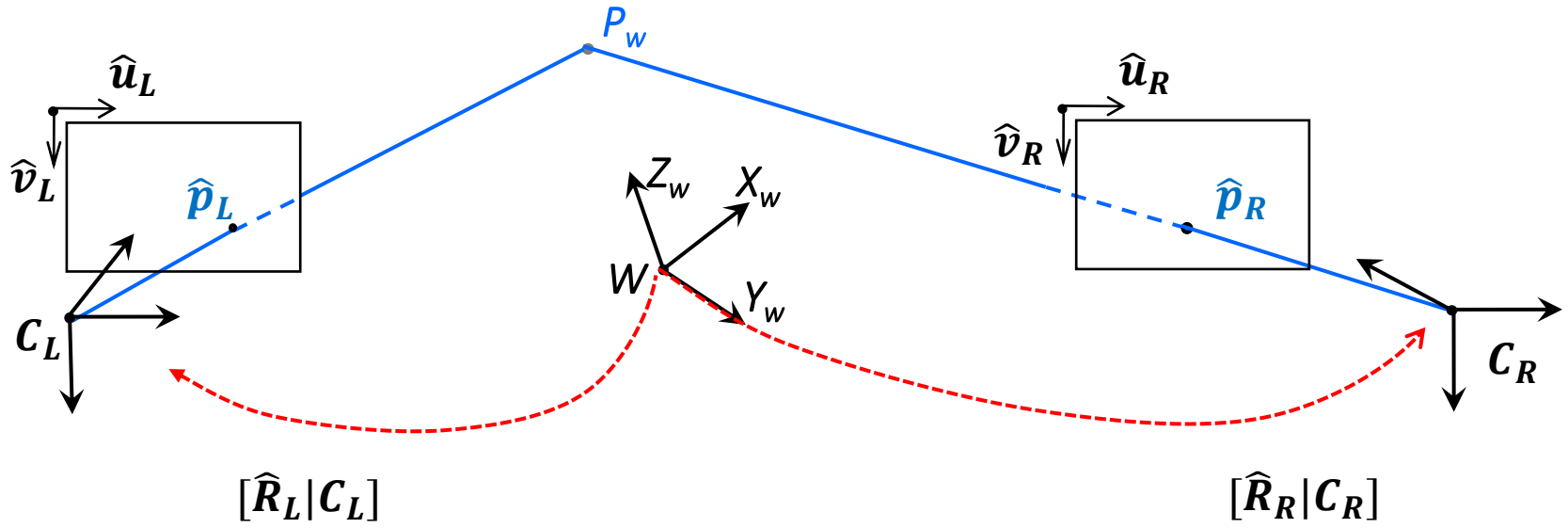
The goal of stereo rectification is to **warp** the **left** and **right camera** images such that their **image planes** are **coplanar (i.e., same $\widehat{R}$)** and their **intrinsic** parameters are **identical (i.e., same $\widehat{K}$)**

$$\lambda_L \begin{bmatrix} u_L \\ v_L \\ 1 \end{bmatrix} = K_L {R_L}^{-1} \left(\begin{bmatrix} X_w \\ Y_w \\ Z_w \end{bmatrix} - C_L \right)$$ **Old Left camera**

$$\rightarrow \hat{\lambda}_L \begin{bmatrix} \hat{u}_L \\ \hat{v}_L \\ 1 \end{bmatrix} = \widehat{\boldsymbol{K}} \widehat{\boldsymbol{R}}^{-1} \left(\begin{bmatrix} X_w \\ Y_w \\ Z_w \end{bmatrix} - C_L \right)$$ **New Left camera**

$$\lambda_R \begin{bmatrix} u_R \\ v_R \\ 1 \end{bmatrix} = K_R {R_R}^{-1} \left(\begin{bmatrix} X_w \\ Y_w \\ Z_w \end{bmatrix} - C_R \right)$$ **Old Right camera**

$$\rightarrow \hat{\lambda}_R \begin{bmatrix} \hat{u}_R \\ \hat{v}_R \\ 1 \end{bmatrix} = \widehat{\boldsymbol{K}} \widehat{\boldsymbol{R}}^{-1} \left(\begin{bmatrix} X_w \\ Y_w \\ Z_w \end{bmatrix} - C_R \right)$$ **New Right camera**



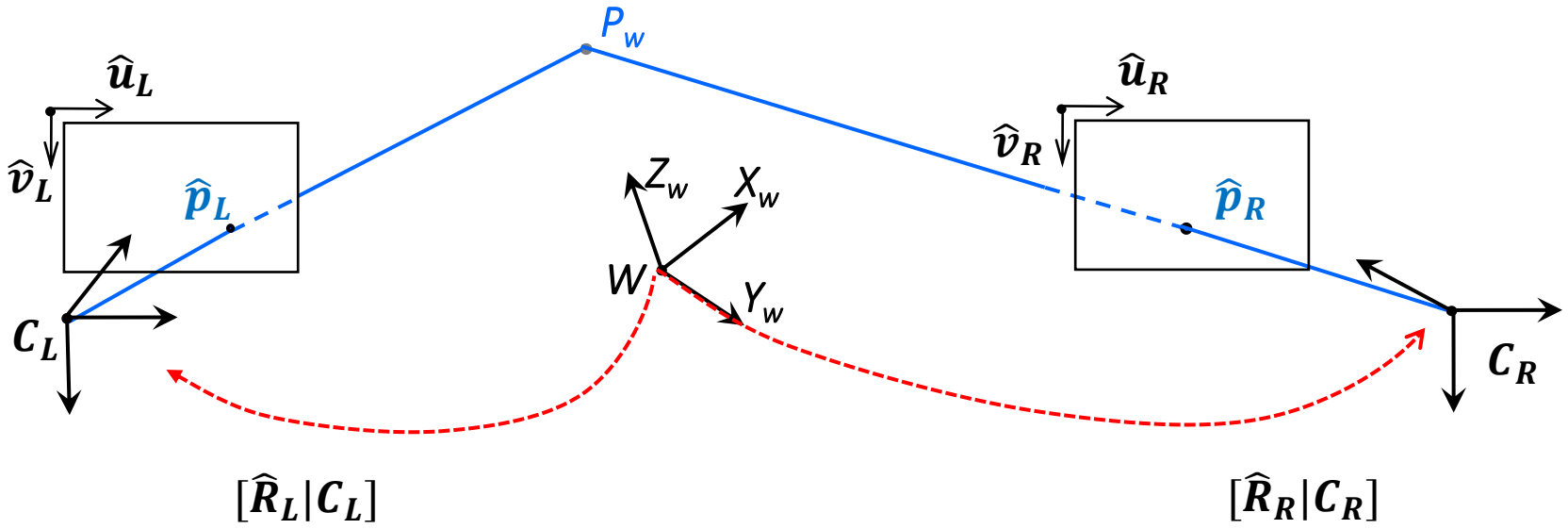
Slide courtesy of Davide Scaramuzza

Stereo Rectification (4/5)

By solving with respect to (X_w, Y_w, Z_w) for each camera, we can compute the Homography that needs to be applied to rectify each camera image:

$$\hat{\lambda}_L \begin{bmatrix} \hat{u}_L \\ \hat{v}_L \\ 1 \end{bmatrix} = \lambda_L \underbrace{\hat{\boldsymbol{K}}\hat{\boldsymbol{R}}^{-1} R_L K_L{}^{-1}}_{\text{Homography of Left Camera}} \begin{bmatrix} u_L \\ v_L \\ 1 \end{bmatrix}$$

$$\hat{\lambda}_R \begin{bmatrix} \hat{u}_R \\ \hat{v}_R \\ 1 \end{bmatrix} = \lambda_R \underbrace{\hat{\boldsymbol{K}}\hat{\boldsymbol{R}}^{-1} R_R K_R{}^{-1}}_{\text{Homography of Right Camera}} \begin{bmatrix} u_R \\ v_R \\ 1 \end{bmatrix}$$



Slide courtesy of Davide Scaramuzza

Stereo Rectification (5/5)

- How do we choose the new $\widehat{\boldsymbol{K}}$? A common choice is to take the arithmetic average of K_L and K_R:

$$\widehat{\boldsymbol{K}} = \frac{K_L + K_R}{2}$$

- How do we choose the new $\widehat{\boldsymbol{R}} = [\widehat{r_1}, \widehat{r_2}, \widehat{r_3}]$, with $\widehat{r_1}, \widehat{r_2}, \widehat{r_3}$ being the column vectors of $\hat{R}$?

  A common choice is as follows:

$$\widehat{r_1} = \frac{C_R - C_L}{\|C_R - C_L\|}$$ This makes the new image planes parallel to the baseline

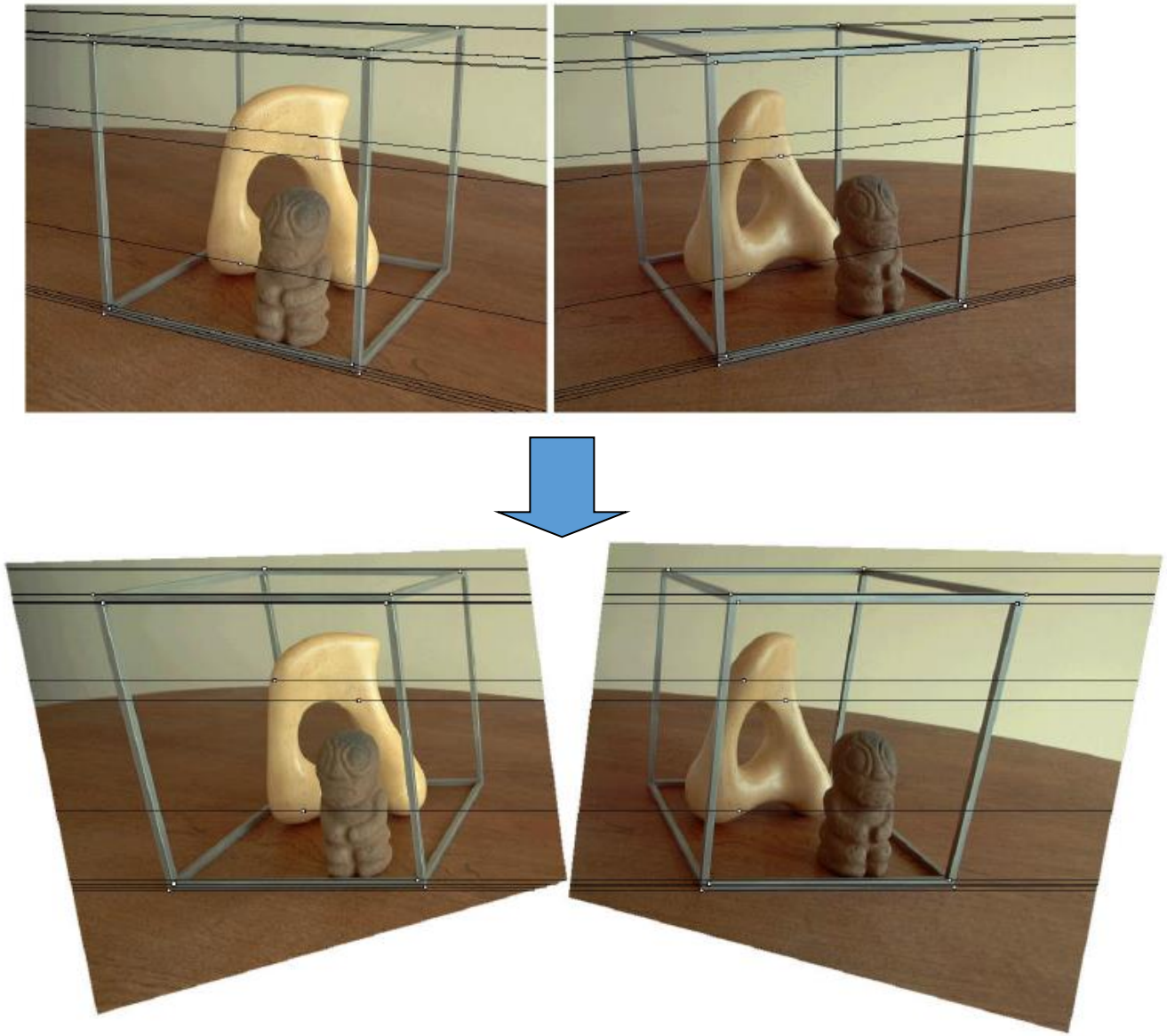
$$\widehat{r_2} = r_{3L} \times \widehat{r_1}$$ where r_{3L} is the 3rd column of the rotation matrix of the left camera, i.e., R_L

$$\widehat{r_3} = \widehat{r_1} \times \widehat{r_2}$$

Fusiello, Trucco, Verri, "A Compact Algorithm for Rectification of Stereo Pairs, International Conference on Computer Vision (ICCV), 1999. PDF.

Slide courtesy of Davide Scaramuzza

Stereo Rectification: Example



Slide courtesy of Davide Scaramuzza

Recap

Recap

- What is the pinhole camera model?
- What are the parameters in a pinhole camera?
- How are the camera parameters determined?
- How can a camera at an arbitrary pose in the world be described?
- What are the actions of a camera on points, lines, and planes in the world?
- What do we mean by the concepts of "points at infinity" and "lines at infinity"?

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Where to go from here

- Watch **Self-guided Learning videos** on **Stereo Vision** on Canvas
  - How does a stereo camera work?
  - How do we reconstruct 3D information from a stereo camera?
  - What assumptions are made by common algorithms of stereo vision and how do we ensure they are valid?

- Attempt to solve Tutorial problems before Friday, see Canvas
  - We will talk through your solutions & questions

- Optional: Watch deep-dive videos on Canvas on camera calibration for more in-depth details

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